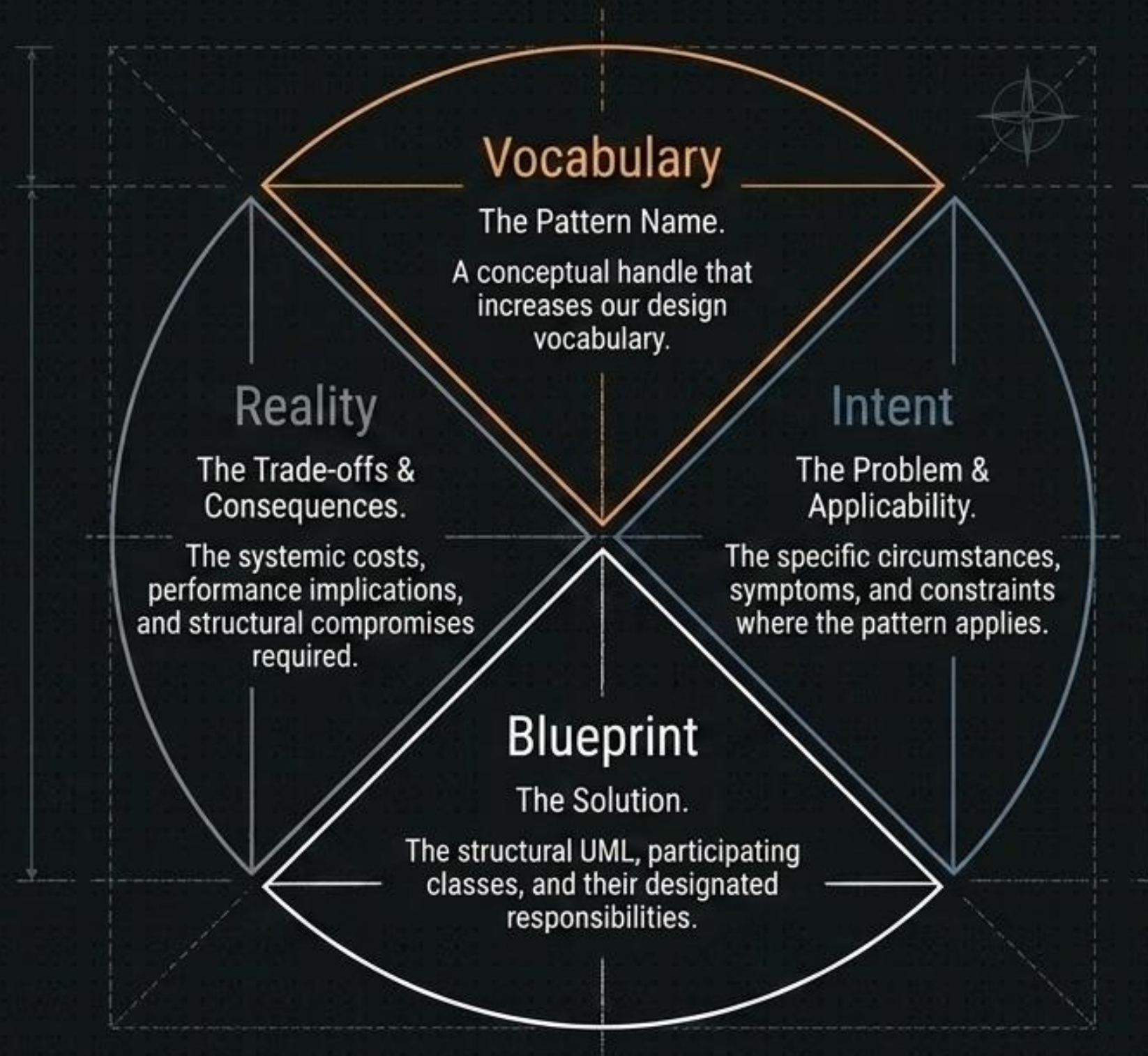


The Architecture of Reusable Software

Core Concepts and Classifications of Design Patterns



The Anatomy of a Design Pattern



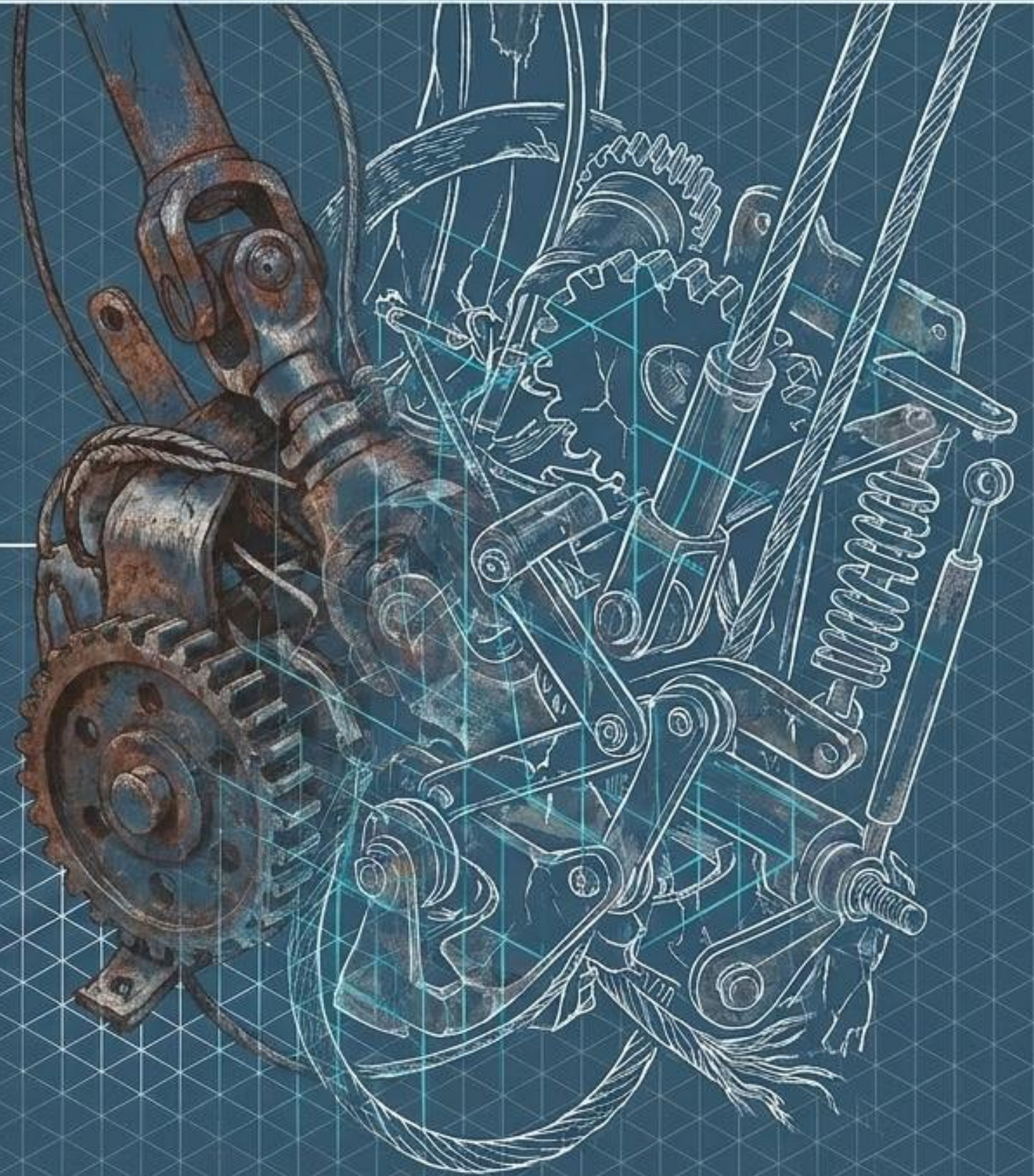
THE BLUEPRINT SOLUTION

Core Definition Panel

Design Patterns are typical solutions to commonly occurring problems in software design. They are not off-the-shelf code you can copy and paste, but conceptual blueprints that can be customized to solve specific design problems in your architecture.

Key Insight

Originating from the collective experience of top-tier designers, these patterns provide a shared vocabulary, allowing developers to create flexible, elegant, and reusable object-oriented systems without rediscovering the solutions themselves.

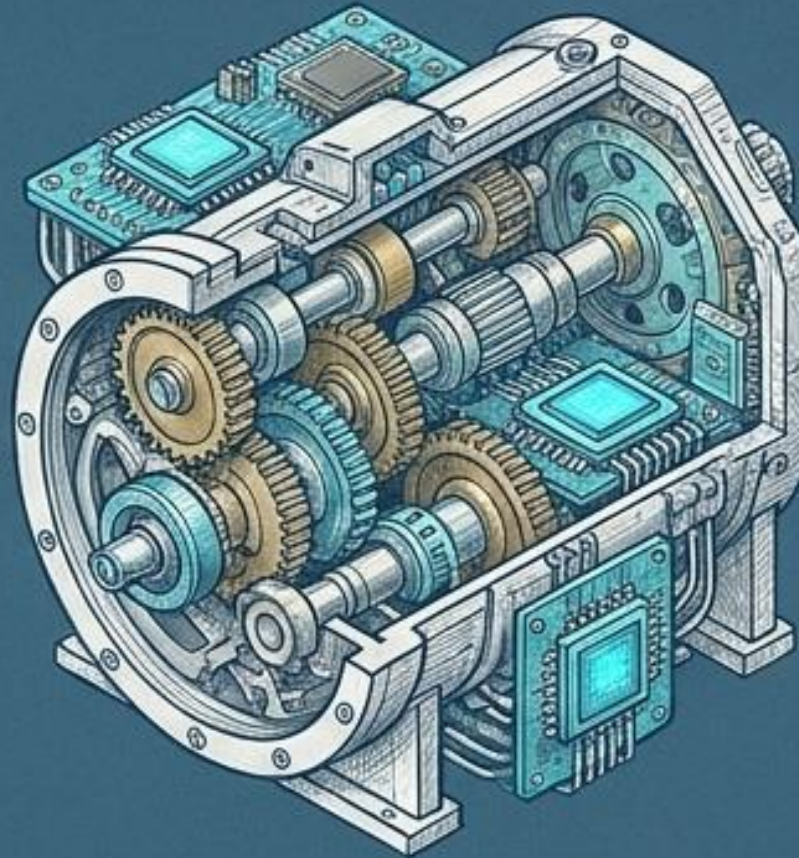


The Developer's Ecosystem



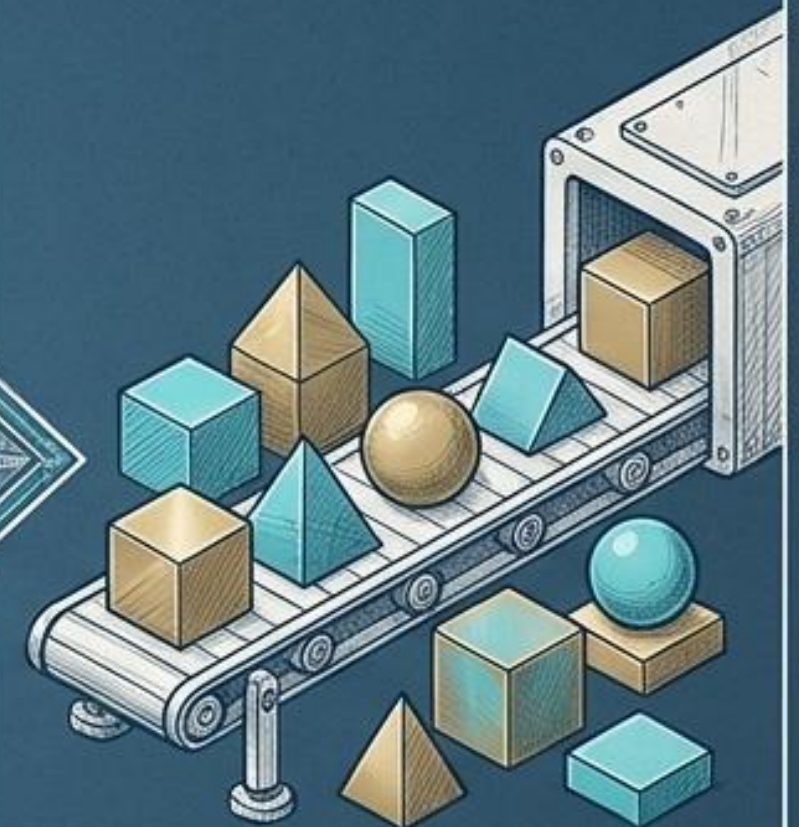
Phase 1: Messy Code

Legacy systems burdened by technical debt and code smells.



Phase 2: The Refactoring Engine

The systematic process of improving code using general principles (like SOLID) to untangle the mess without creating new functionality.



Phase 3: Design Patterns

The target state. Refactored code eventually crystallizes into these proven architectural blueprints.

The Gang of Four Classification Matrix

		
Creational	Structural	Behavioral
Core Intent: How objects are made.	Core Intent: How objects are assembled.	Core Intent: How objects communicate.
What it Abstracts: The instantiation process.	What it Abstracts: Object composition and interfaces.	What it Abstracts: Control flow and assignment of responsibilities.
Real-World Analogy: A high-tech manufacturing plant.	Real-World Analogy: Architectural scaffolding and load-bearing trusses.	Real-World Analogy: A complex switchboard or neural network.

Pillar 1: Creational Patterns

Creational patterns separate the system from how its objects are created, composed, and represented.

The Why

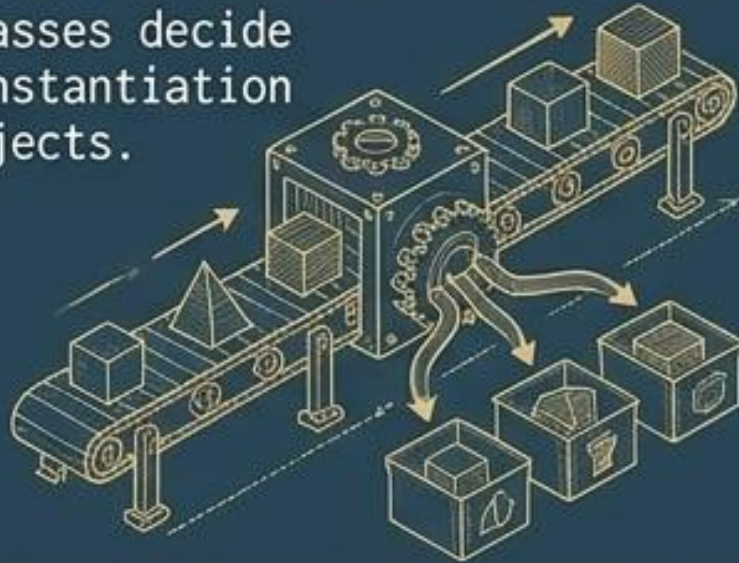
By hiding the complex logic of creation, these patterns make systems independent of specific classes, allowing the software to scale in complexity without tightly coupling the instantiation process to the core business logic.



The Creational Catalog

Factory Method

Subclasses decide the instantiation of objects.



Abstract Factory

Creates families of related objects without specifying concrete classes.



Builder

Step-by-step construction of complex objects.



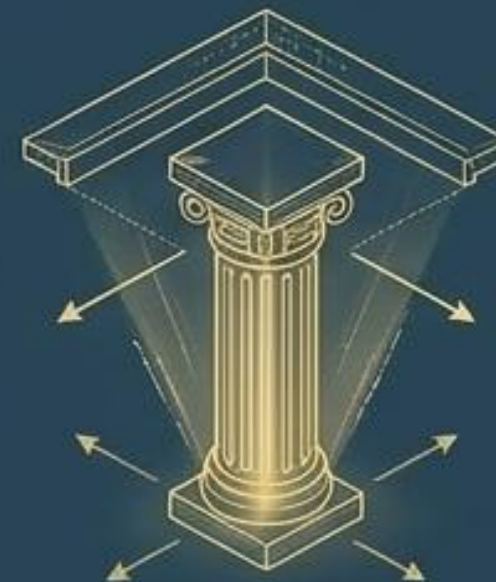
Prototype

Creating new objects by cloning an existing instance.



Singleton

Ensures a class has only one single, globally accessible instance.



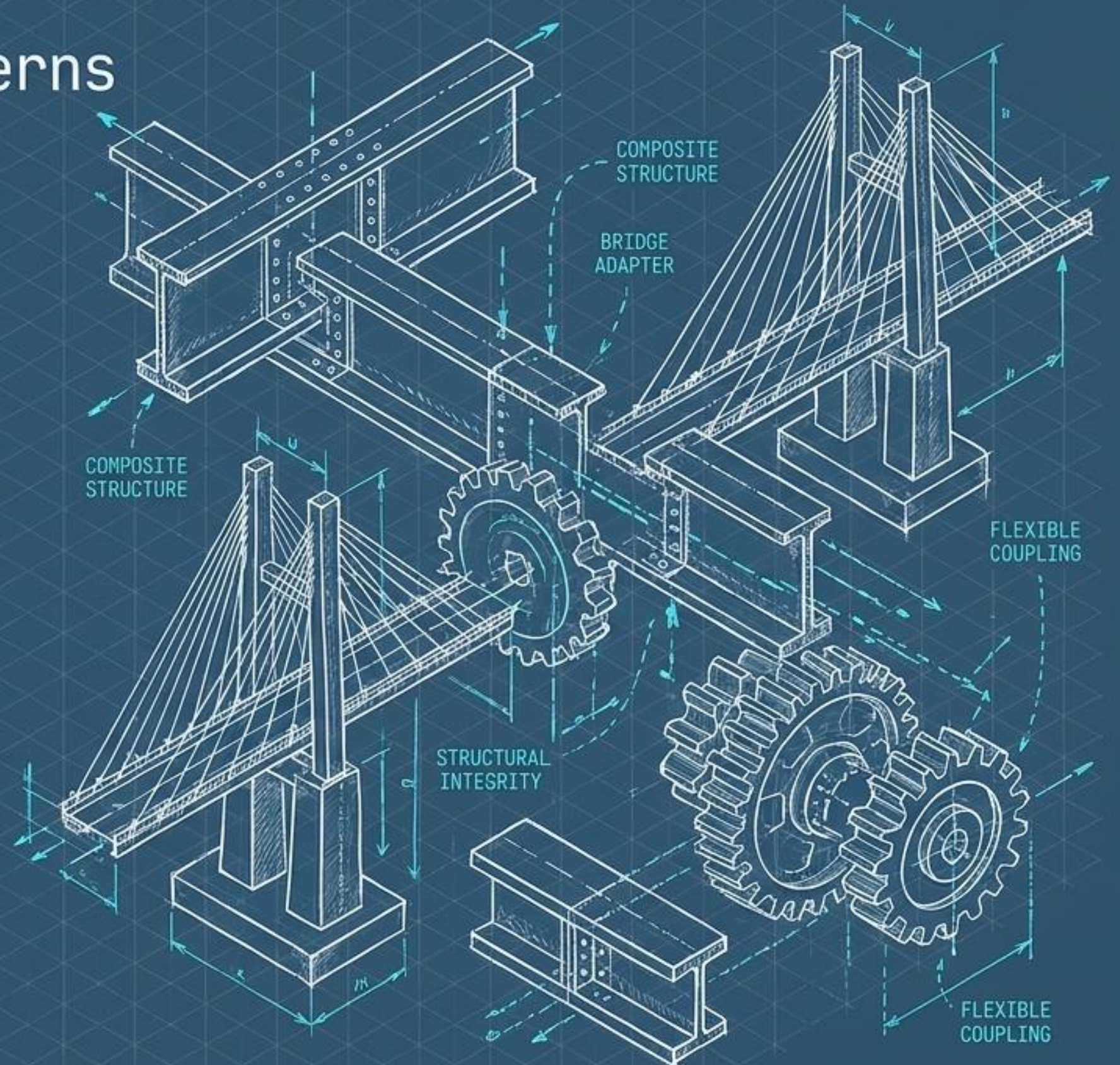
Pillar 2: Structural Patterns

Structural patterns explain how to assemble objects and classes into larger, more complex structures.

The Why

They ensure that even as individual pieces of a system grow and change, the overarching structure remains flexible, efficient, and structurally sound.

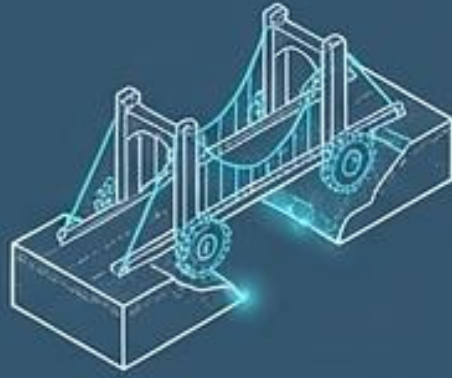
They are the glue and the scaffolding of object-oriented architecture.



The Structural Catalog

Bridge

Separates abstraction from implementation.



Composite

Composes objects into tree structures.



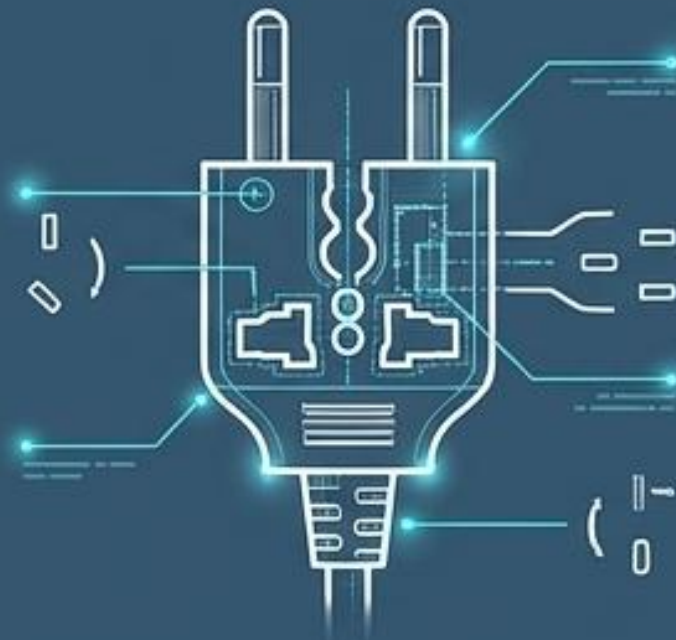
Decorator

Attaches new behaviors dynamically.



Adapter

Makes incompatible interfaces collaborate.



Facade

Simplified interface to a complex subsystem.



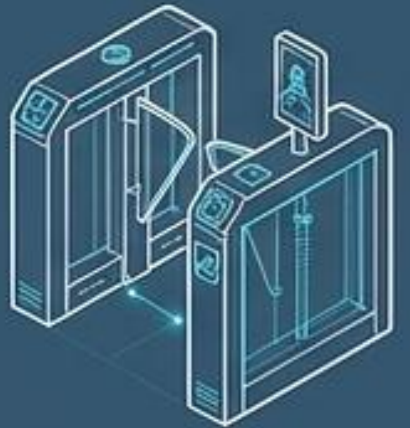
Flyweight

Shares state to support massive numbers of objects efficiently.



Proxy

Placeholder to control access to another object.

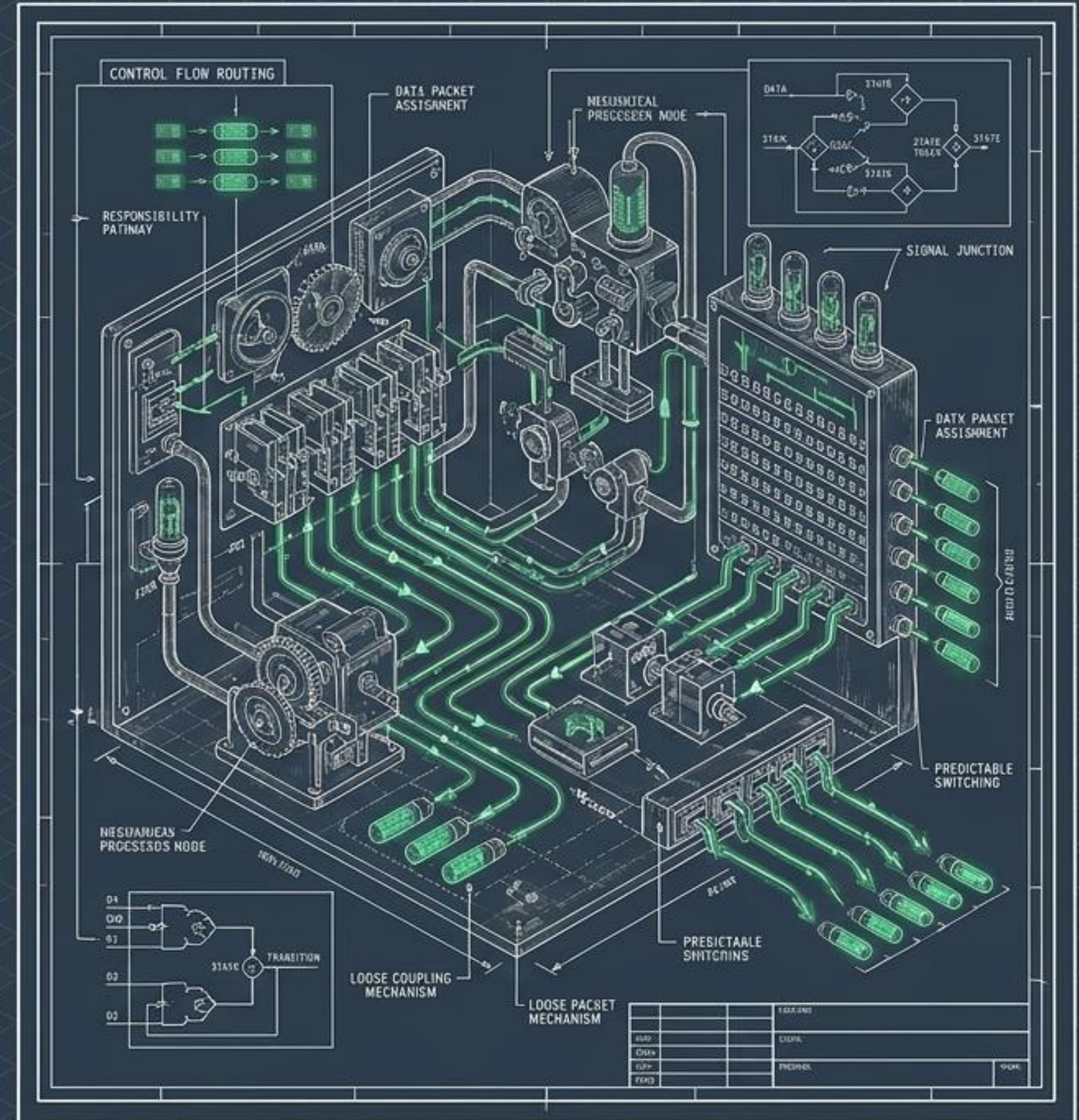


Pillar 3: Behavioral Patterns

Behavioral patterns focus entirely on algorithms and the assignment of responsibilities between objects.

The Why

They dictate not just what objects are, but how they talk to each other. They untangle the communication web, ensuring that control flow between distinct entities remains predictable, scalable, and loosely coupled.



The Behavioral Catalog

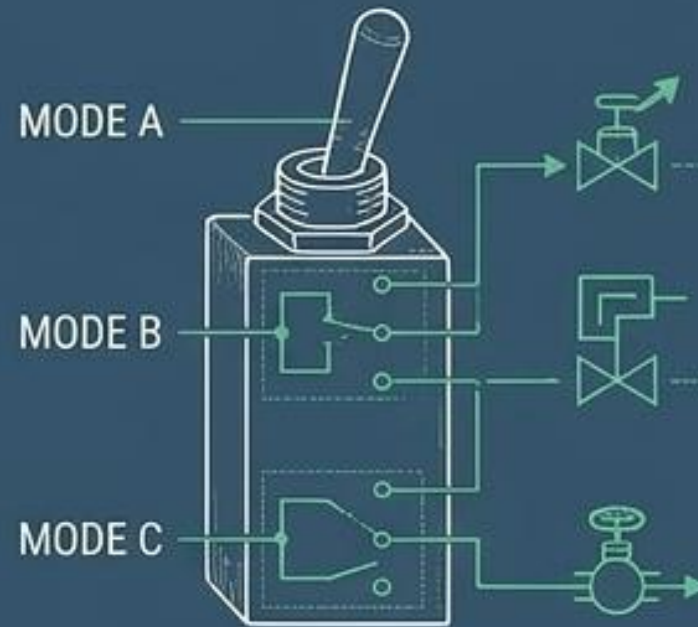
Observer

Defines a subscription mechanism to notify objects of state changes.



State

Alters an object's behavior when its internal state changes.



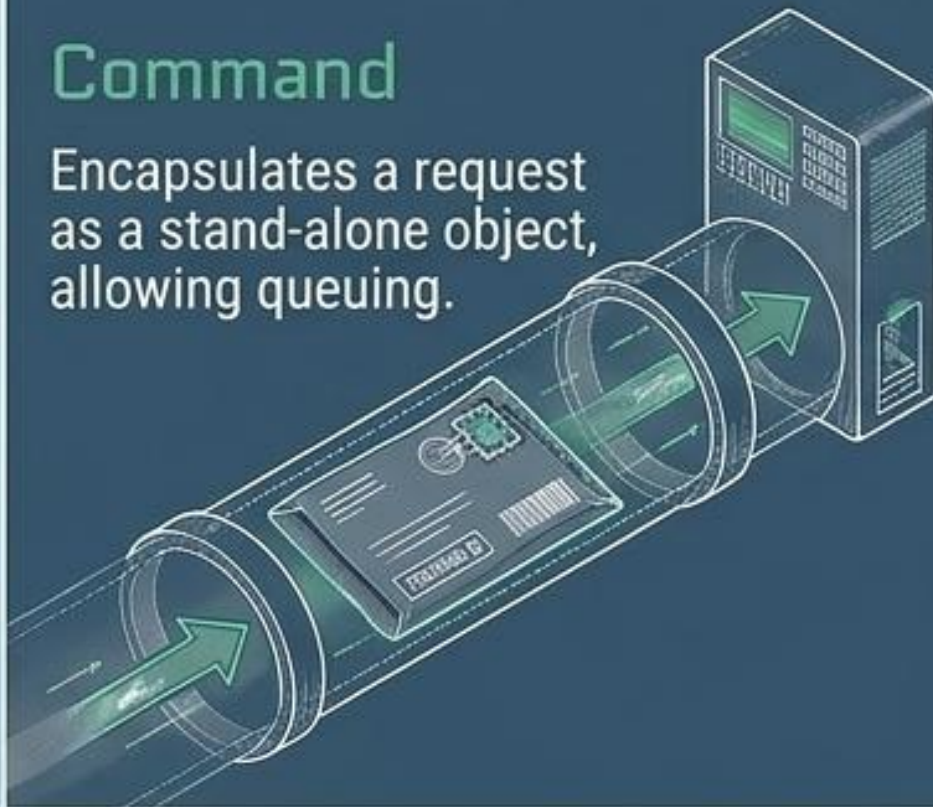
Strategy

Encapsulates a family of algorithms, making them interchangeable.



Command

Encapsulates a request as a stand-alone object, allowing queuing.

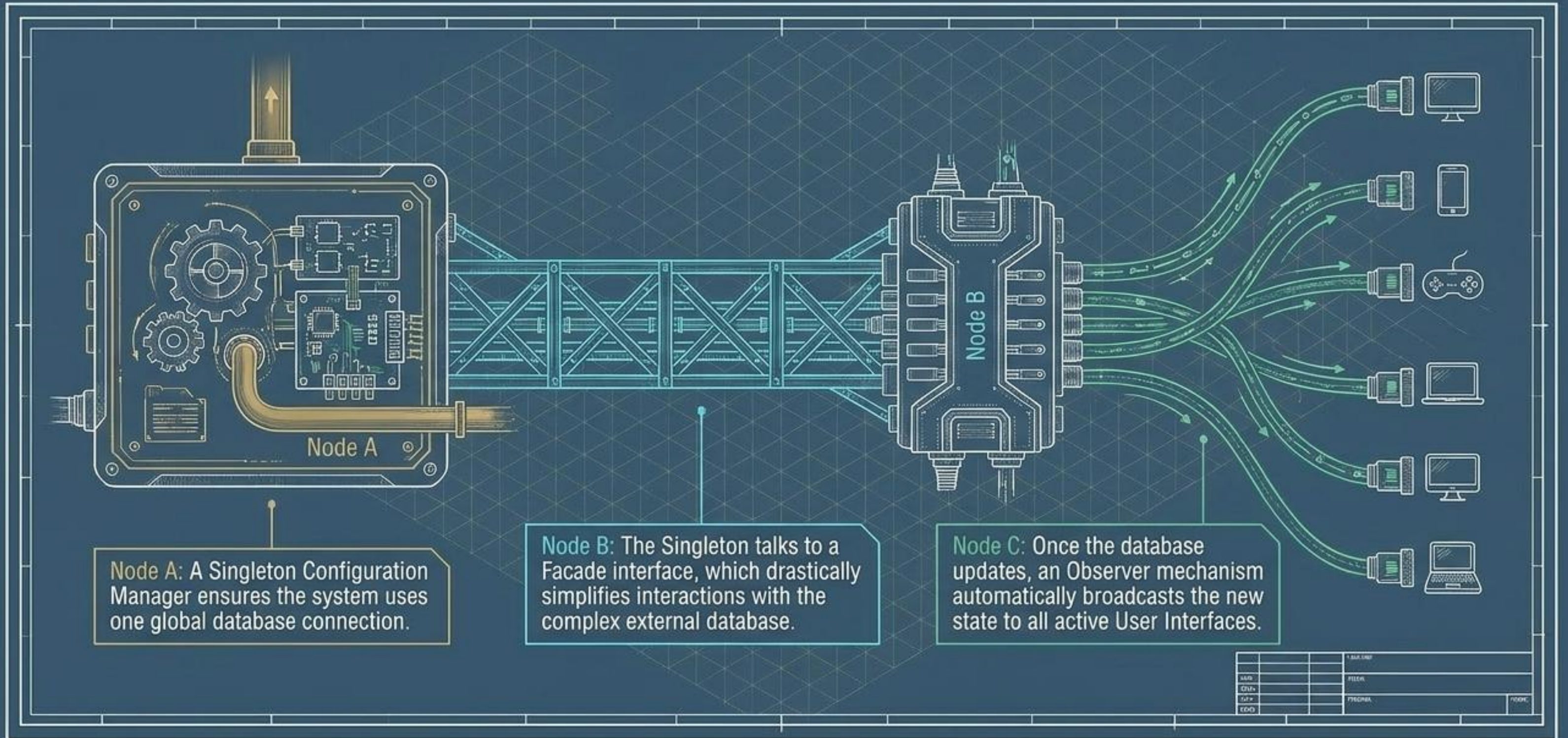


Iterator

Traverses elements of a collection without exposing underlying representation.



Patterns in Concert



The True Value of Patterns

A Shared Vocabulary

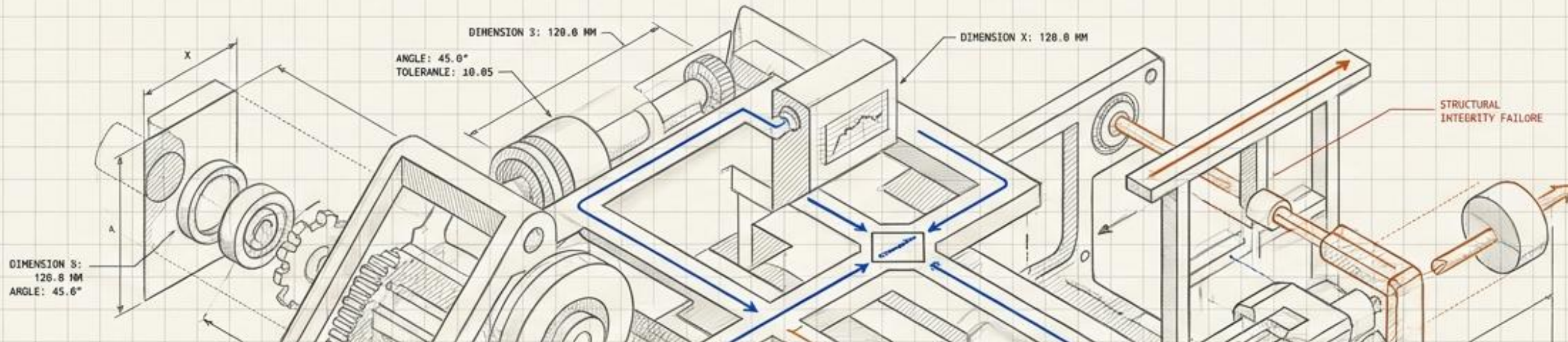
Saying “Just use a Factory” instantly communicates a dozen complex architectural decisions in four words.

Eradication of Reinvention

Leveraging 20 years of collective software engineering experience to solve recurring problems immediately.

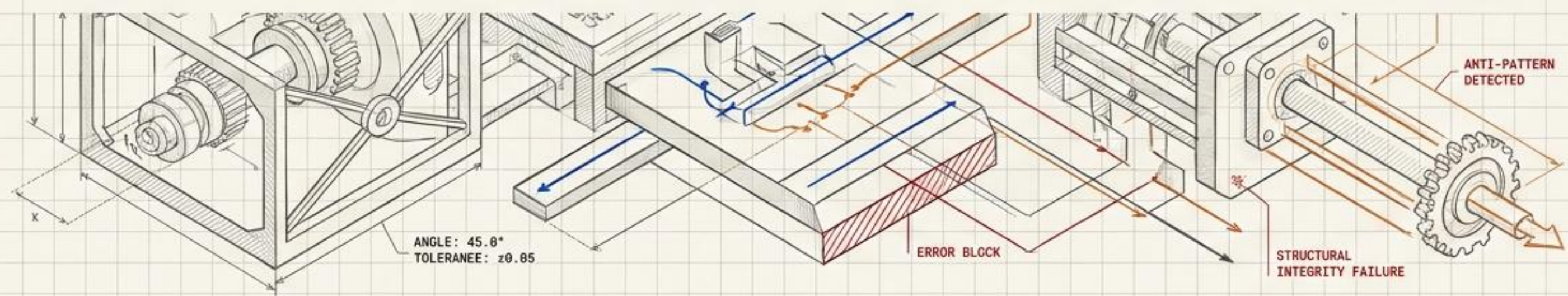
Resilient Architecture

Future-proofing codebases against changing requirements through intentional, reusable object-oriented design.

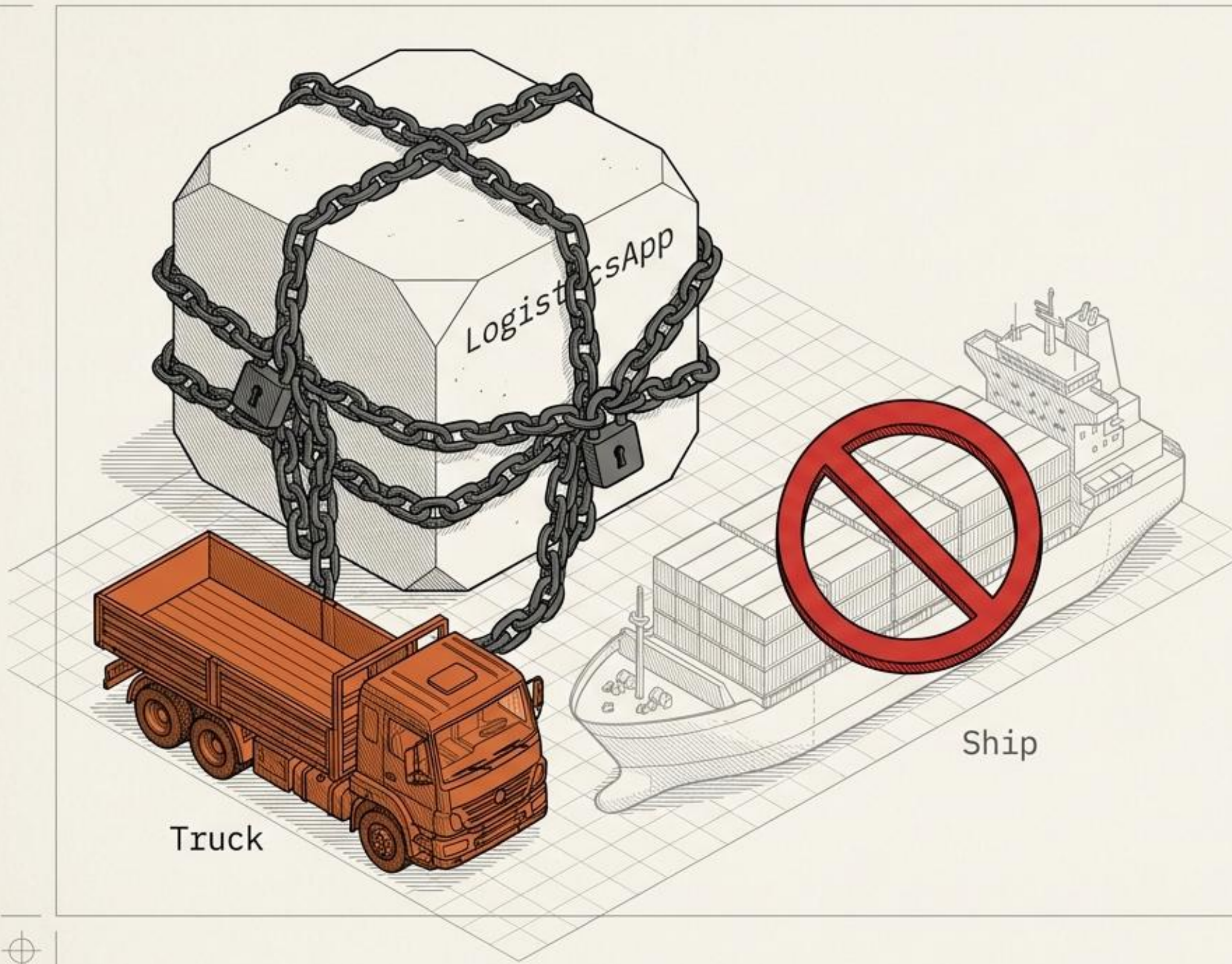


CREATIONAL DESIGN PATTERNS

Architecting Object Creation: A Diagnostic Blueprint



FACTORY METHOD • ABSTRACT FACTORY • BUILDER • PROTOTYPE • SINGLETON



The Rigid Dependency

SYMPTOM

Core business logic is entangled with the new keyword.

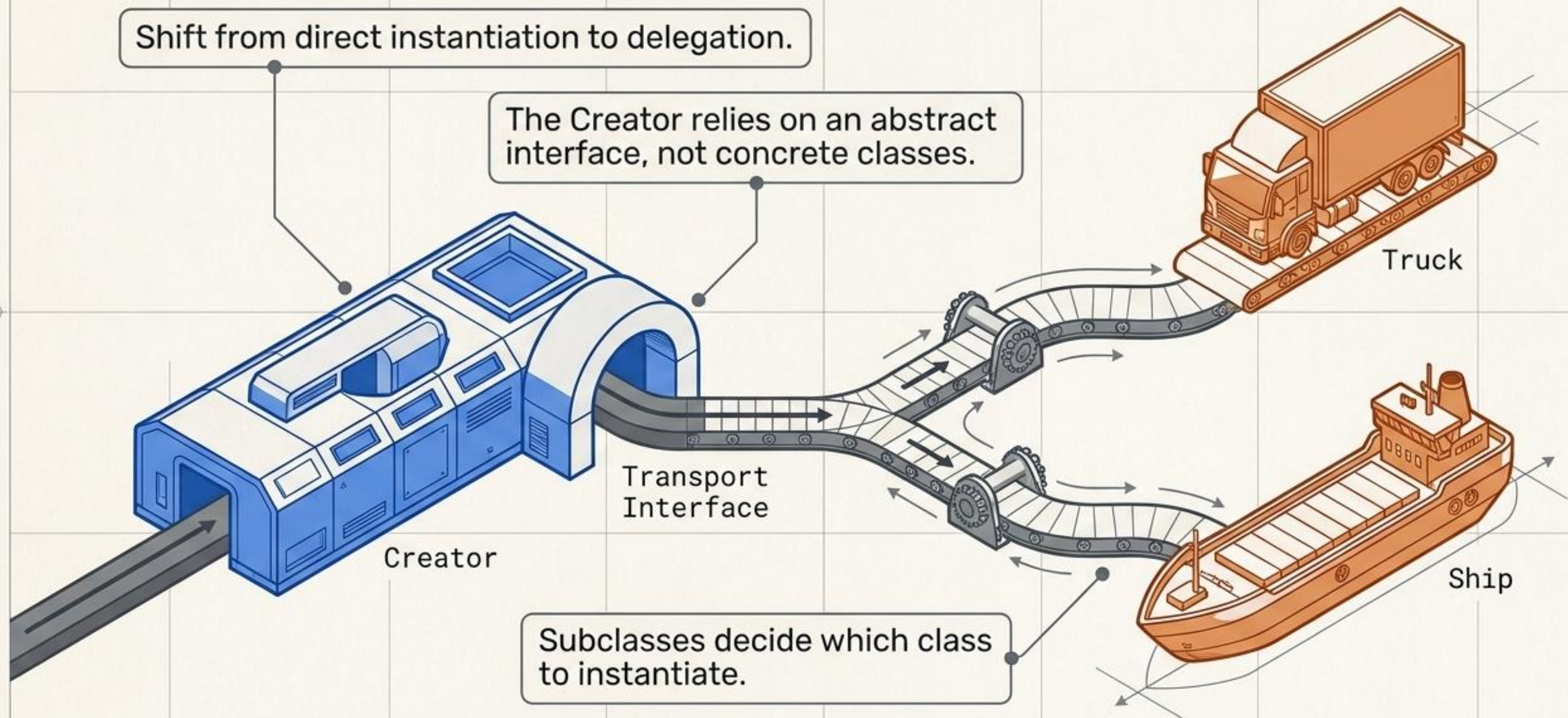
CONSTRAINT

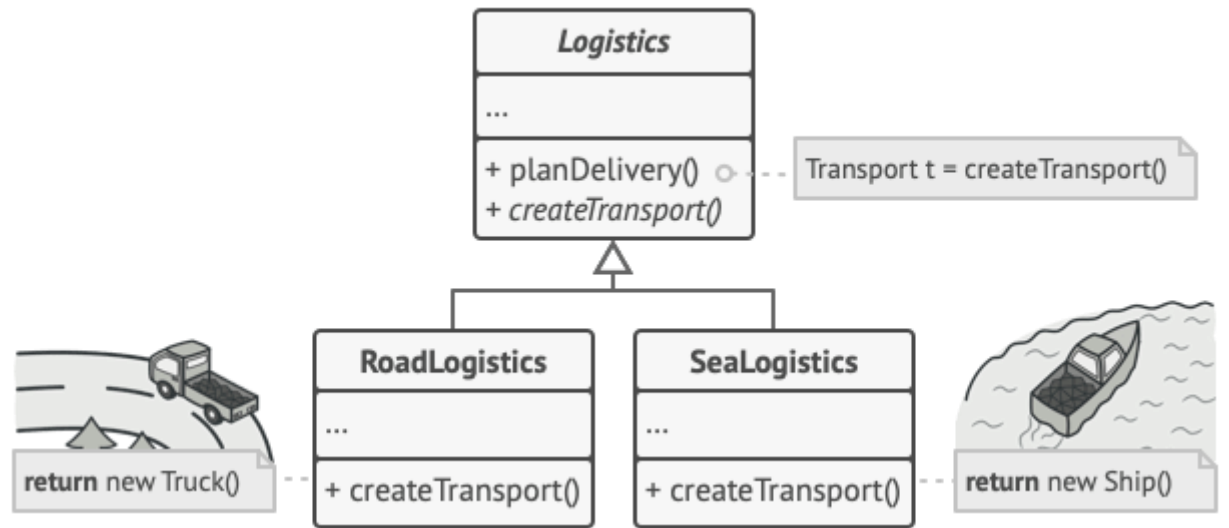
The system is brittle. Adding new product types requires modifying existing, tested code.

RESULT

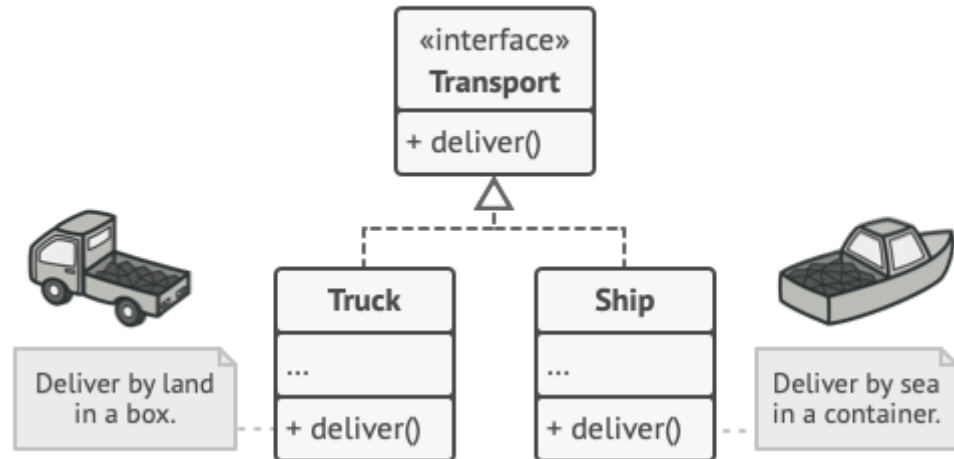
Closed to extension, violating the Open/Closed Principle.

The Virtual Constructor

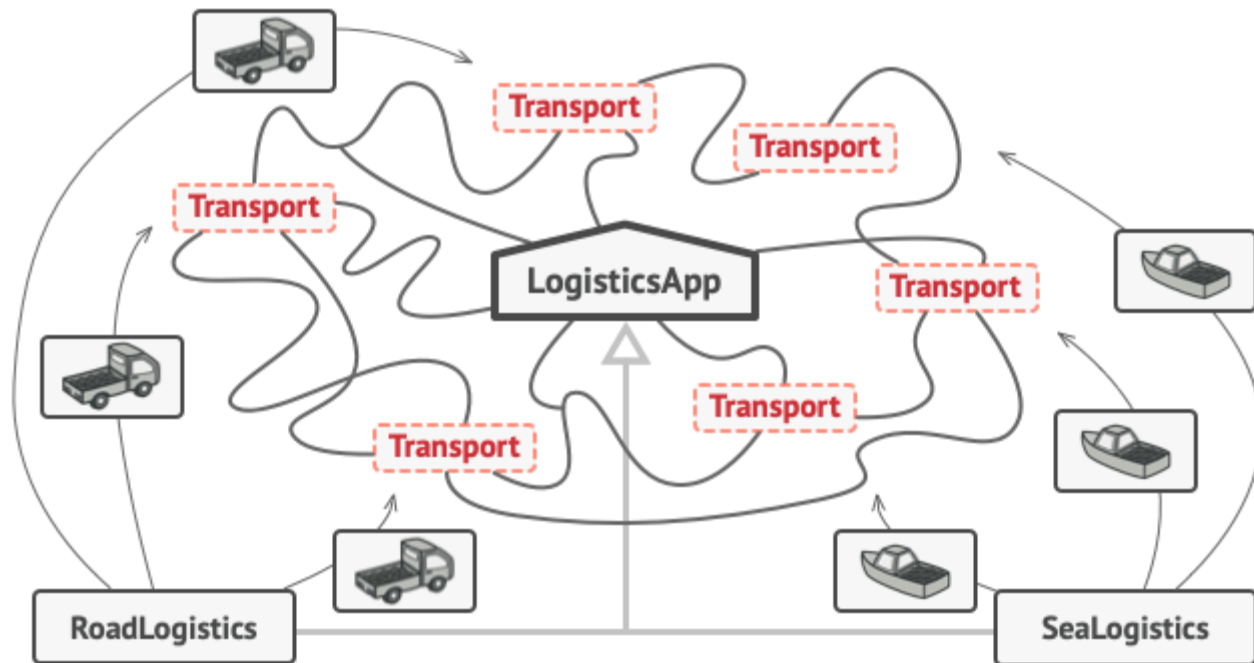




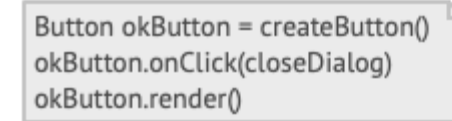
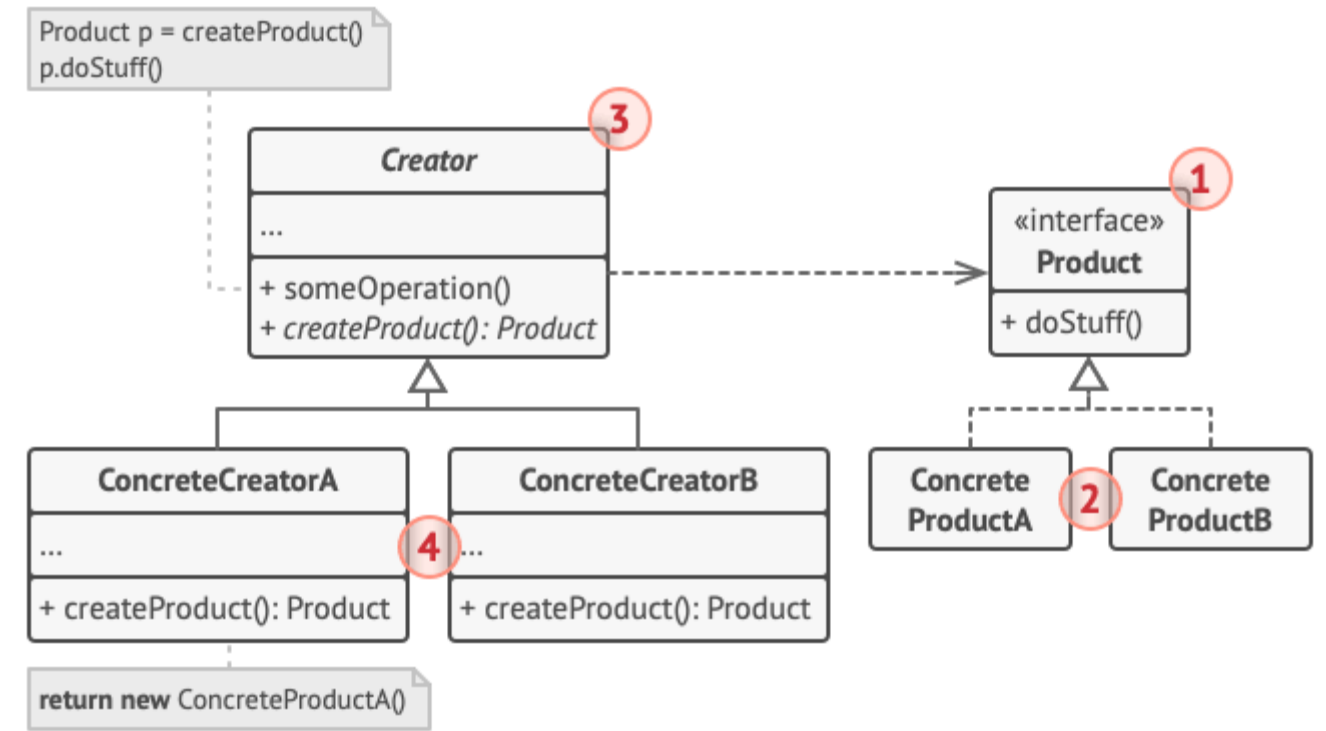
Subclasses can alter the class of objects being returned by the factory method.



All products must follow the same interface.



As long as all product classes implement a common interface, you can pass their objects to the client code without breaking it.



The cross-platform dialog example.


```

// The creator class declares the factory method that must
// return an object of a product class. The creator's subclasses
// usually provide the implementation of this method.
class Dialog is
    // The creator may also provide some default implementation
    // of the factory method.
    abstract method createButton():Button

    // Note that, despite its name, the creator's primary
    // responsibility isn't creating products. It usually
    // contains some core business logic that relies on product
    // objects returned by the factory method. Subclasses can
    // indirectly change that business logic by overriding the
    // factory method and returning a different type of product
    // from it.
    method render() is
        // Call the factory method to create a product object.
        Button okButton = createButton()
        // Now use the product.
        okButton.onClick(closeDialog)
        okButton.render()

// Concrete creators override the factory method to change the
// resulting product's type.
class WindowsDialog extends Dialog is
    method createButton():Button is
        return new WindowsButton()

class WebDialog extends Dialog is
    method createButton():Button is
        return new HTMLButton()

// The product interface declares the operations that all
// concrete products must implement.
interface Button is
    method render()
    method onClick(f)

```

```

// The product interface declares the operations that all
// concrete products must implement.
interface Button is
    method render()
    method onClick(f)

// Concrete products provide various implementations of the
// product interface.
class WindowsButton implements Button is
    method render(a, b) is
        // Render a button in Windows style.
    method onClick(f) is
        // Bind a native OS click event.

class HTMLButton implements Button is
    method render(a, b) is
        // Return an HTML representation of a button.
    method onClick(f) is
        // Bind a web browser click event.

class Application is
    field dialog: Dialog

    // The application picks a creator's type depending on the
    // current configuration or environment settings.
    method initialize() is
        config = readApplicationConfigFile()

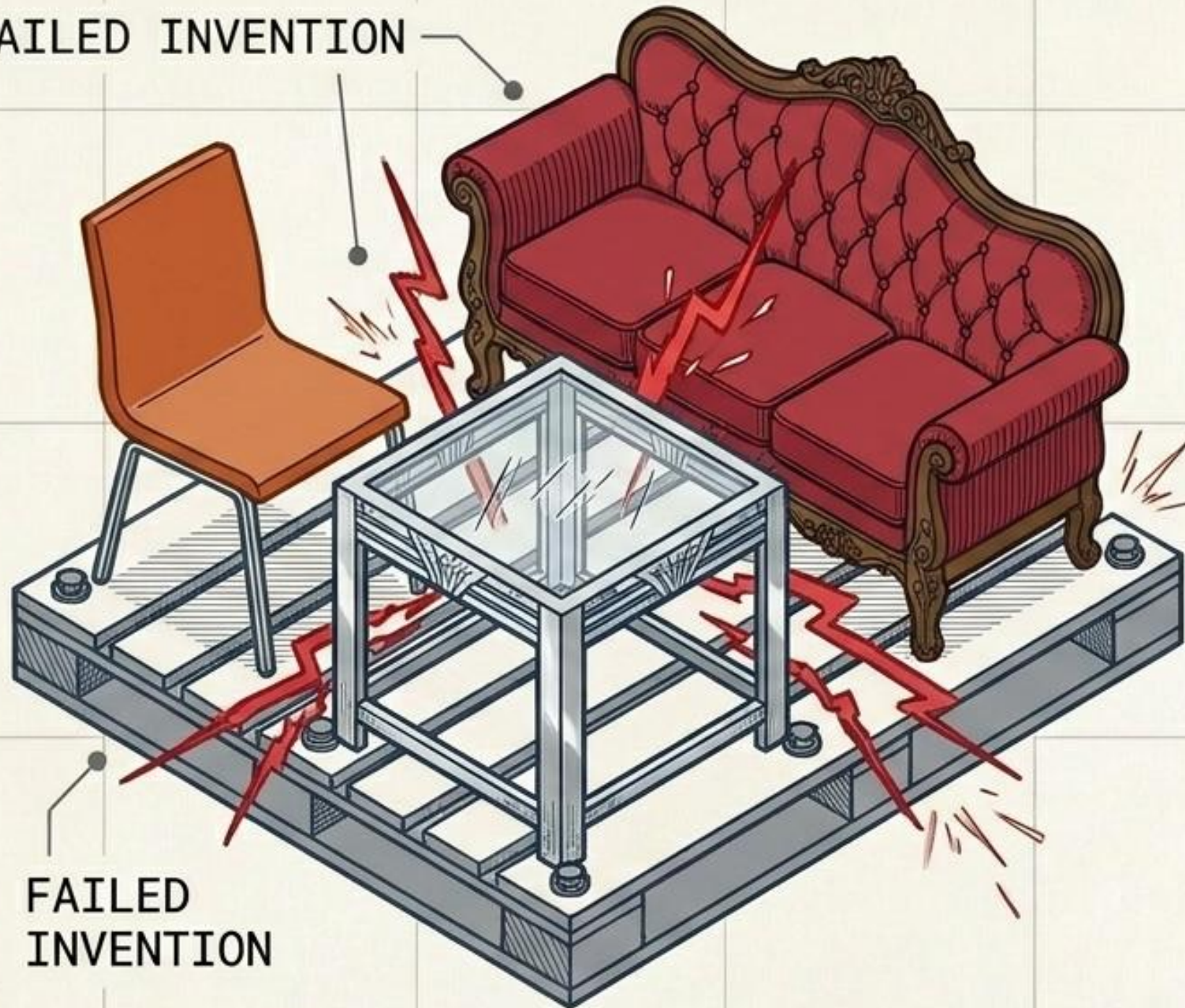
        if (config.OS == "Windows") then
            dialog = new WindowsDialog()
        else if (config.OS == "Web") then
            dialog = new WebDialog()
        else
            throw new Exception("Error! Unknown operating system.")

// The client code works with an instance of a concrete
// creator, albeit through its base interface. As long as
// the client keeps working with the creator via the base
// interface, you can pass it any creator's subclass.
method main() is
    this.initialize()
    dialog.render()

```


The Mismatched Family

FAILED INVENTION



FAILED
INVENTION

SYMPTOM

Instantiating related objects from different families randomly throughout the codebase.

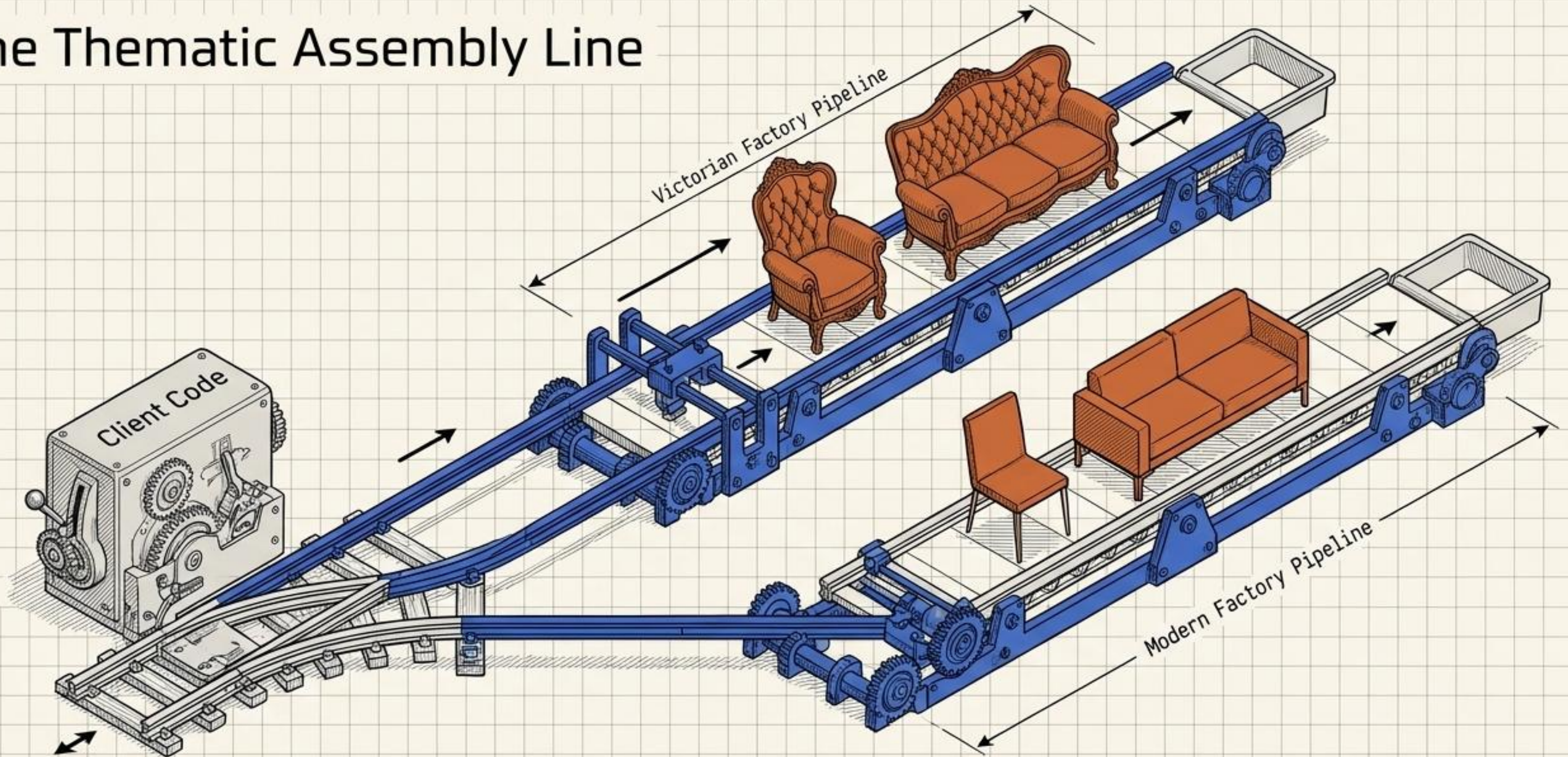
CONSTRAINT

Products must be guaranteed to match their counterparts, but concrete classes are hardcoded.

RESULT

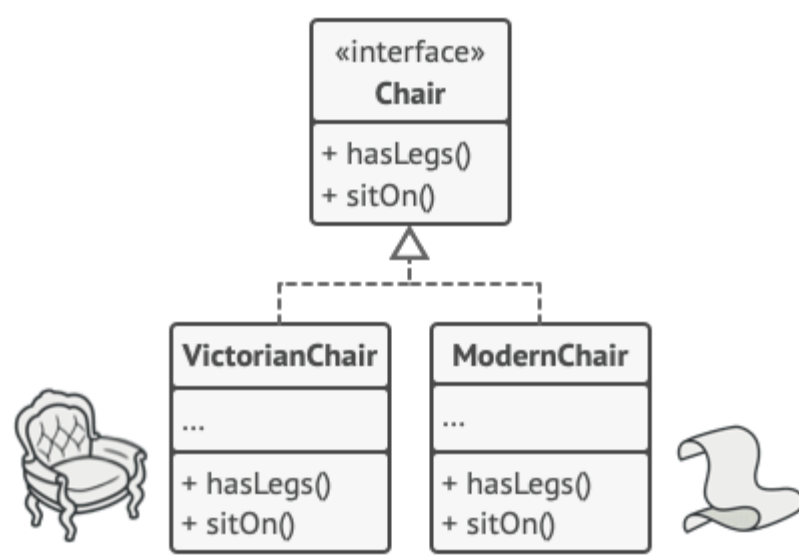
Visual and functional inconsistencies; updating a product family breaks client code.

The Thematic Assembly Line

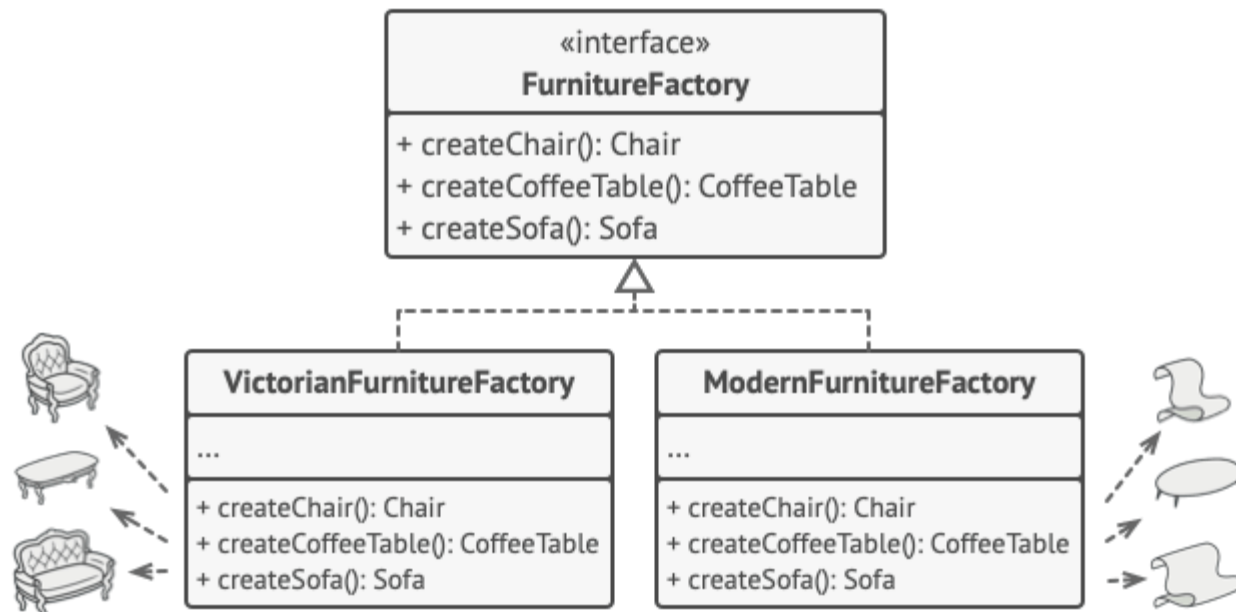


ARCHITECTURAL LABEL

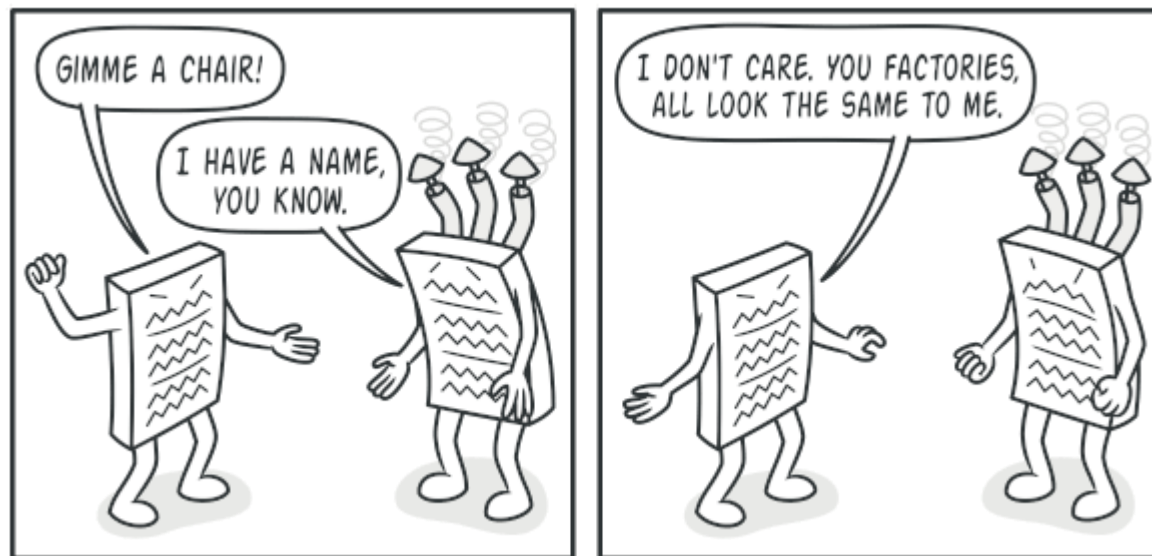
Encapsulate individual factories into a single "super-factory" interface. The client code works with any concrete factory/product variant through their abstract interfaces, guaranteeing 10000% compatibility.



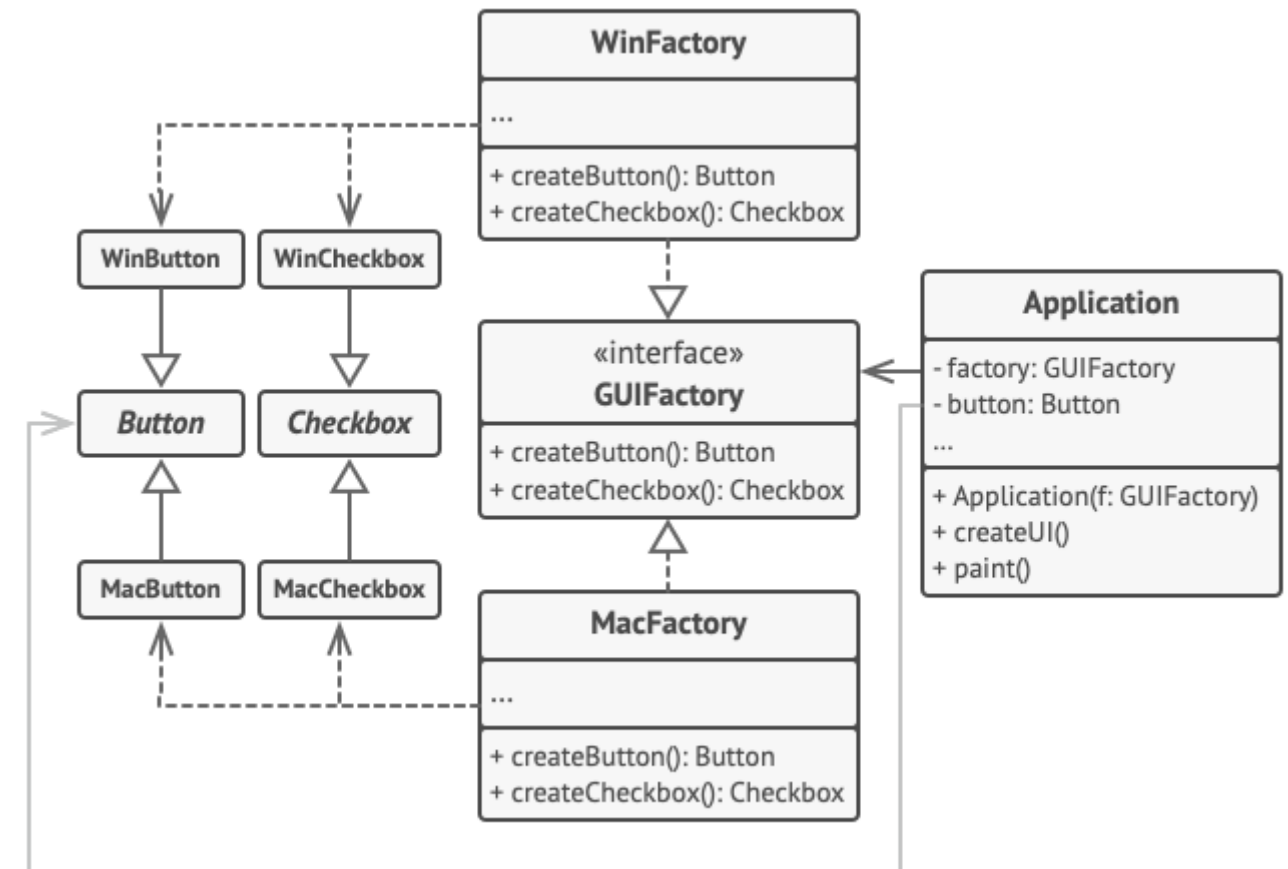
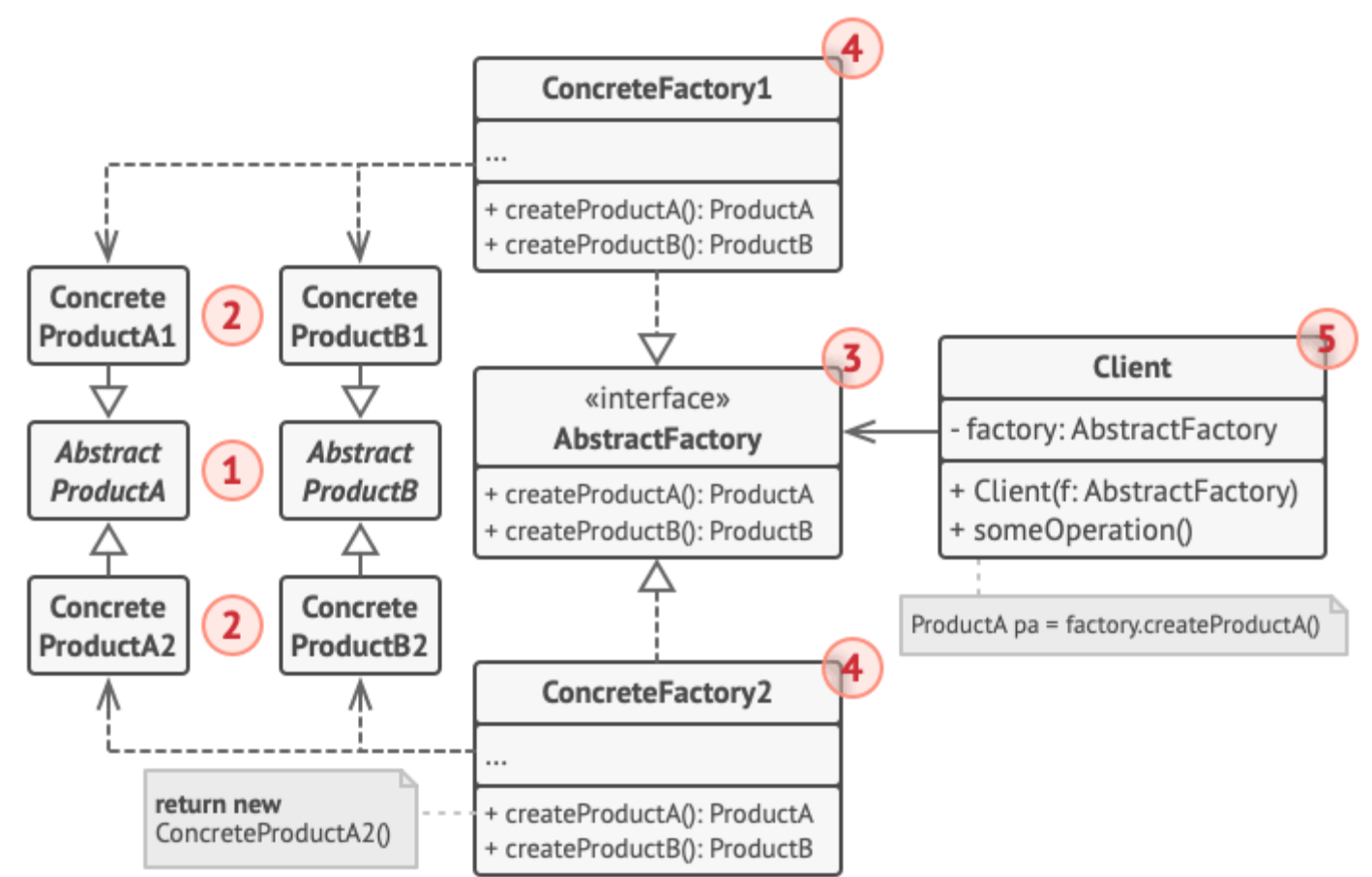
All variants of the same object must be moved to a single class hierarchy.



Each concrete factory corresponds to a specific product variant.



The client shouldn't care about the concrete class of the factory it works with.



The cross-platform UI classes example.


```

// The abstract factory interface declares a set of methods that
// return different abstract products. These products are called
// a family and are related by a high-level theme or concept.
// Products of one family are usually able to collaborate among
// themselves. A family of products may have several variants,
// but the products of one variant are incompatible with the
// products of another variant.
interface GUIFactory is
    method createButton():Button
    method createCheckbox():Checkbox

// Concrete factories produce a family of products that belong
// to a single variant. The factory guarantees that the
// resulting products are compatible. Signatures of the concrete
// factory's methods return an abstract product, while inside
// the method a concrete product is instantiated.
class WinFactory implements GUIFactory is
    method createButton():Button is
        return new WinButton()
    method createCheckbox():Checkbox is
        return new WinCheckbox()

// Each concrete factory has a corresponding product variant.
class MacFactory implements GUIFactory is
    method createButton():Button is
        return new MacButton()
    method createCheckbox():Checkbox is
        return new MacCheckbox()

// Each distinct product of a product family should have a base
// interface. All variants of the product must implement this
// interface.
interface Button is
    method paint()

// Concrete products are created by corresponding concrete
// factories.
class WinButton implements Button is
    method paint() is
        // Render a button in Windows style.

class MacButton implements Button is
    method paint() is
        // Render a button in macOS style.

```

```

// Here's the base interface of another product. All products
// can interact with each other, but proper interaction is
// possible only between products of the same concrete variant.
interface Checkbox is
    method paint()

class WinCheckbox implements Checkbox is
    method paint() is
        // Render a checkbox in Windows style.

class MacCheckbox implements Checkbox is
    method paint() is
        // Render a checkbox in macOS style.

// The client code works with factories and products only
// through abstract types: GUIFactory, Button and Checkbox. This
// lets you pass any factory or product subclass to the client
// code without breaking it.
class Application is
    private field factory: GUIFactory
    private field button: Button
    constructor Application(factory: GUIFactory) is
        this.factory = factory
    method createUI() is
        this.button = factory.createButton()
    method paint() is
        button.paint()

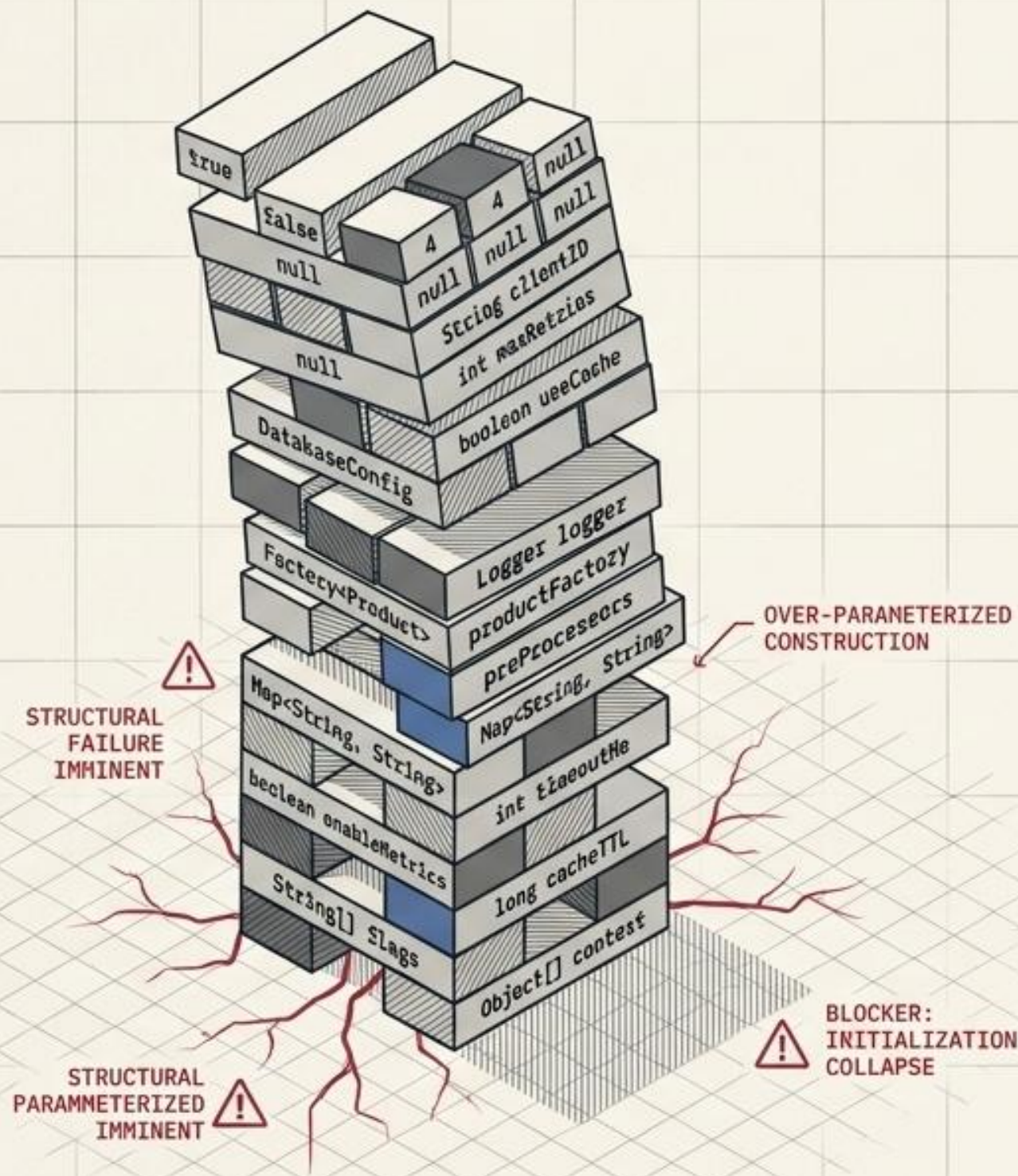
// The application picks the factory type depending on the
// current configuration or environment settings and creates it
// at runtime (usually at the initialization stage).
class ApplicationConfigurator is
    method main() is
        config = readApplicationConfigFile()

        if (config.OS == "Windows") then
            factory = new WinFactory()
        else if (config.OS == "Mac") then
            factory = new MacFactory()
        else
            throw new Exception("Error! Unknown operating system.")

        Application app = new Application(factory)

```


The Telescoping Monolith



SYMPTOM

The Telescoping Constructor anti-pattern—dozens of parameters, mostly nulls.

CONSTRAINT

Creating a complex object requires massive initialization code buried inside a monster subclass hierarchy or gigantic constructor.

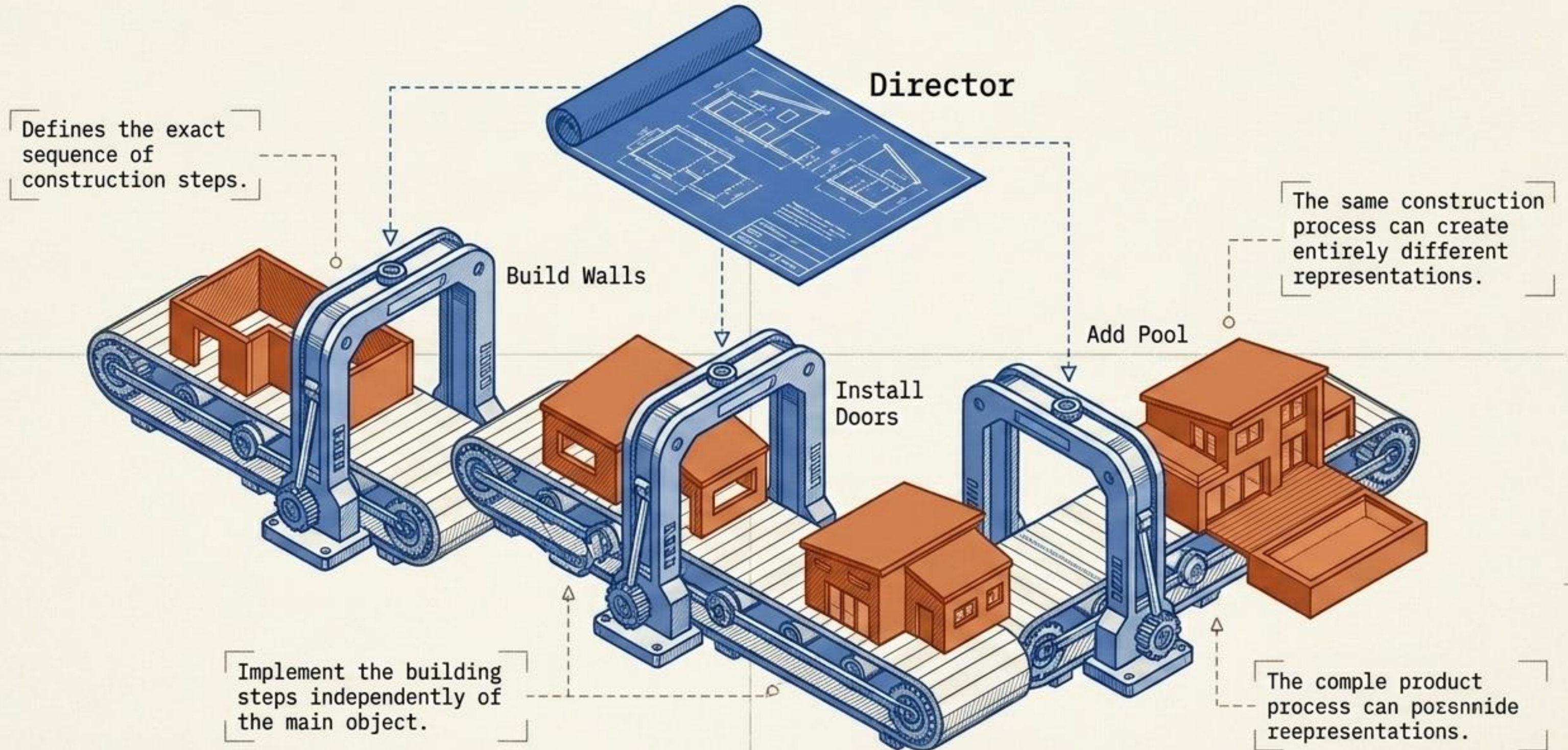
RESULT

Unreadable, error-prone initialization that cannot be easily varied.

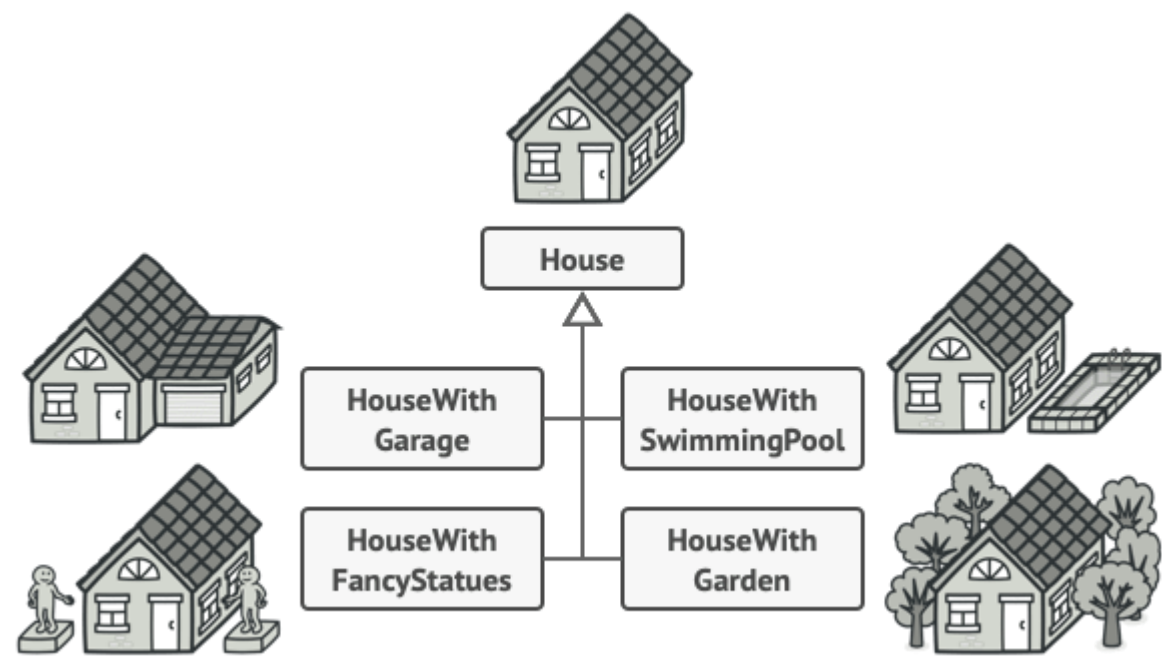
ARCHITECTURAL NOTE

This anti-pattern often necessitates a builder or factory to abstract away complex initialization, improving readability and maintainability by reducing the cognitive load of construction.

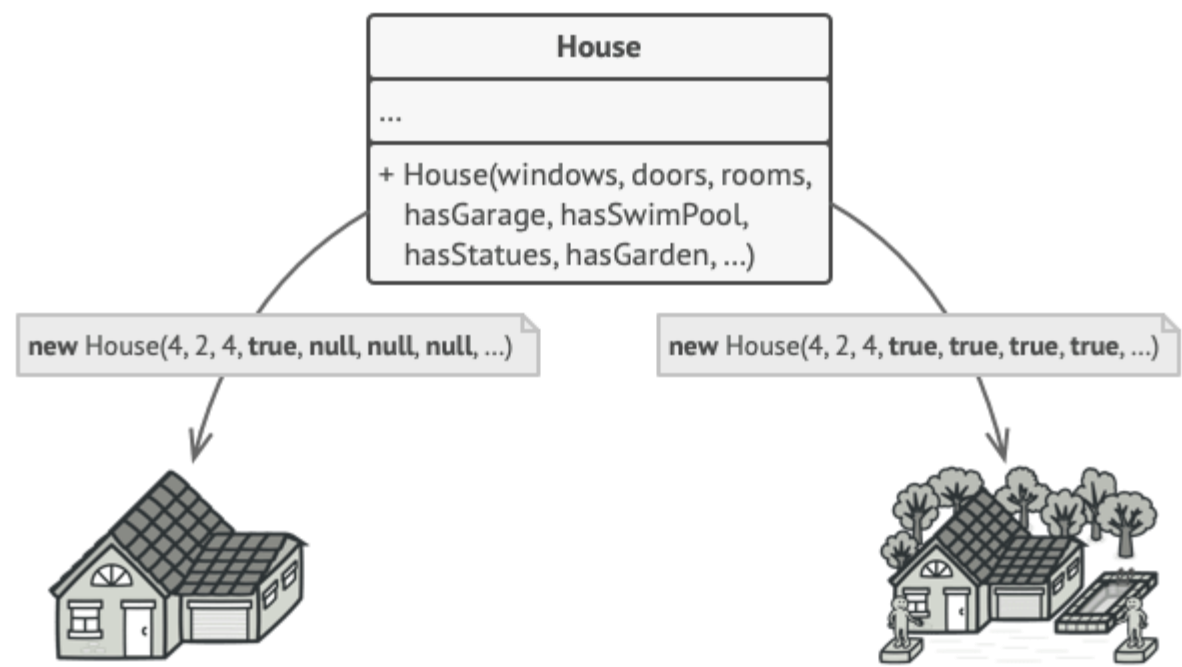
Orchestrated Assembly Steps



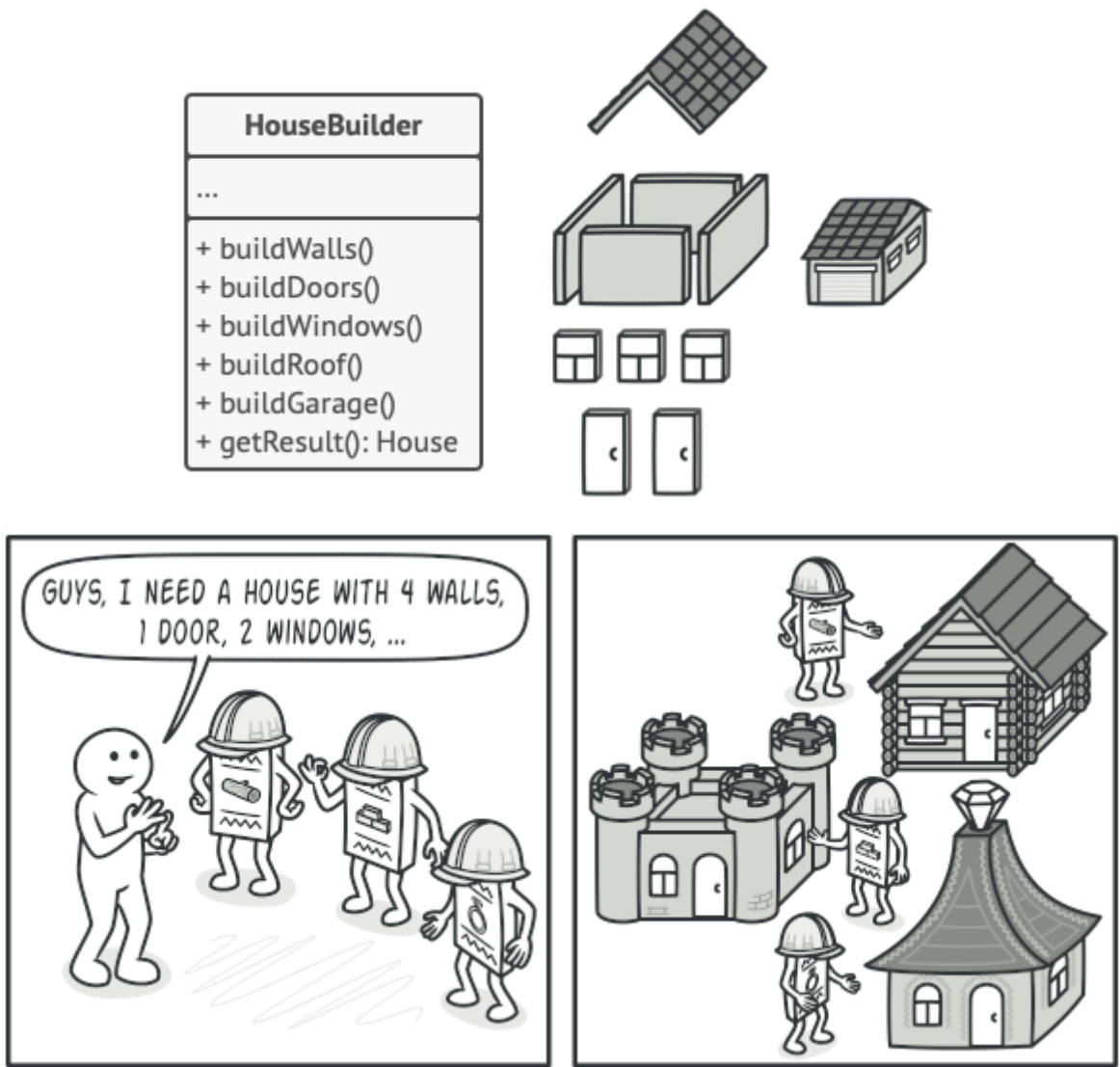
The Builder pattern lets you construct complex objects step by step. The Builder doesn't allow other objects to access the product while it's being built.



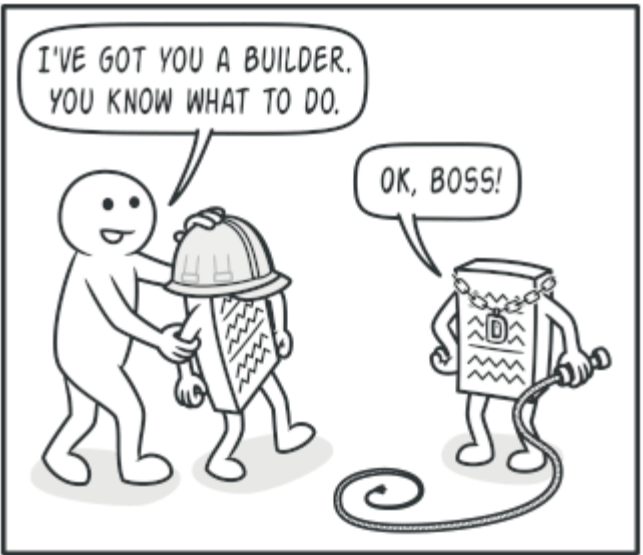
You might make the program too complex by creating a subclass for every possible configuration of an object.



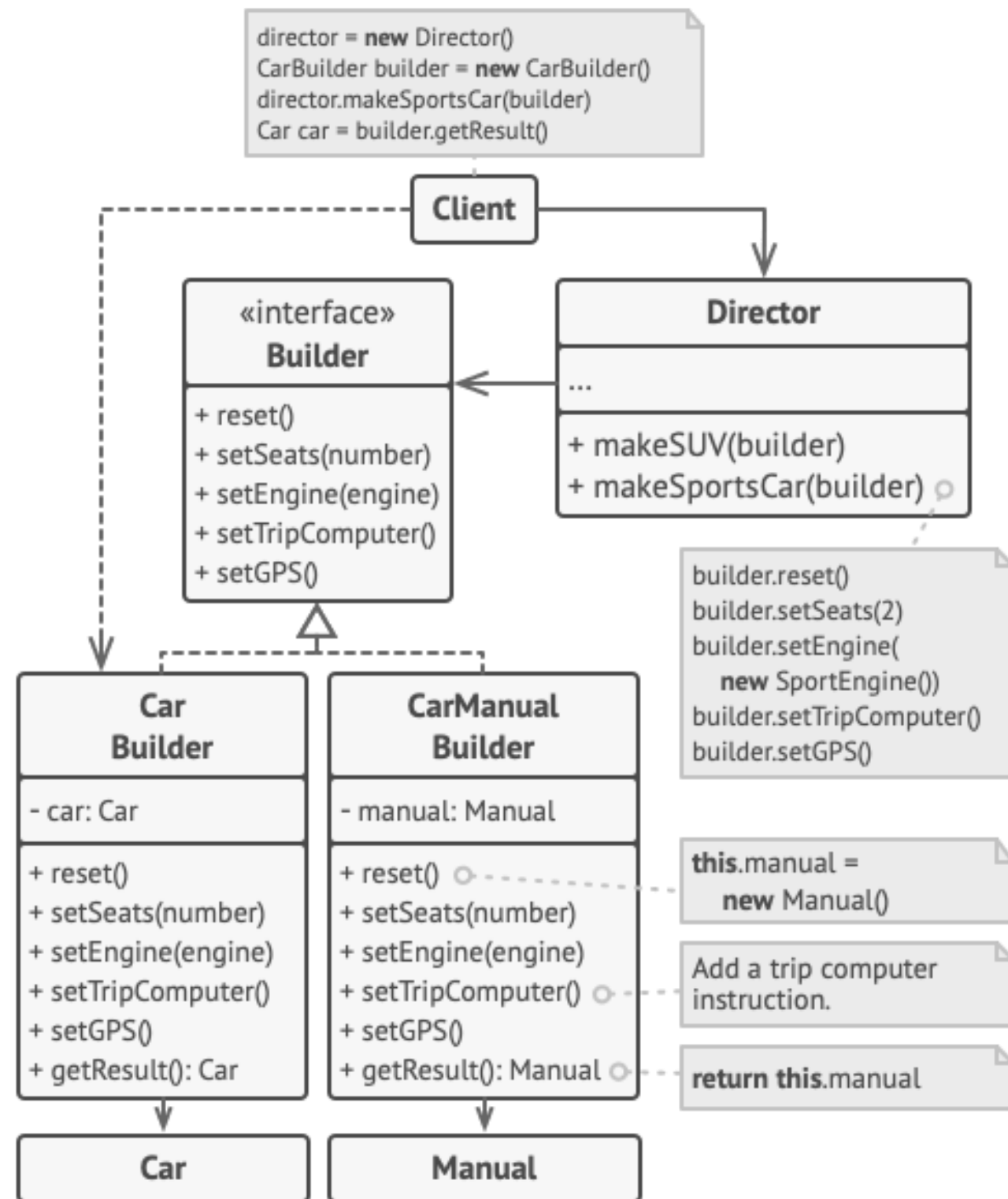
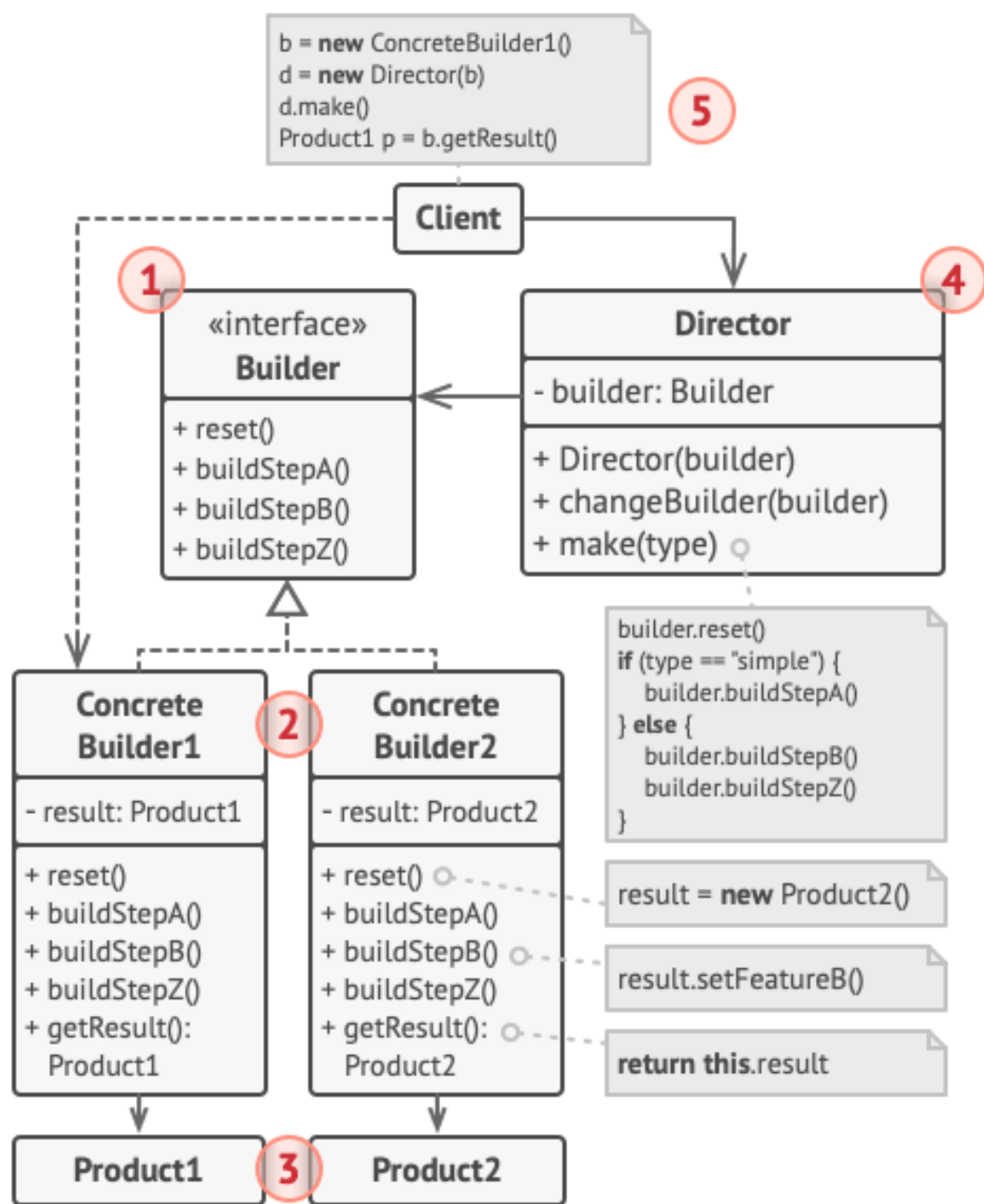
The constructor with lots of parameters has its downside: not all the parameters are needed at all times.



Different builders execute the same task in various ways.



The director knows which building steps to execute to get a working product.




```
// Using the Builder pattern makes sense only when your products
// are quite complex and require extensive configuration. The
// following two products are related, although they don't have
// a common interface.
class Car is
    // A car can have a GPS, trip computer and some number of
    // seats. Different models of cars (sports car, SUV,
    // cabriolet) might have different features installed or
    // enabled.

class Manual is
    // Each car should have a user manual that corresponds to
    // the car's configuration and describes all its features.

// The builder interface specifies methods for creating the
// different parts of the product objects.
interface Builder is
    method reset()
    method setSeats(...)
    method setEngine(...)
    method setTripComputer(...)
    method setGPS(...)

// Unlike other creational patterns, builder lets you construct
// products that don't follow the common interface.
class CarManualBuilder implements Builder is
    private field manual:Manual

    constructor CarManualBuilder() is
        this.reset()

    method reset() is
        this.manual = new Manual()

    method setSeats(...) is
        // Document car seat features.

    method setEngine(...) is
        // Add engine instructions.

    method setTripComputer(...) is
        // Add trip computer instructions.

    method setGPS(...) is
        // Add GPS instructions.

    method getProduct():Manual is
        // Return the manual and reset the builder.
```

```
// The concrete builder classes follow the builder interface and
// provide specific implementations of the building steps. Your
// program may have several variations of builders, each
// implemented differently.
class CarBuilder implements Builder is
    private field car:Car

    // A fresh builder instance should contain a blank product
    // object which it uses in further assembly.
    constructor CarBuilder() is
        this.reset()

    // The reset method clears the object being built.
    method reset() is
        this.car = new Car()

    // All production steps work with the same product instance.
    method setSeats(...) is
        // Set the number of seats in the car.

    method setEngine(...) is
        // Install a given engine.

    method setTripComputer(...) is
        // Install a trip computer.

    method setGPS(...) is
        // Install a global positioning system.

    // Concrete builders are supposed to provide their own
    // methods for retrieving results. That's because various
    // types of builders may create entirely different products
    // that don't all follow the same interface. Therefore such
    // methods can't be declared in the builder interface (at
    // least not in a statically-typed programming language).
    //
    // Usually, after returning the end result to the client, a
    // builder instance is expected to be ready to start
    // producing another product. That's why it's a usual
    // practice to call the reset method at the end of the
    // `getProduct` method body. However, this behavior isn't
    // mandatory, and you can make your builder wait for an
    // explicit reset call from the client code before disposing
    // of the previous result.
    method getProduct():Car is
        product = this.car
        this.reset()
        return product
```

```
// The director is only responsible for executing the building
// steps in a particular sequence. It's helpful when producing
// products according to a specific order or configuration.
// Strictly speaking, the director class is optional, since the
// client can control builders directly.
class Director is
    // The director works with any builder instance that the
    // client code passes to it. This way, the client code may
    // alter the final type of the newly assembled product.
    // The director can construct several product variations
    // using the same building steps.
    method constructSportsCar(builder: Builder) is
        builder.reset()
        builder.setSeats(2)
        builder.setEngine(new SportEngine())
        builder.setTripComputer(true)
        builder.setGPS(true)

    method constructSUV(builder: Builder) is
        // ...
```

```
// The client code creates a builder object, passes it to the
// director and then initiates the construction process. The end
// result is retrieved from the builder object.
class Application is

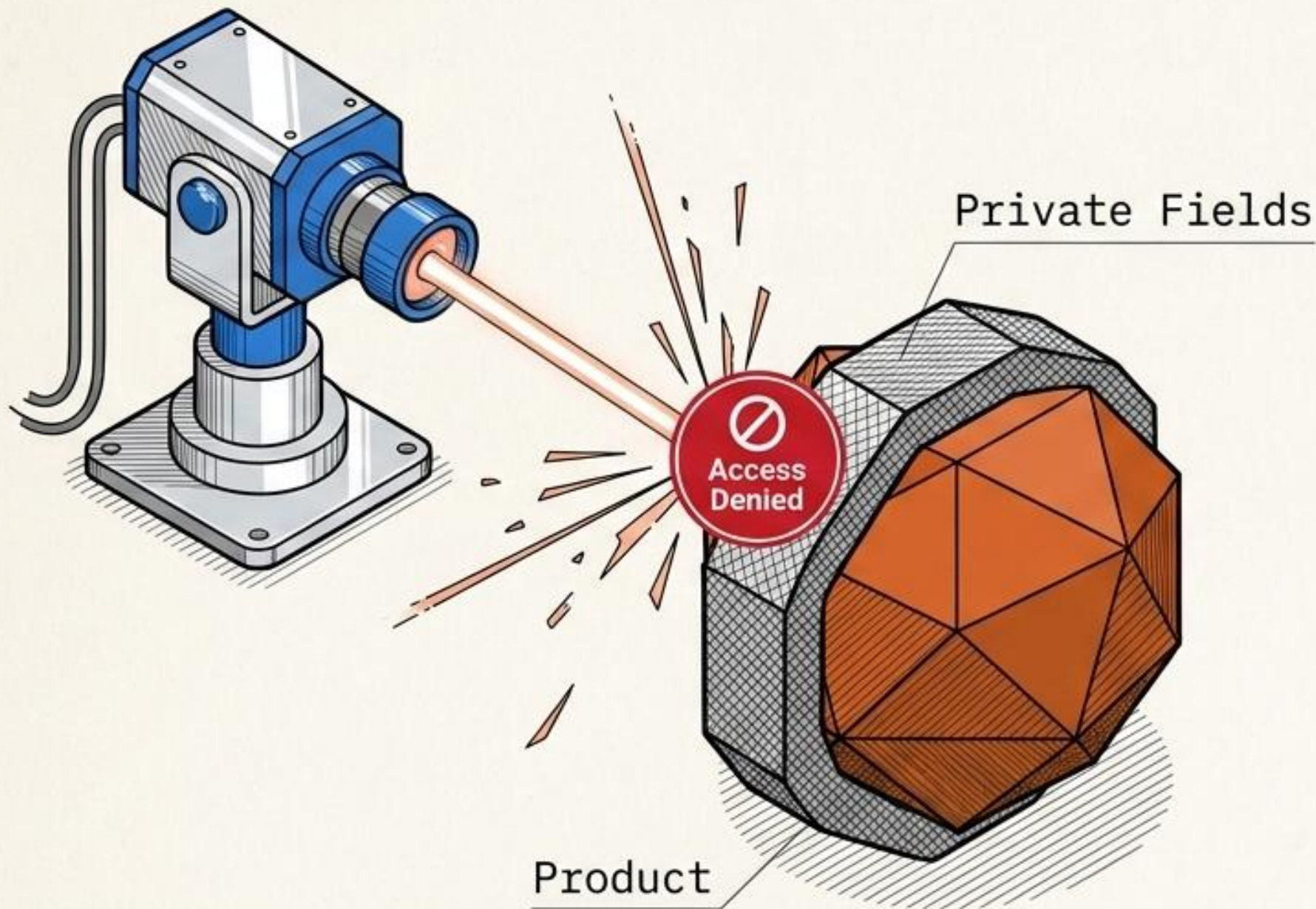
    method makeCar() is
        director = new Director()

        CarBuilder builder = new CarBuilder()
        director.constructSportsCar(builder)
        Car car = builder.getProduct()

        CarManualBuilder builder = new CarManualBuilder()
        director.constructSportsCar(builder)

    // The final product is often retrieved from a builder
    // object since the director isn't aware of and not
    // dependent on concrete builders and products.
    Manual manual = builder.getProduct()
```


The Costly From-Scratch Build



SYMPTOM

Relying on external classes to duplicate an object.

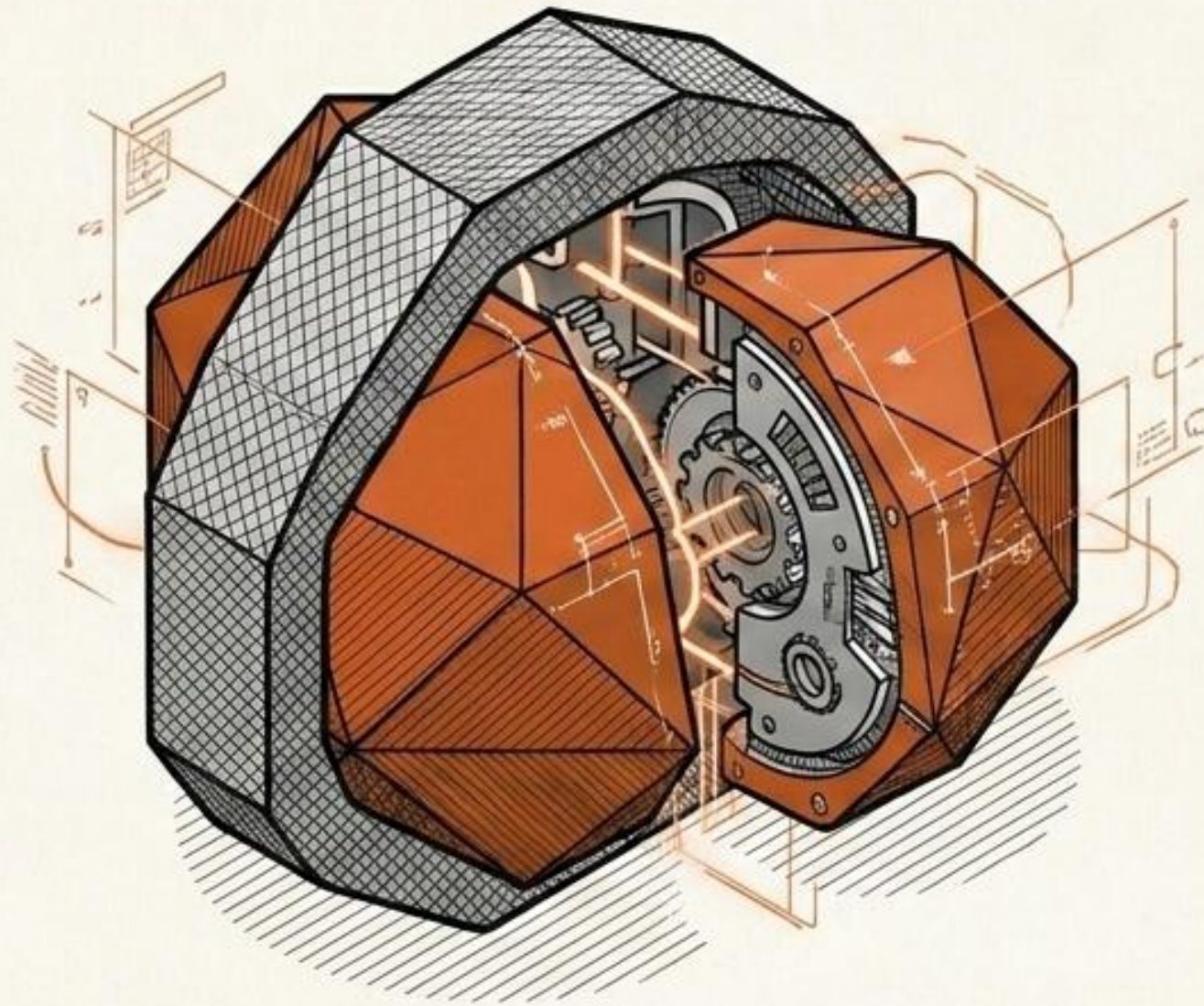
CONSTRAINT

External code cannot see or copy an object's private state or fields. Furthermore, initializing a complex object from scratch consumes massive CPU cycles.

RESULT

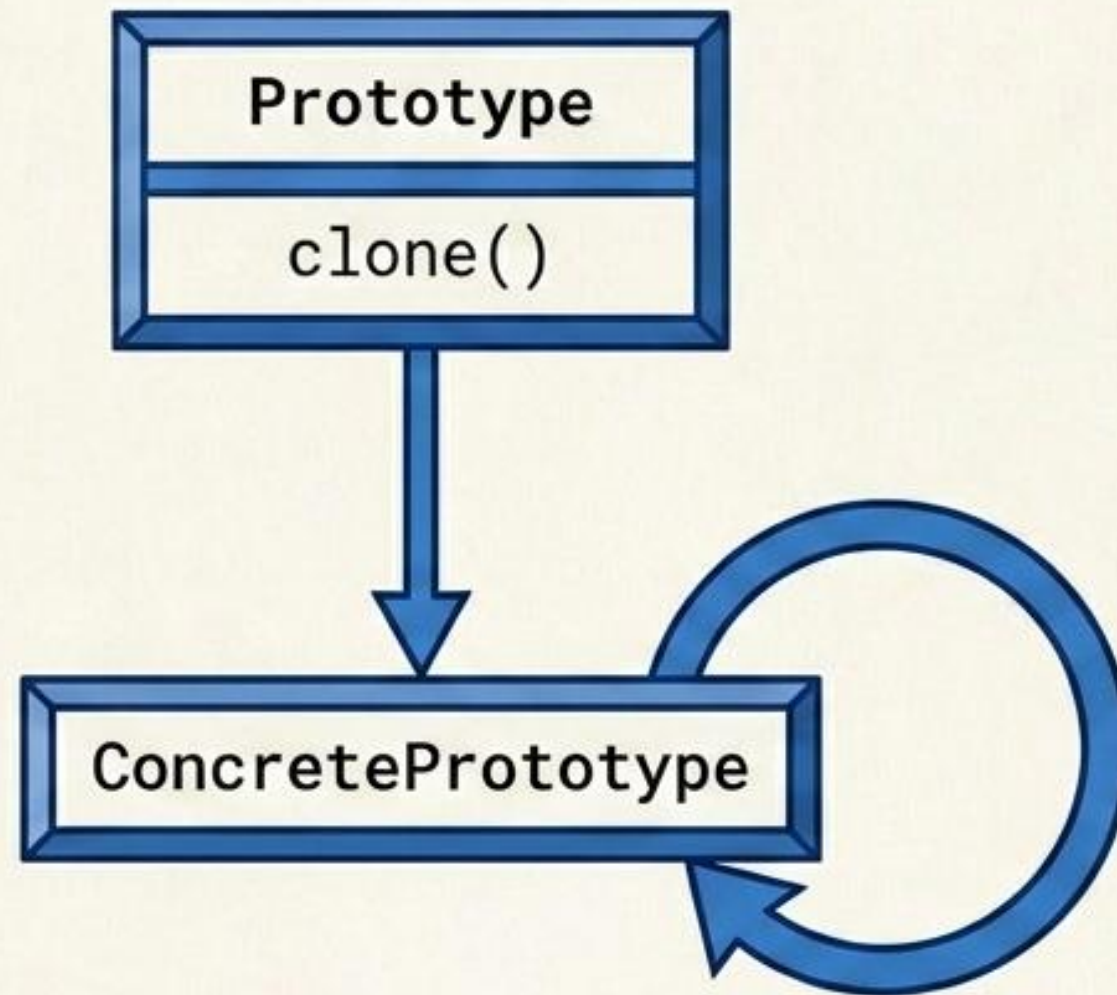
Incomplete copies and tight coupling to concrete classes.

Cellular Mitosis

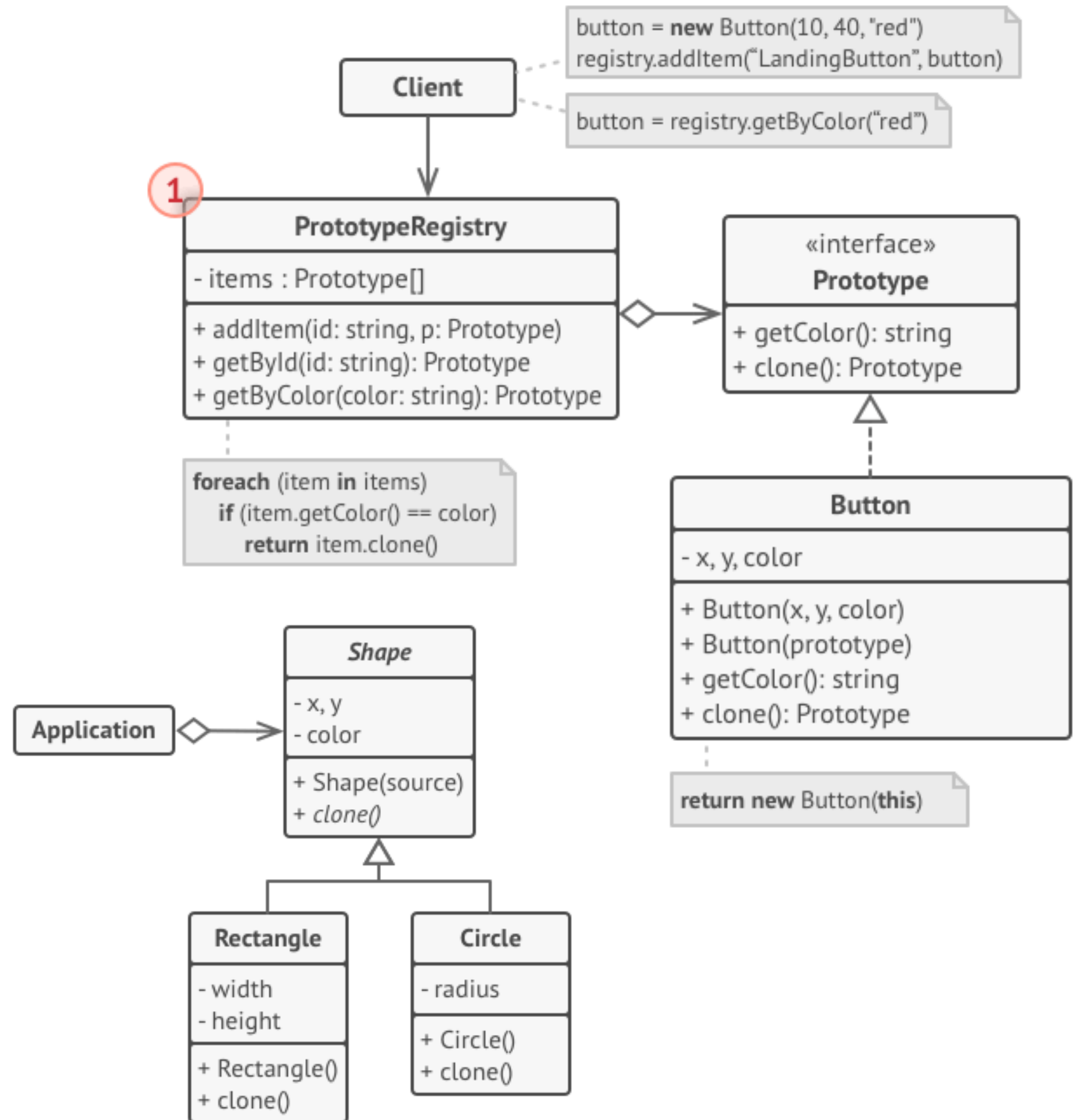
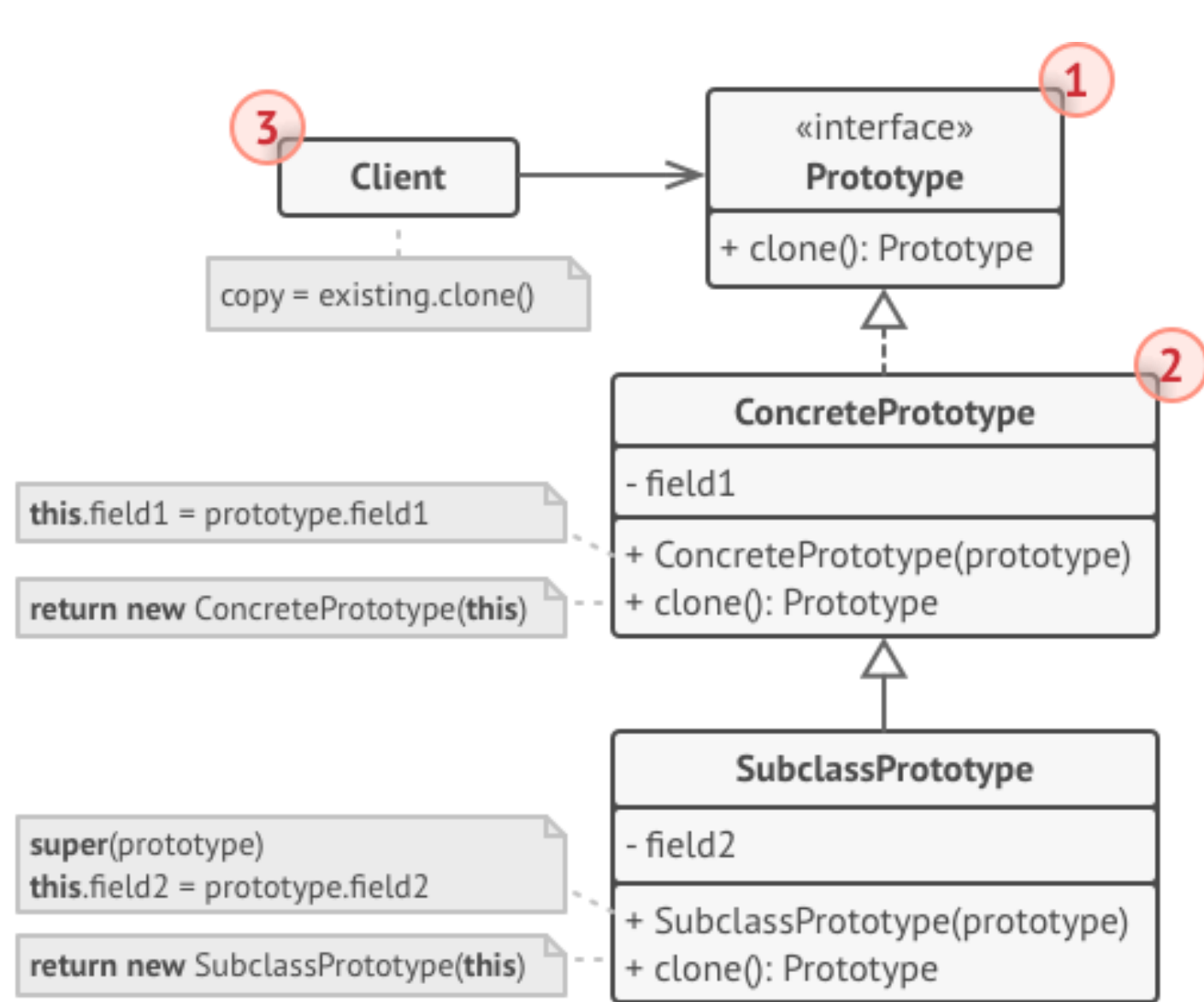


Delegate the cloning process to the actual objects being cloned. An object with full access to its own state creates an exact replica of itself without exposing its internal structure to the client.

Prototype Schematic



A unified cloning interface lets you copy objects without coupling your code to the classes of those objects.



Cloning a set of objects that belong to a class hierarchy.


```
// Base prototype.
abstract class Shape is
    field X: int
    field Y: int
    field color: string

    // A regular constructor.
    constructor Shape() is
        // ...

    // The prototype constructor. A fresh object is initialized
    // with values from the existing object.
    constructor Shape(source: Shape) is
        this()
        this.X = source.X
        this.Y = source.Y
        this.color = source.color

    // The clone operation returns one of the Shape subclasses.
    abstract method clone():Shape

// Concrete prototype. The cloning method creates a new object
// in one go by calling the constructor of the current class and
// passing the current object as the constructor's argument.
// Performing all the actual copying in the constructor helps to
// keep the result consistent: the constructor will not return a
// result until the new object is fully built; thus, no object
// can have a reference to a partially-built clone.
class Rectangle extends Shape is
    field width: int
    field height: int

    constructor Rectangle(source: Rectangle) is
        // A parent constructor call is needed to copy private
        // fields defined in the parent class.
        super(source)
        this.width = source.width
        this.height = source.height

    method clone():Shape is
        return new Rectangle(this)

class Circle extends Shape is
    field radius: int

    constructor Circle(source: Circle) is
        super(source)
        this.radius = source.radius

    method clone():Shape is
        return new Circle(this)
```

```
// Somewhere in the client code.
class Application is
    field shapes: array of Shape

    constructor Application() is
        Circle circle = new Circle()
        circle.X = 10
        circle.Y = 10
        circle.radius = 20
        shapes.add(circle)

        Circle anotherCircle = circle.clone()
        shapes.add(anotherCircle)
        // The `anotherCircle` variable contains an exact copy
        // of the `circle` object.

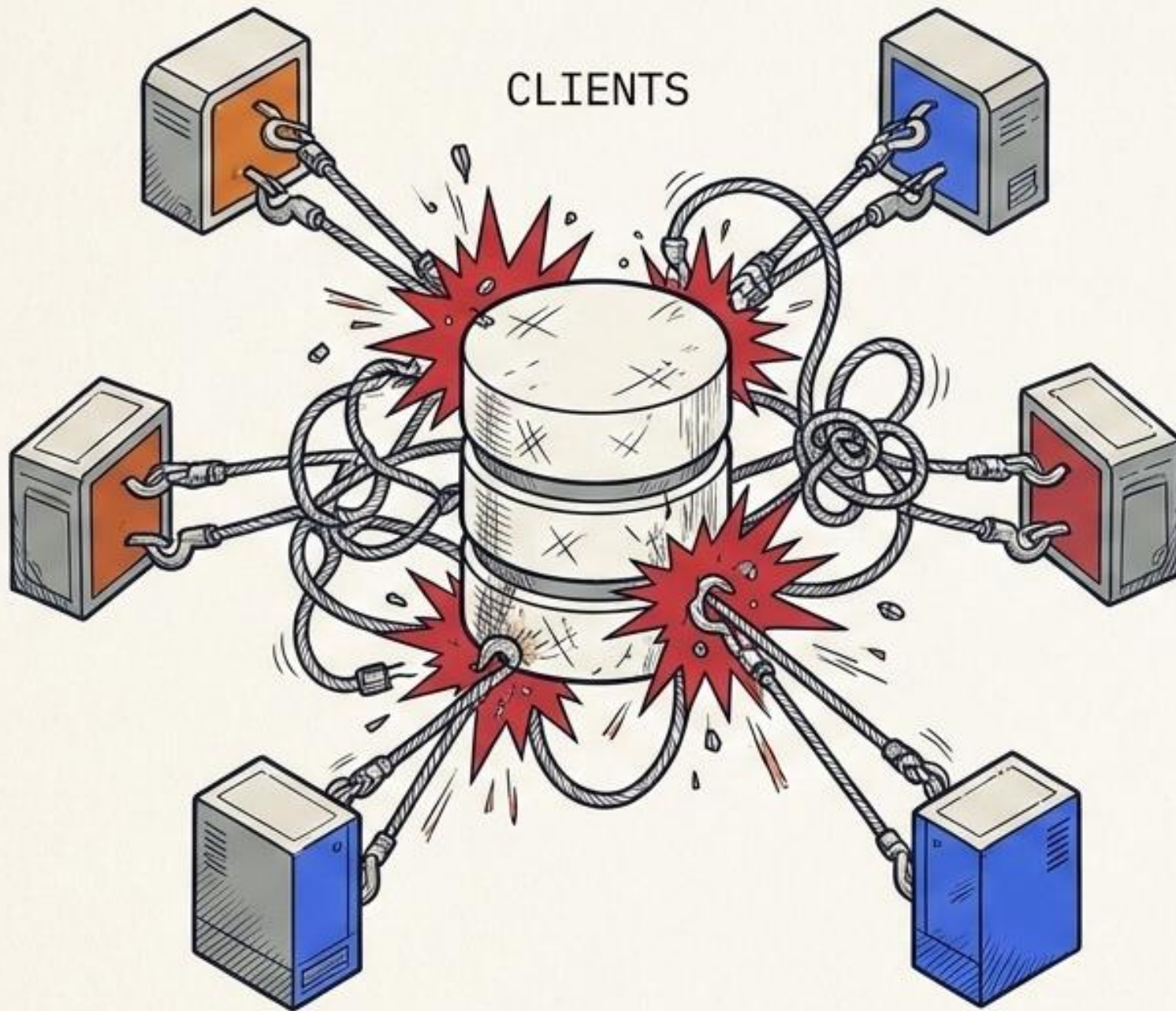
        Rectangle rectangle = new Rectangle()
        rectangle.width = 10
        rectangle.height = 20
        shapes.add(rectangle)

    method businessLogic() is
        // Prototype rocks because it lets you produce a copy of
        // an object without knowing anything about its type.
        Array shapesCopy = new Array of Shapes.

        // For instance, we don't know the exact elements in the
        // shapes array. All we know is that they are all
        // shapes. But thanks to polymorphism, when we call the
        // `clone` method on a shape the program checks its real
        // class and runs the appropriate clone method defined
        // in that class. That's why we get proper clones
        // instead of a set of simple Shape objects.
        foreach (s in shapes) do
            shapesCopy.add(s.clone())

        // The `shapesCopy` array contains exact copies of the
        // `shape` array's children.
```


The Chaotic Resource Grab



SYMPTOM

Multiple instances of a resource-heavy class colliding, destroying shared state.

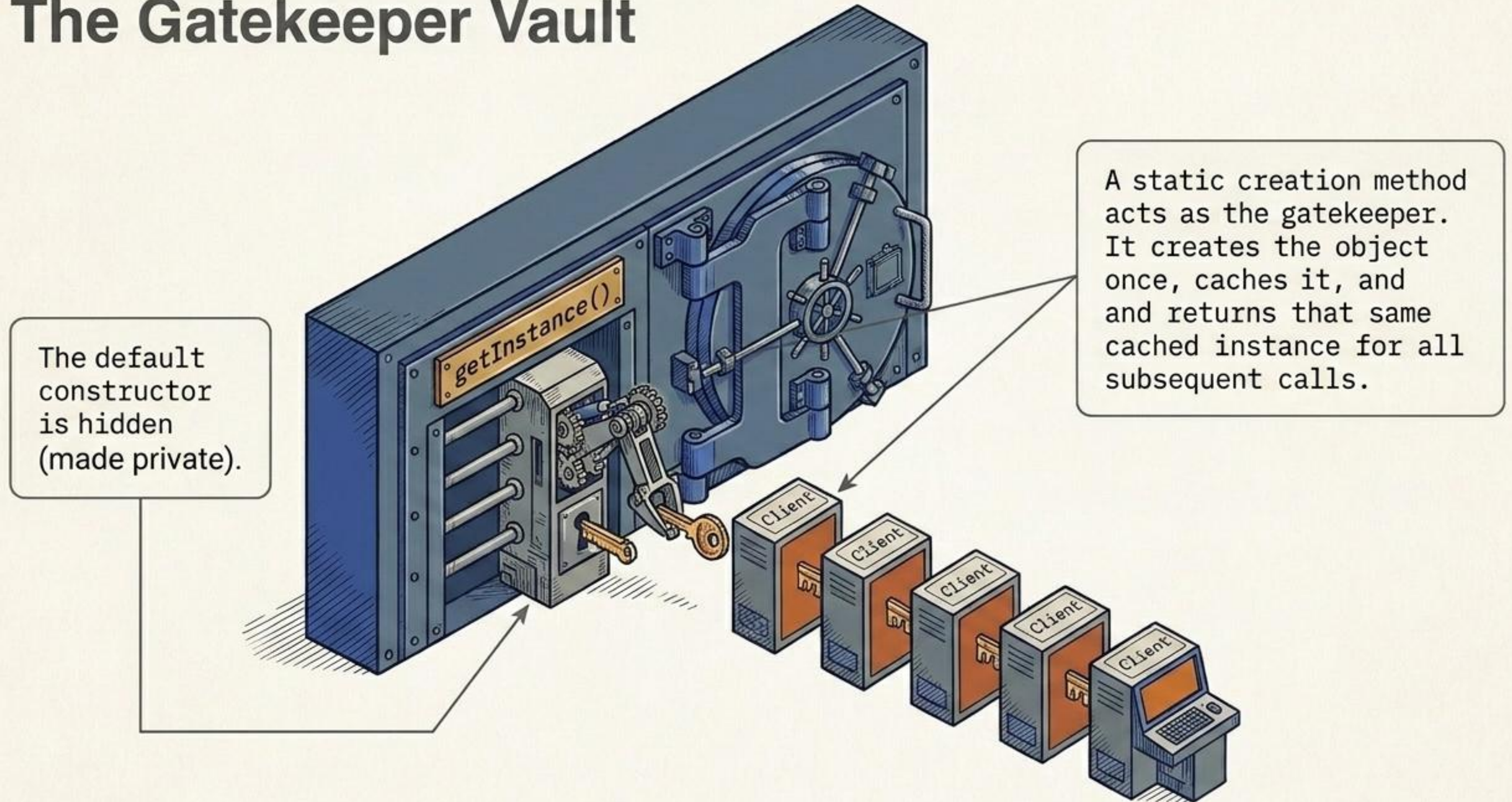
CONSTRAINT

You need strict control over a shared resource (like a database or file system), but you also need a global access point that won't be overridden.

RESULT

Race conditions, corrupted data, and unpredictable system behavior.

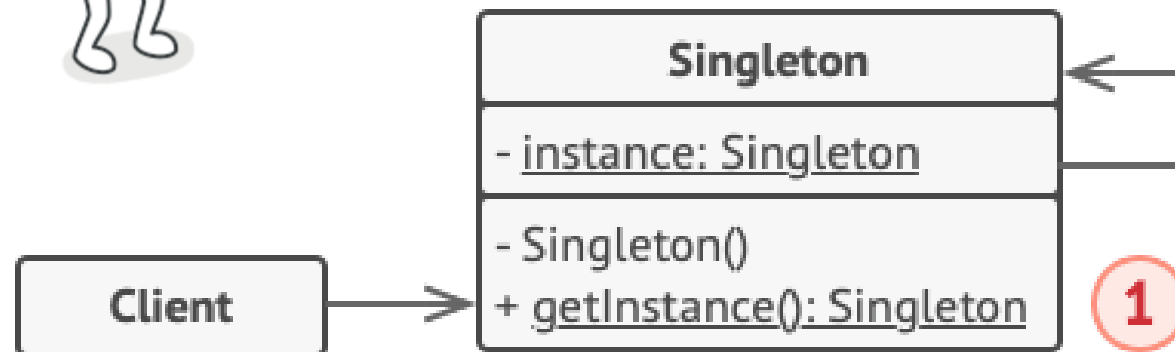
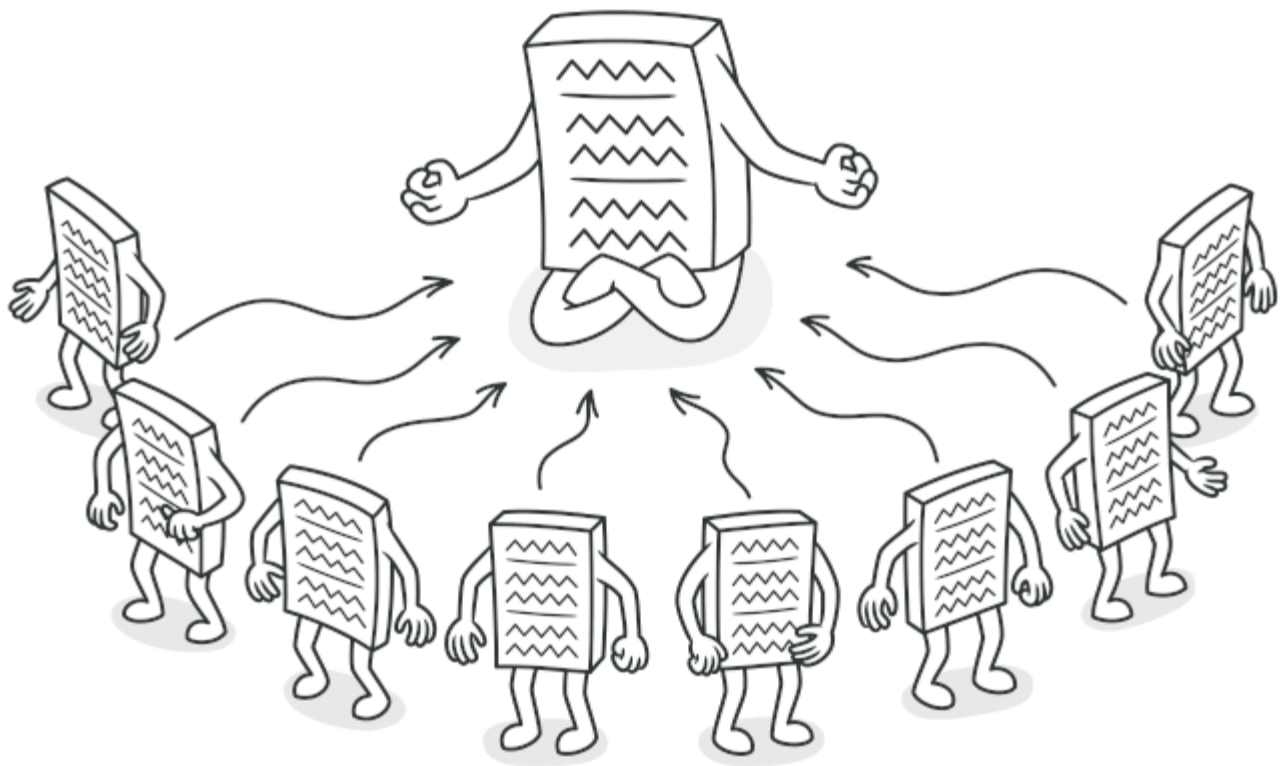
The Gatekeeper Vault



Singleton Schematic



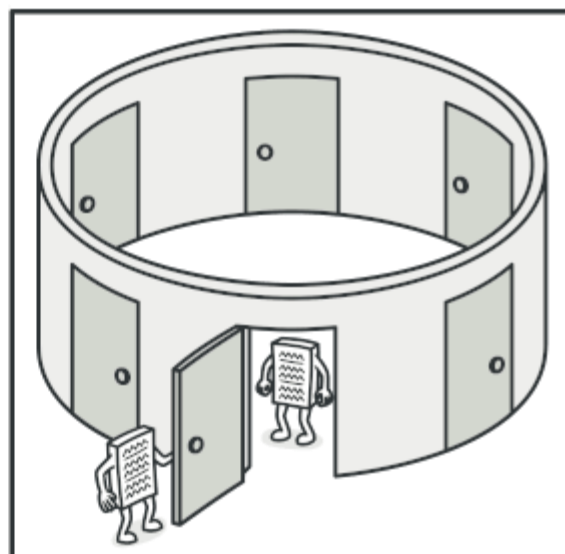
Solves two problems at once: Guarantees a single instance exists, and provides a global, strictly controlled access point to it.



```

if (instance == null) {
    // Note: if you're creating an app with
    // multithreading support, you should
    // place a thread lock here.
    instance = new Singleton()
}
return instance

```



```

// The Database class defines the `getInstance` method that lets
// clients access the same instance of a database connection
// throughout the program.

```

```

class Database is

```

```

    // The field for storing the singleton instance should be
    // declared static.

```

```

    private static field instance: Database

```

```

    // The singleton's constructor should always be private to
    // prevent direct construction calls with the `new`
    // operator.

```

```

    private constructor Database() is

```

```

        // Some initialization code, such as the actual
        // connection to a database server.
        // ...

```

```

    // The static method that controls access to the singleton
    // instance.

```

```

    public static method getInstance() is

```

```

        if (Database.instance == null) then
            acquireThreadLock() and then
                // Ensure that the instance hasn't yet been
                // initialized by another thread while this one
                // has been waiting for the lock's release.

```

```

                if (Database.instance == null) then
                    Database.instance = new Database()

```

```

            return Database.instance

```

```

    // Finally, any singleton should define some business logic
    // which can be executed on its instance.

```

```

    public method query(sql) is

```

```

        // For instance, all database queries of an app go
        // through this method. Therefore, you can place
        // throttling or caching logic here.
        // ...

```

```

class Application is

```

```

    method main() is

```

```

        Database foo = Database.getInstance()

```

```

        foo.query("SELECT ...")

```

```

        // ...

```

```

        Database bar = Database.getInstance()

```

```

        bar.query("SELECT ...")

```

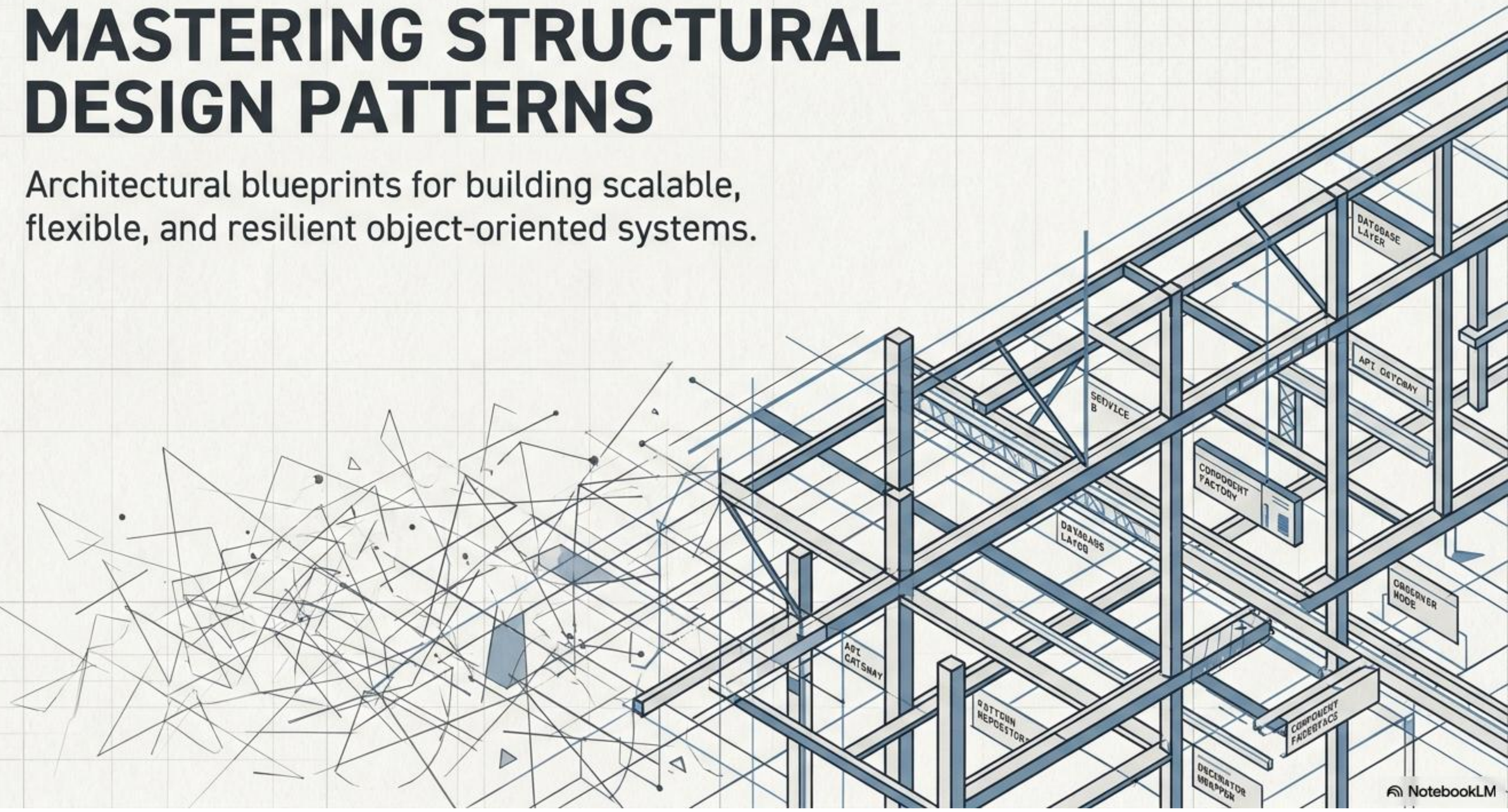
```

        // The variable `bar` will contain the same object as
        // the variable `foo`.

```


MASTERING STRUCTURAL DESIGN PATTERNS

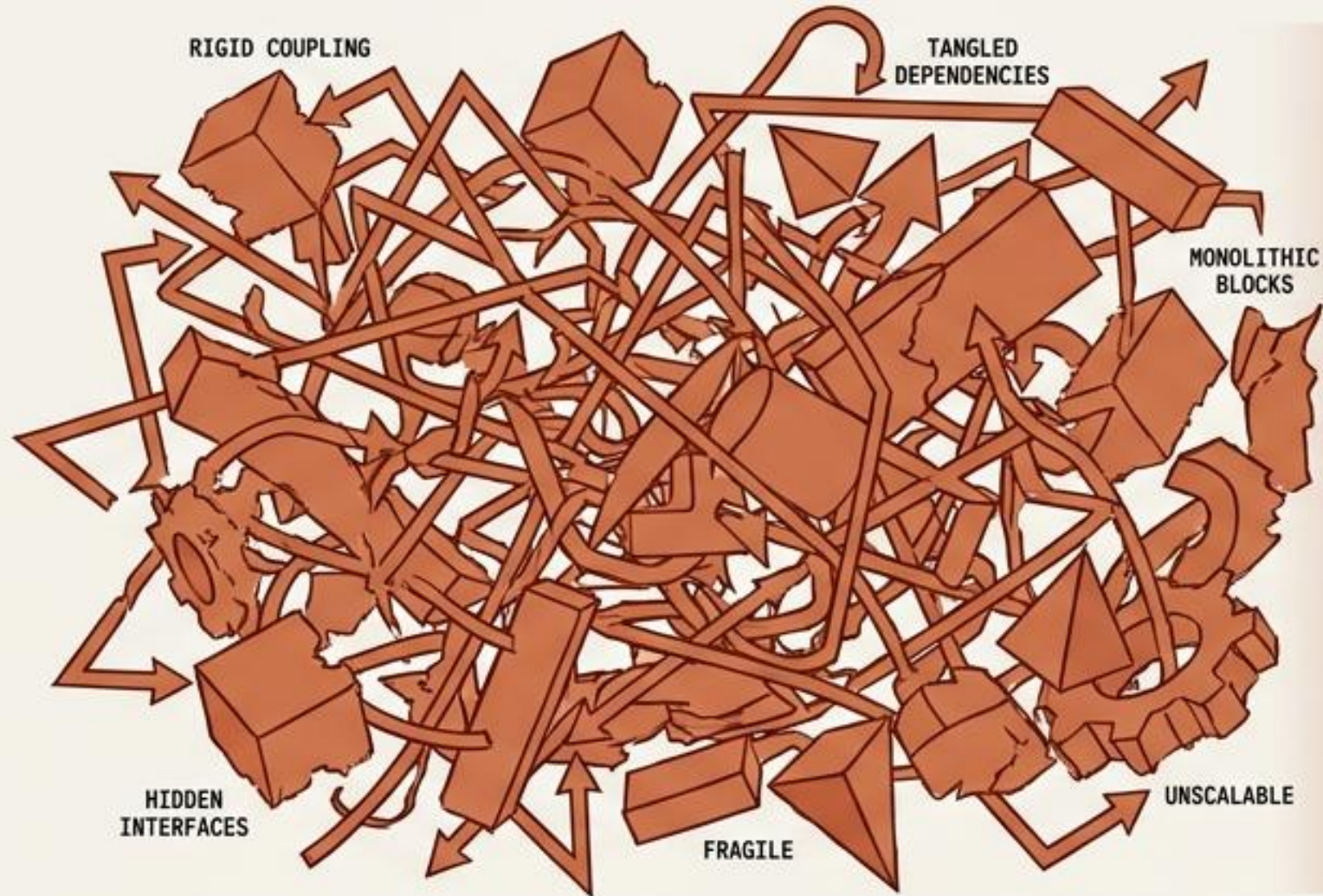
Architectural blueprints for building scalable, flexible, and resilient object-oriented systems.



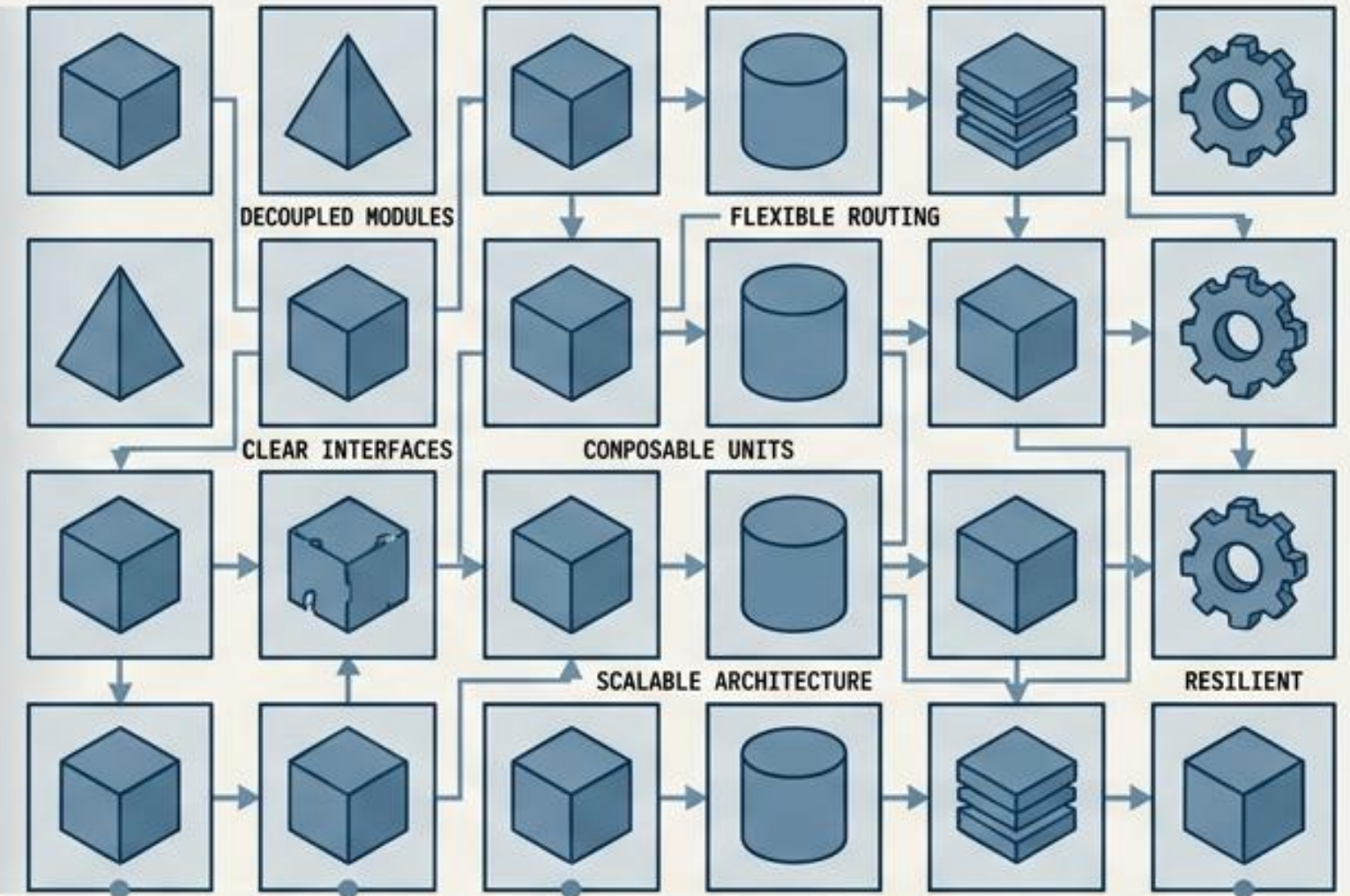
THE ANATOMY OF COMPOSITION

Structural patterns define how to assemble objects into larger, flexible architectures. They resolve systemic friction by focusing on three core architectural principles.

SYSTEMIC FRICTION



STRUCTURAL COMPOSITION



1. DECOUPLING:

Separating rigid interfaces from their underlying implementations.

2. COMPOSITION:

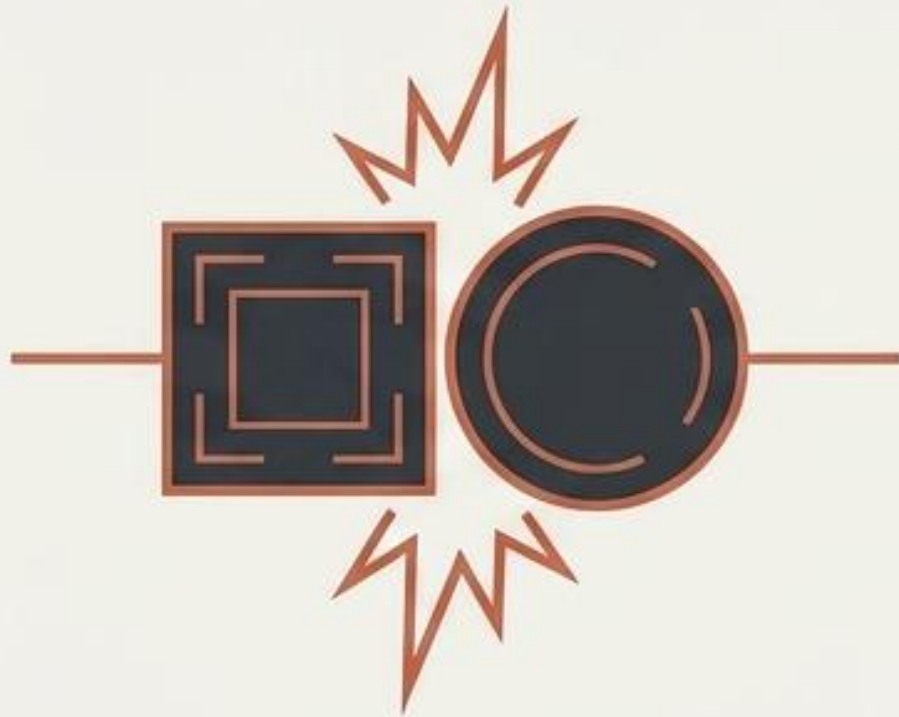
Building complex, scalable behaviors from simple, isolated pieces.

3. WRAPPING:

Enhancing objects dynamically without altering their core DNA.

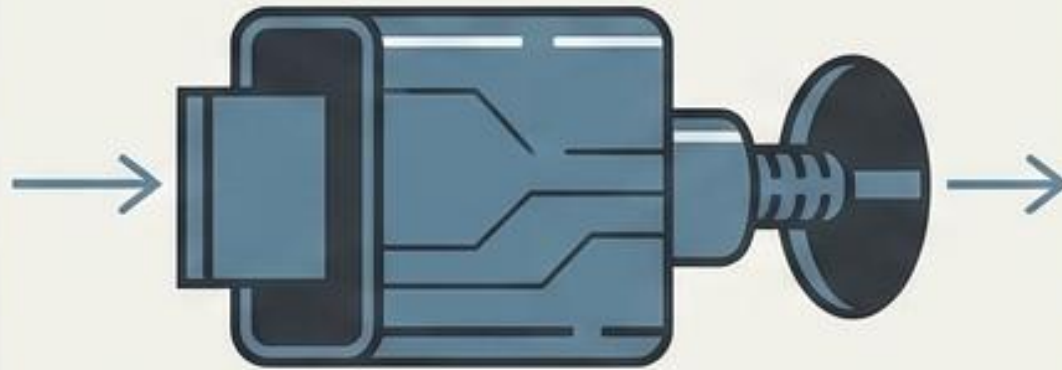
THE FRICTION

Incompatible interfaces prevent distinct systems from communicating.

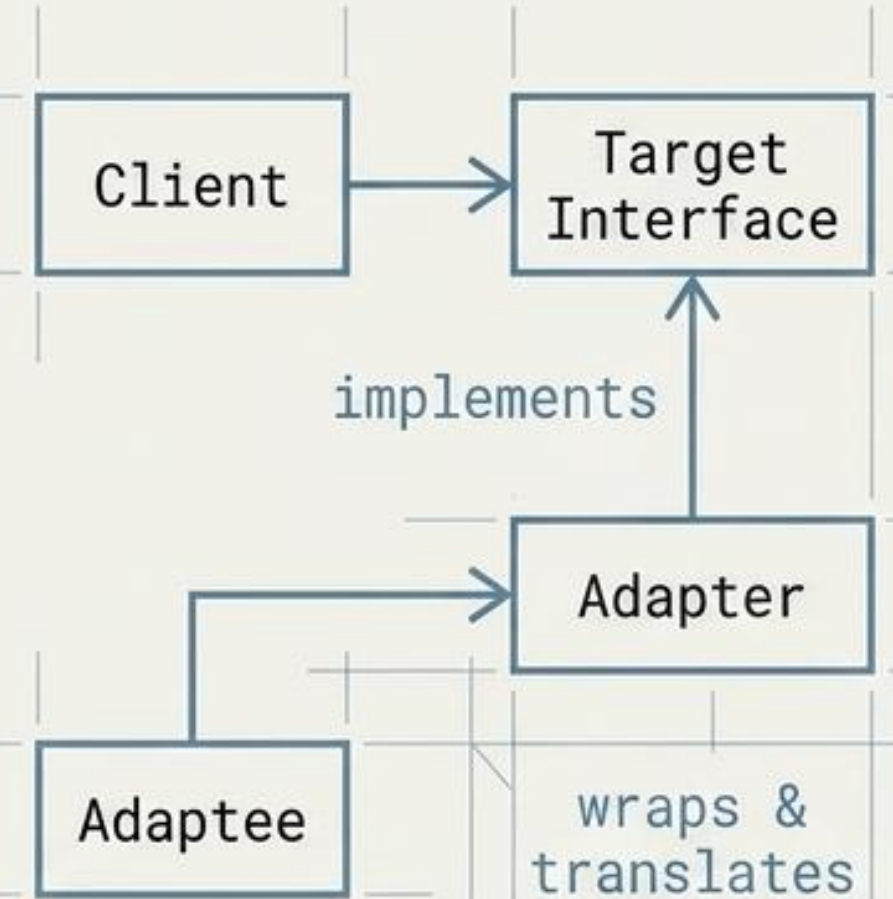


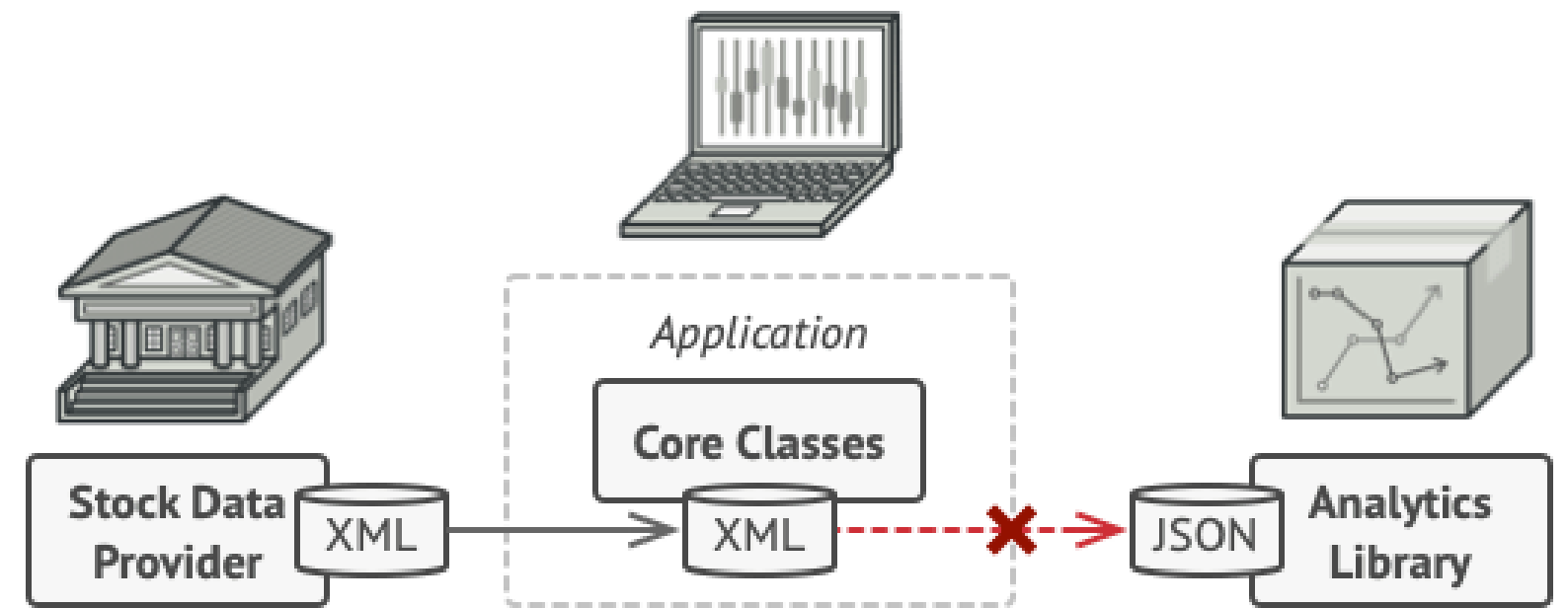
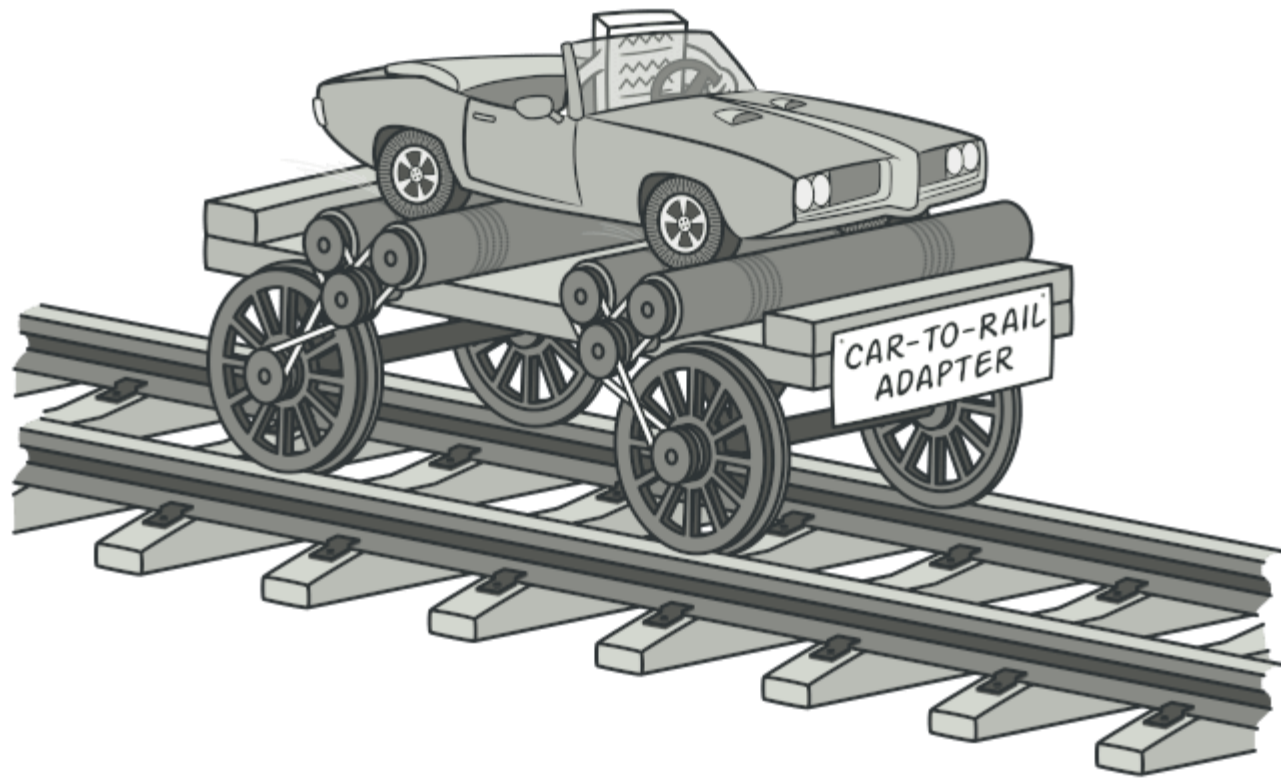
THE MECHANISM

Create a wrapper class that catches calls from the client and translates them into a format the existing system understands.



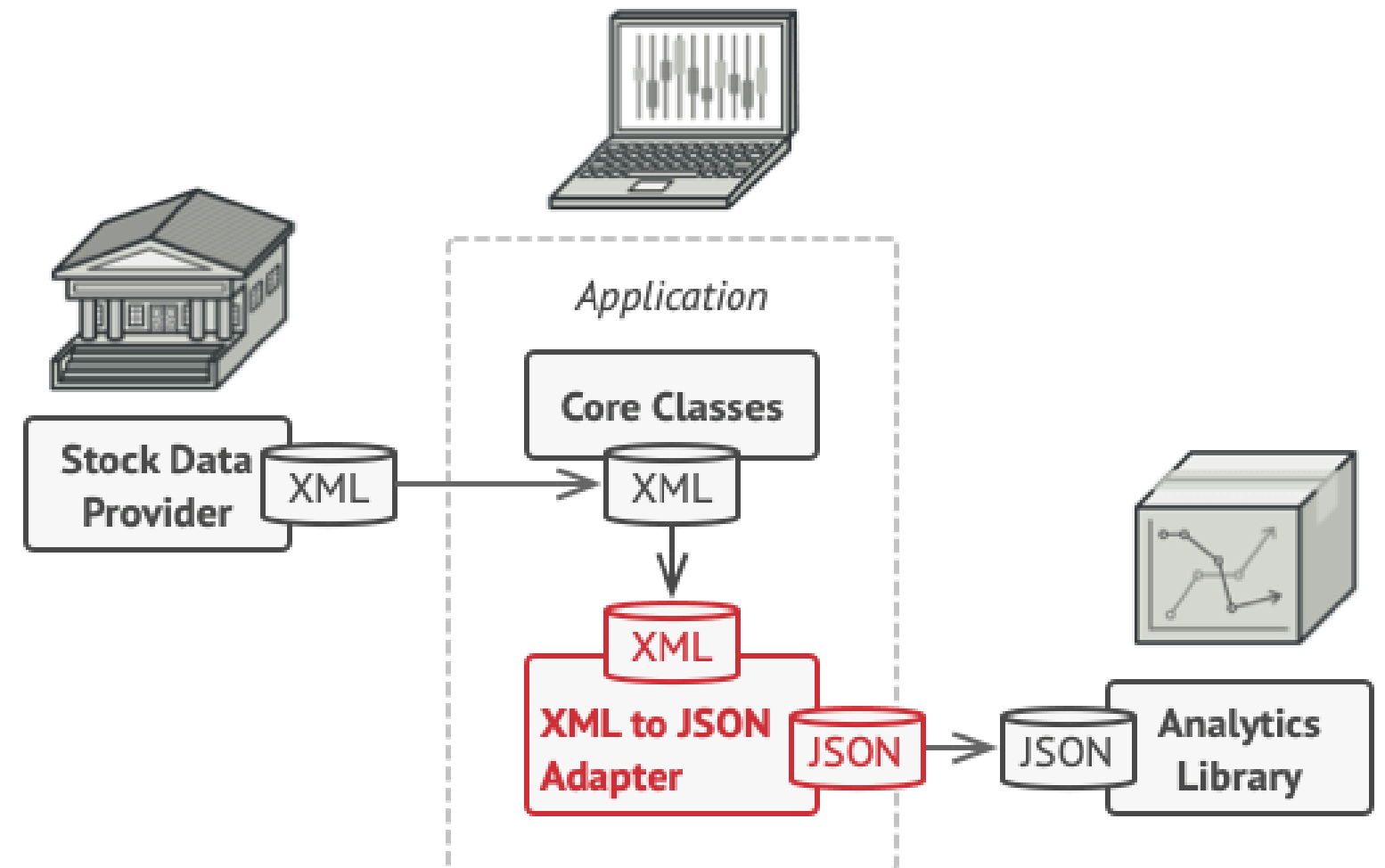
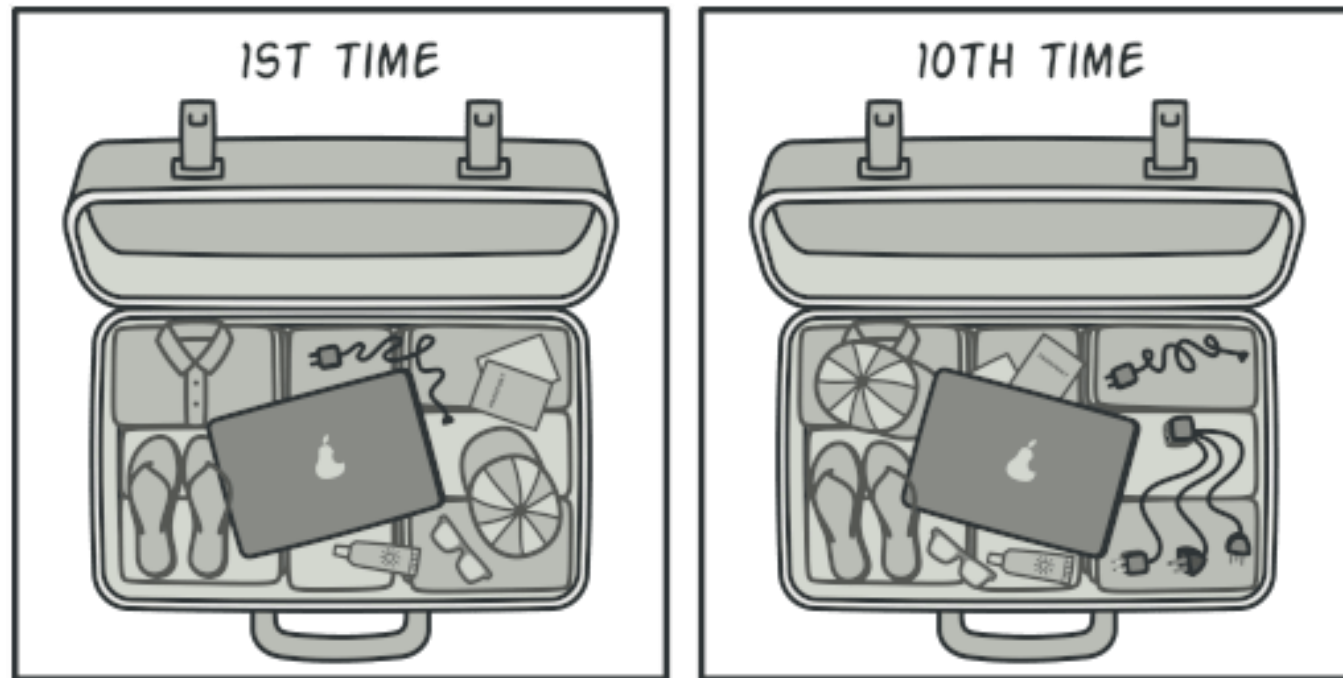
THE SCHEMATIC: ADAPTER

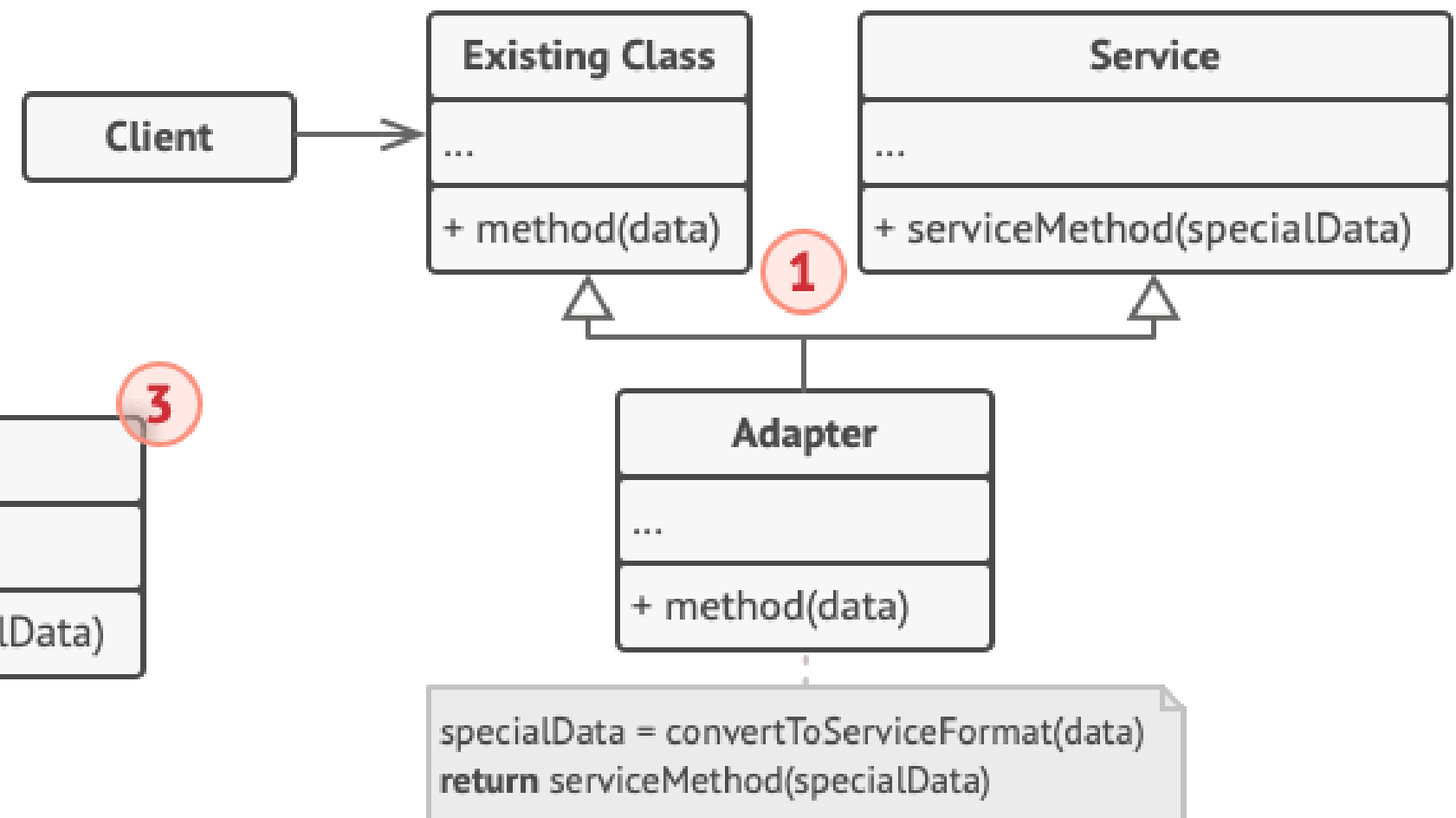
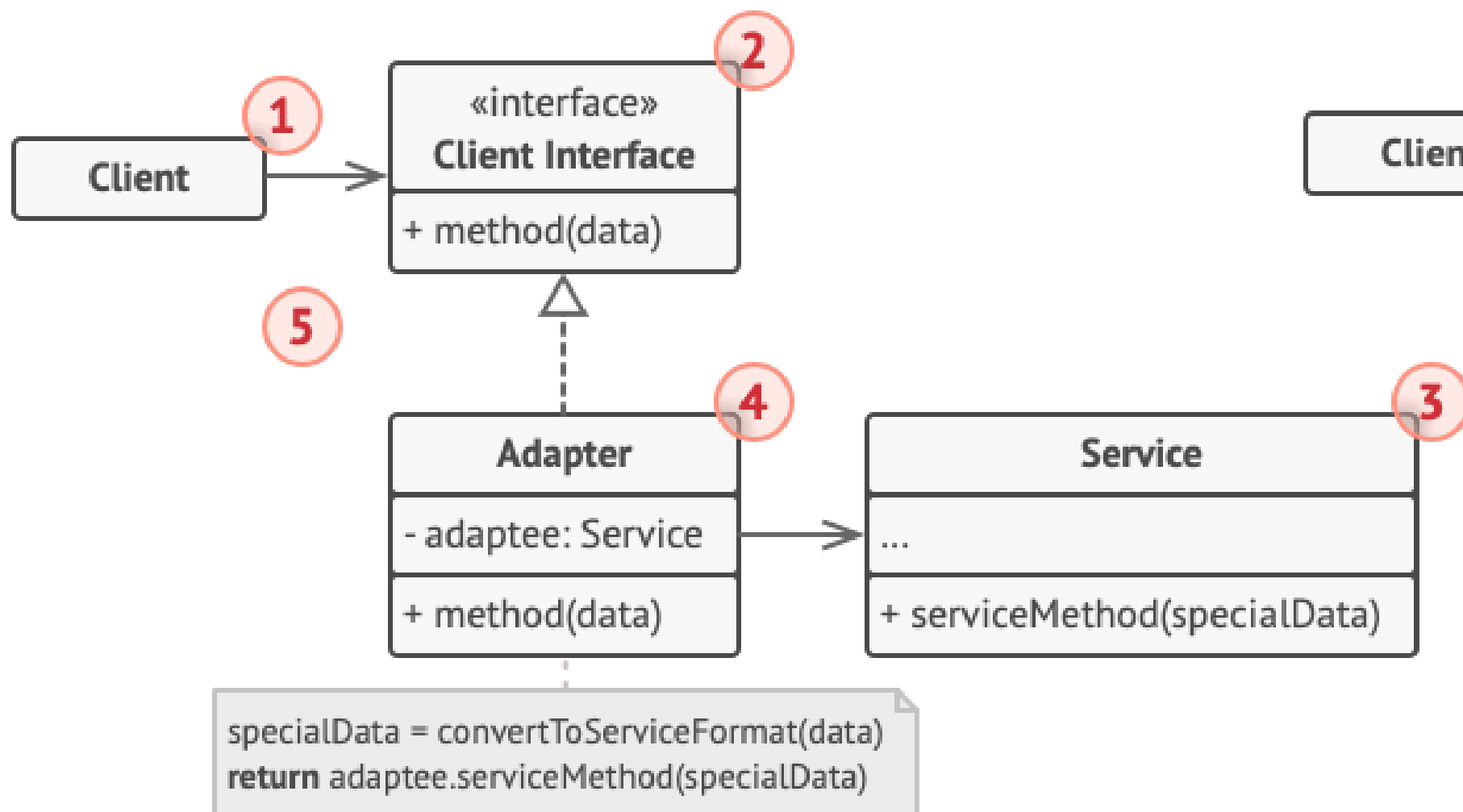




You can't use the analytics library "as is" because it expects the data in a format that's incompatible with your app.

TRAVELING ABROAD






```
// Say you have two classes with compatible interfaces:
// RoundHole and RoundPeg.
class RoundHole is
    constructor RoundHole(radius) { ... }

    method getRadius() is
        // Return the radius of the hole.

    method fits(peg: RoundPeg) is
        return this.getRadius() >= peg.getRadius()

class RoundPeg is
    constructor RoundPeg(radius) { ... }

    method getRadius() is
        // Return the radius of the peg.

// But there's an incompatible class: SquarePeg.
class SquarePeg is
    constructor SquarePeg(width) { ... }

    method getWidth() is
        // Return the square peg width.

// An adapter class lets you fit square pegs into round holes.
// It extends the RoundPeg class to let the adapter objects act
// as round pegs.
class SquarePegAdapter extends RoundPeg is
    // In reality, the adapter contains an instance of the
    // SquarePeg class.
    private field peg: SquarePeg

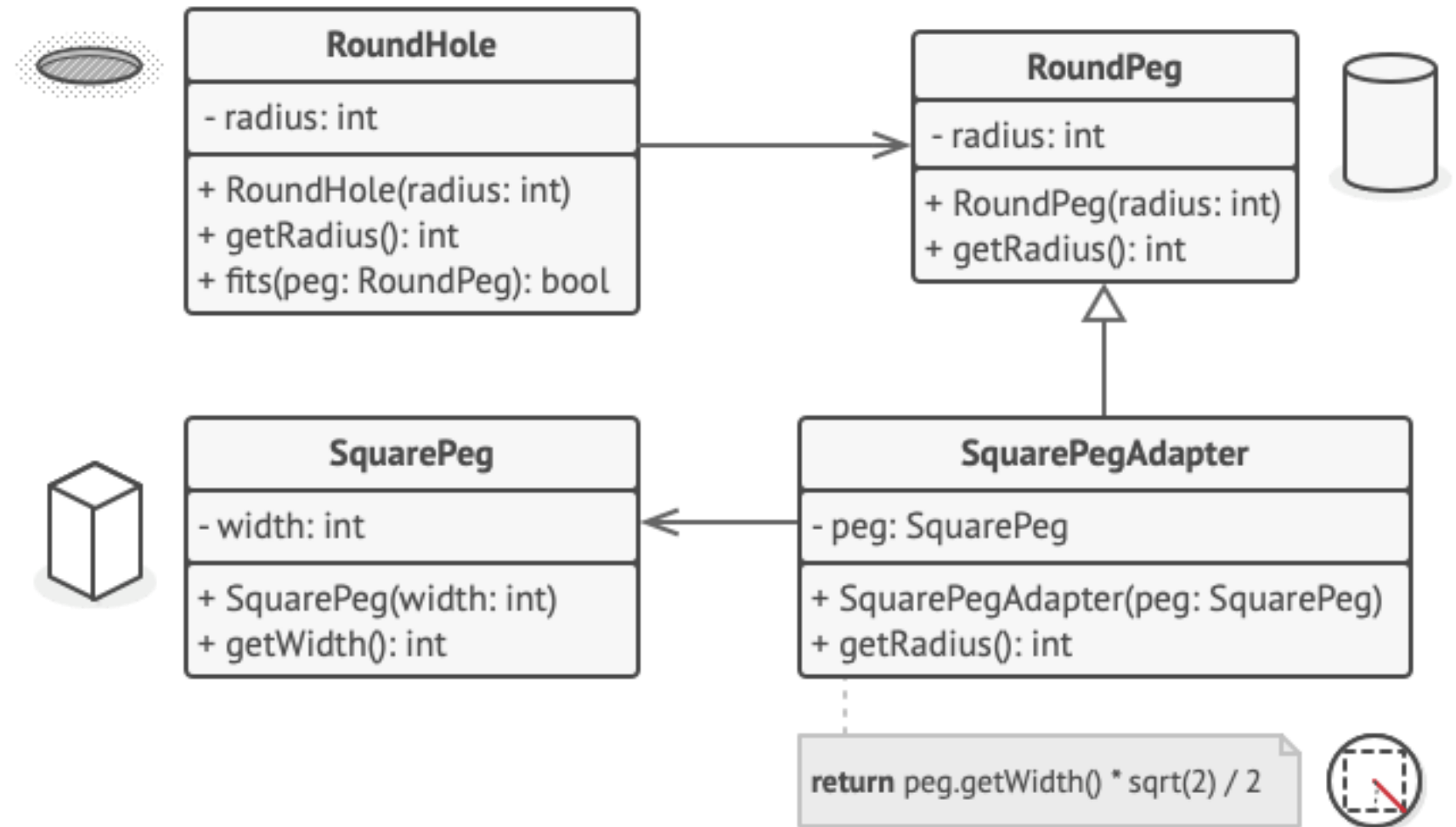
    constructor SquarePegAdapter(peg: SquarePeg) is
        this.peg = peg

    method getRadius() is
        // The adapter pretends that it's a round peg with a
        // radius that could fit the square peg that the adapter
        // actually wraps.
        return peg.getWidth() * Math.sqrt(2) / 2

// Somewhere in client code.
hole = new RoundHole(5)
rpeg = new RoundPeg(5)
hole.fits(rpeg) // true

small_sqpeg = new SquarePeg(5)
large_sqpeg = new SquarePeg(10)
hole.fits(small_sqpeg) // this won't compile (incompatible types)

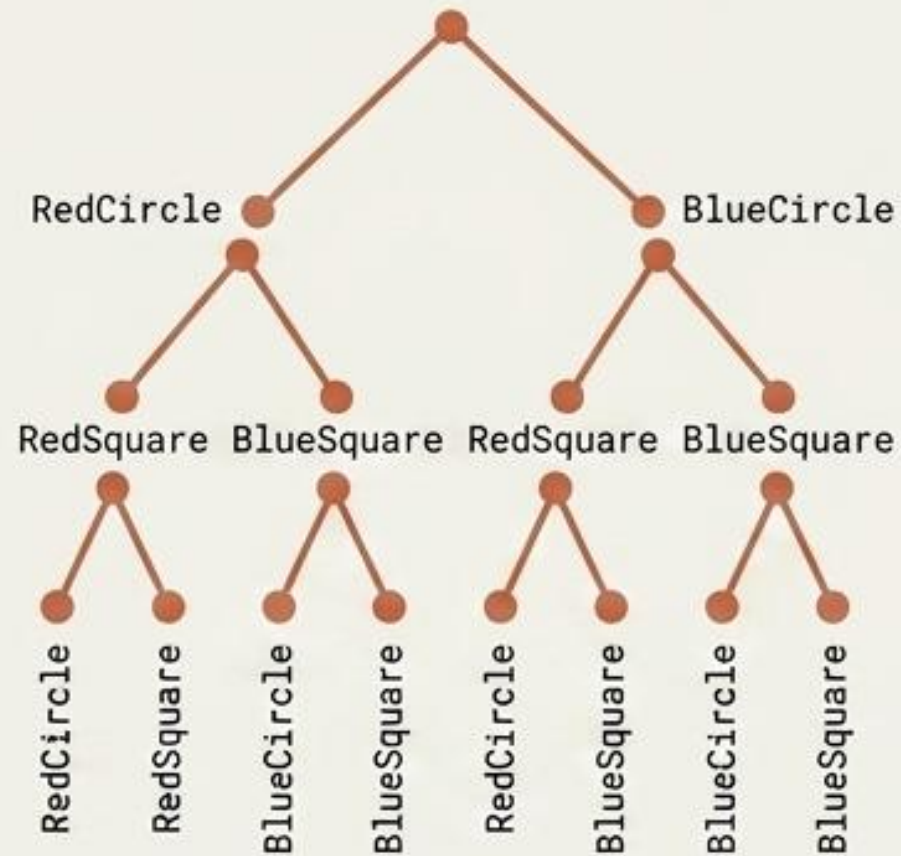
small_sqpeg_adapter = new SquarePegAdapter(small_sqpeg)
large_sqpeg_adapter = new SquarePegAdapter(large_sqpeg)
hole.fits(small_sqpeg_adapter) // true
hole.fits(large_sqpeg_adapter) // false
```



Adapting square pegs to round holes.

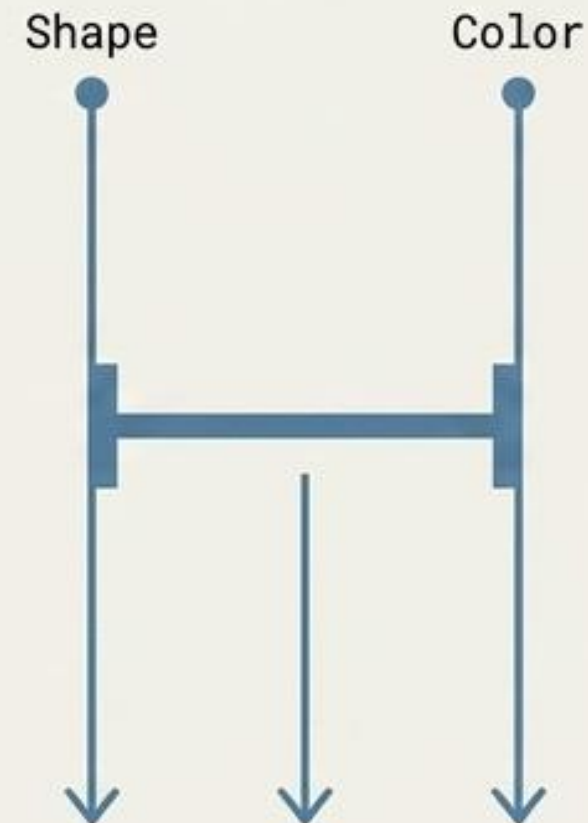
THE FRICTION

Cartesian explosion: hierarchy grows exponentially when extending both shapes and colors.

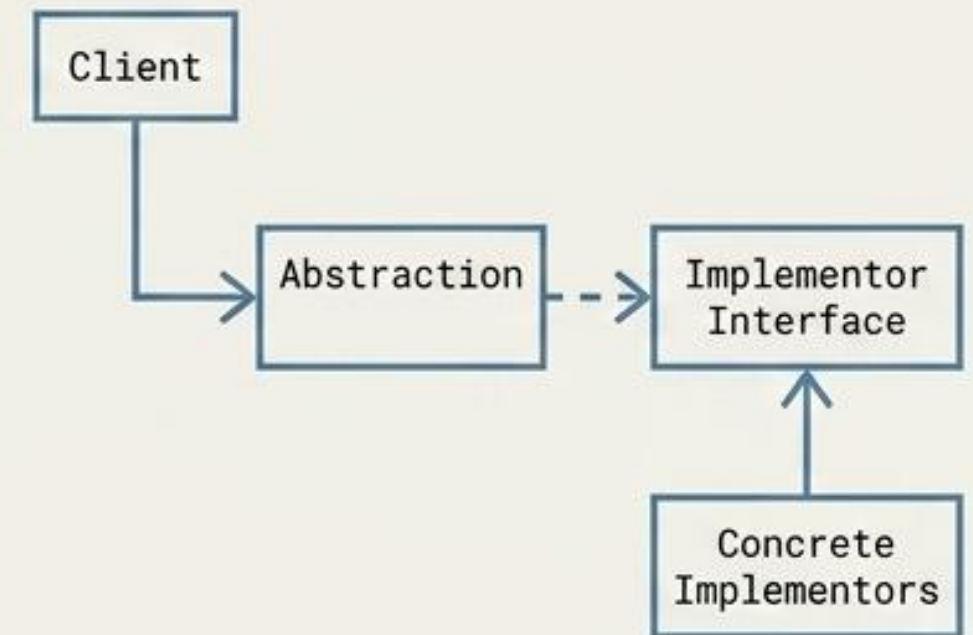


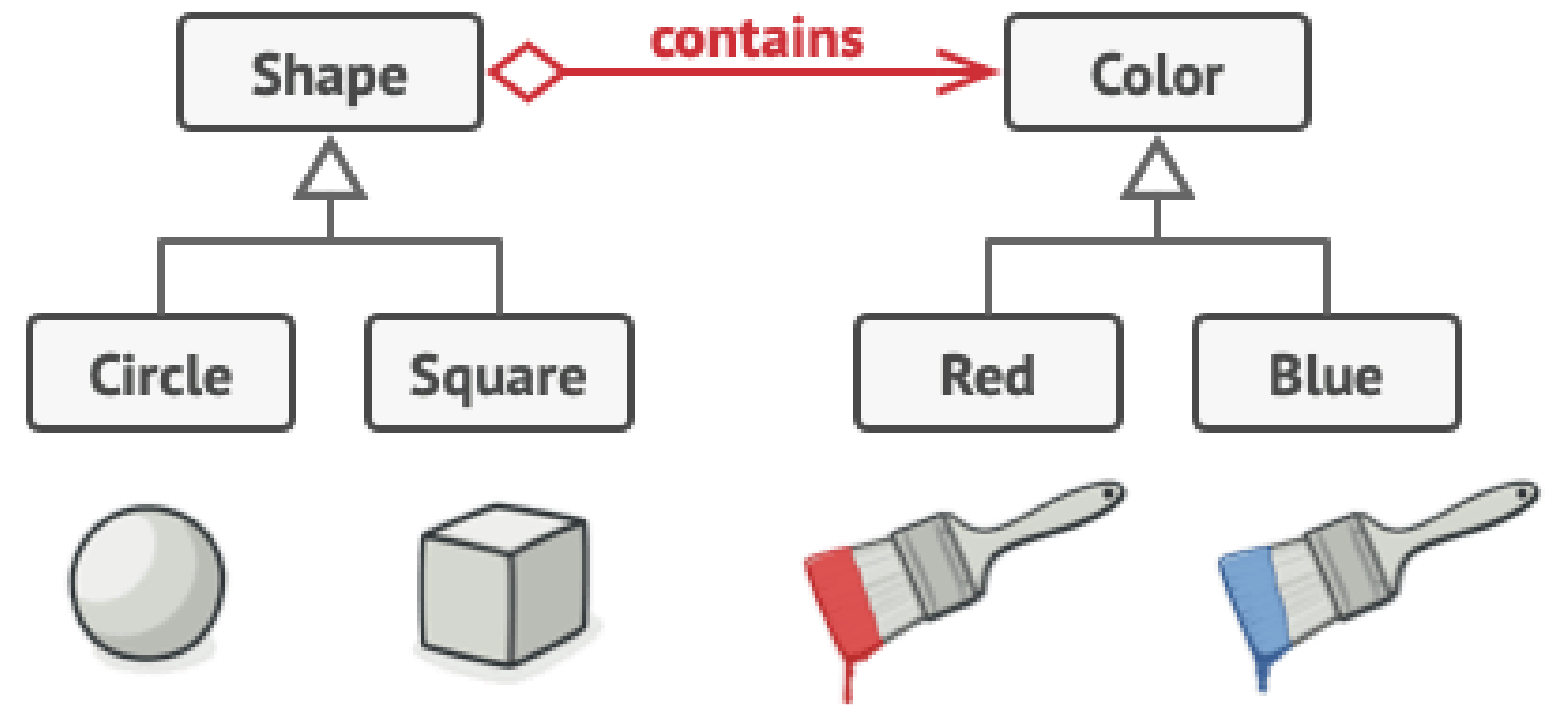
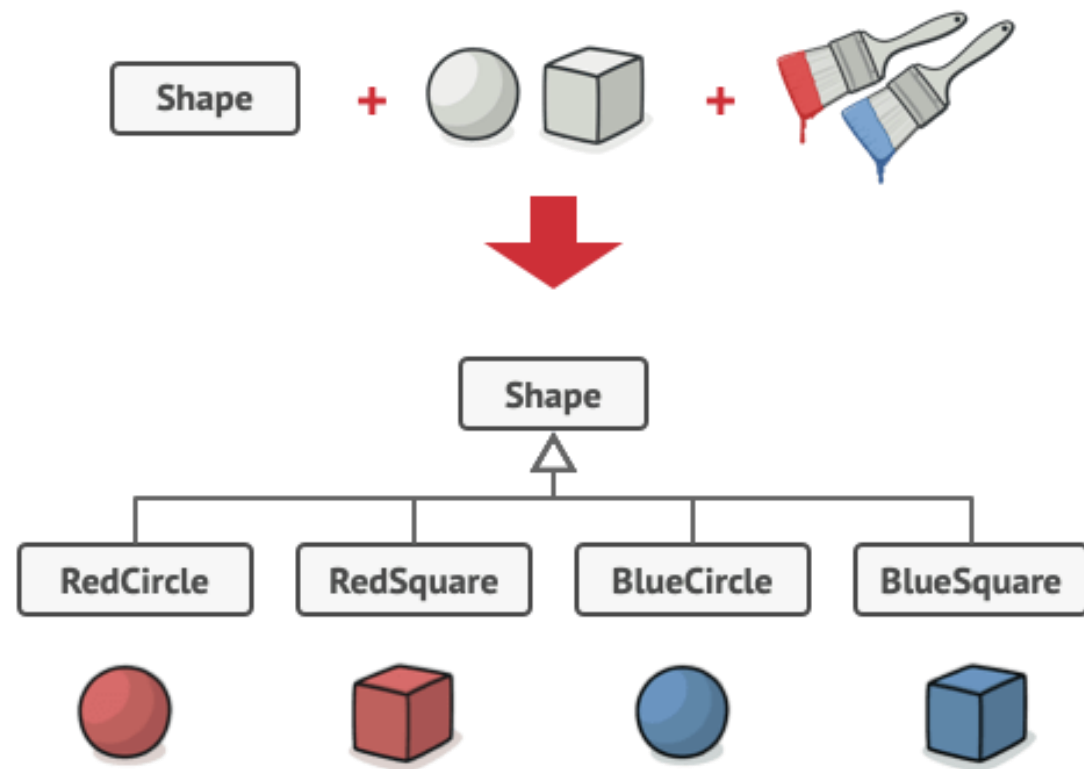
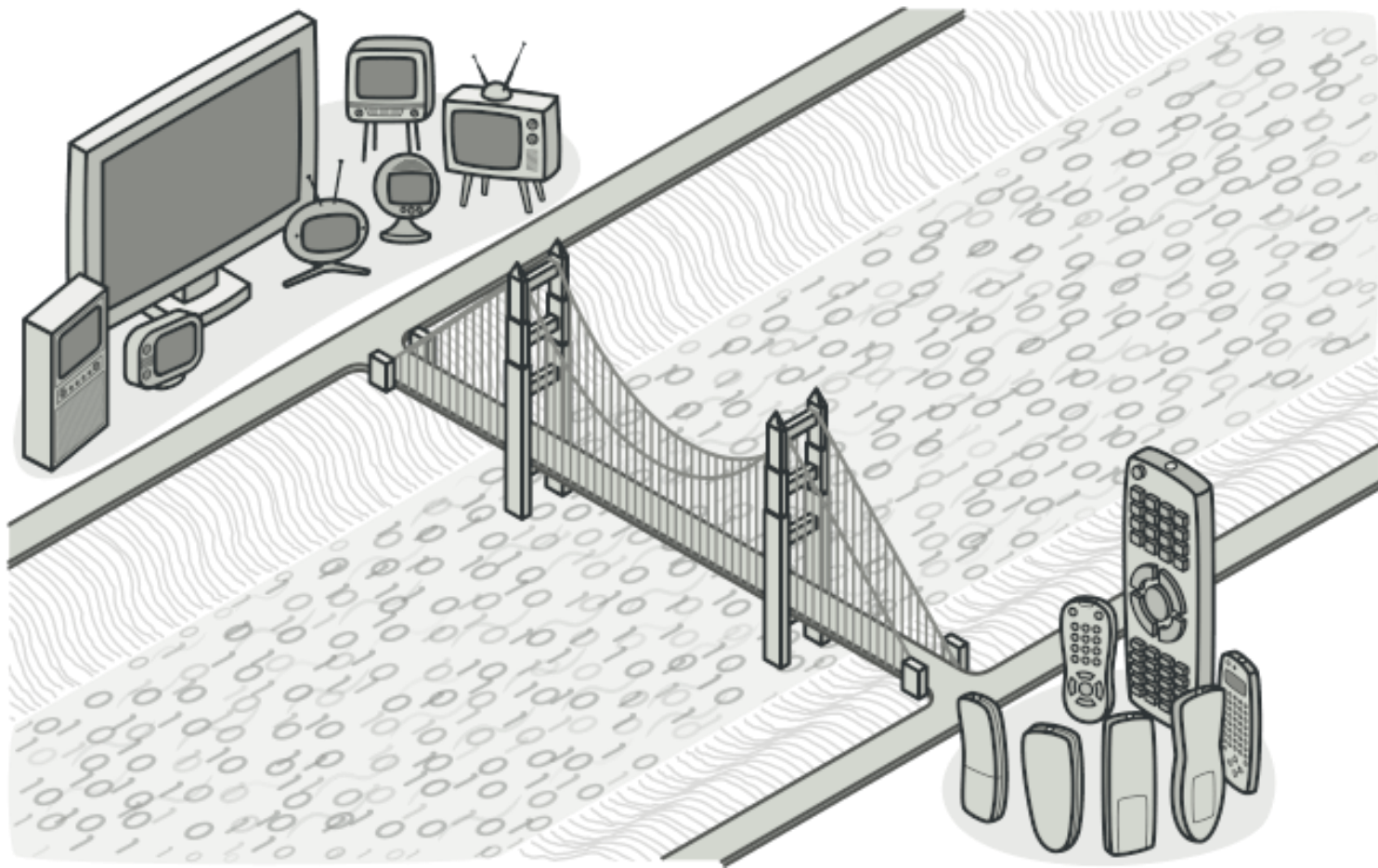
THE MECHANISM

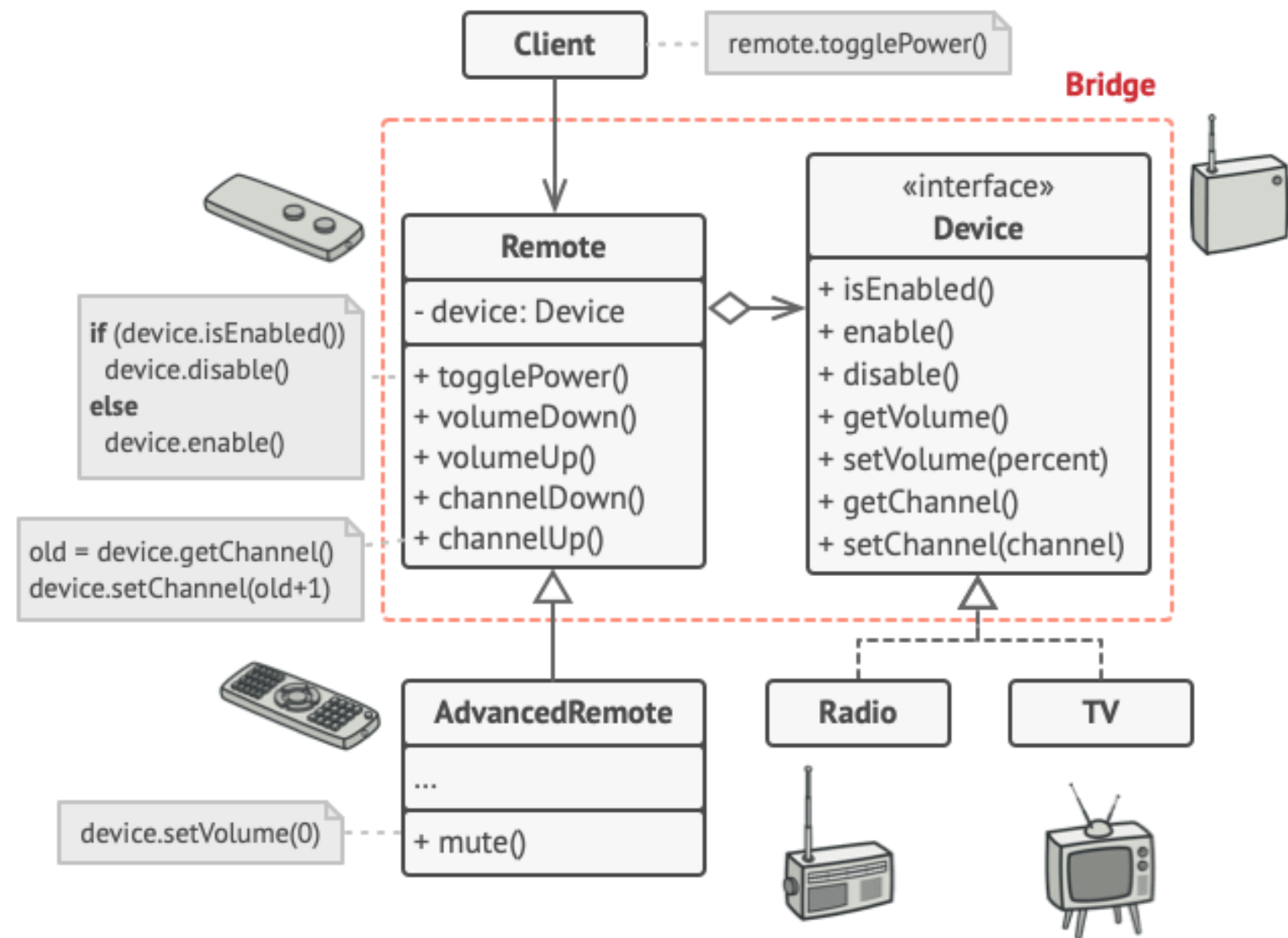
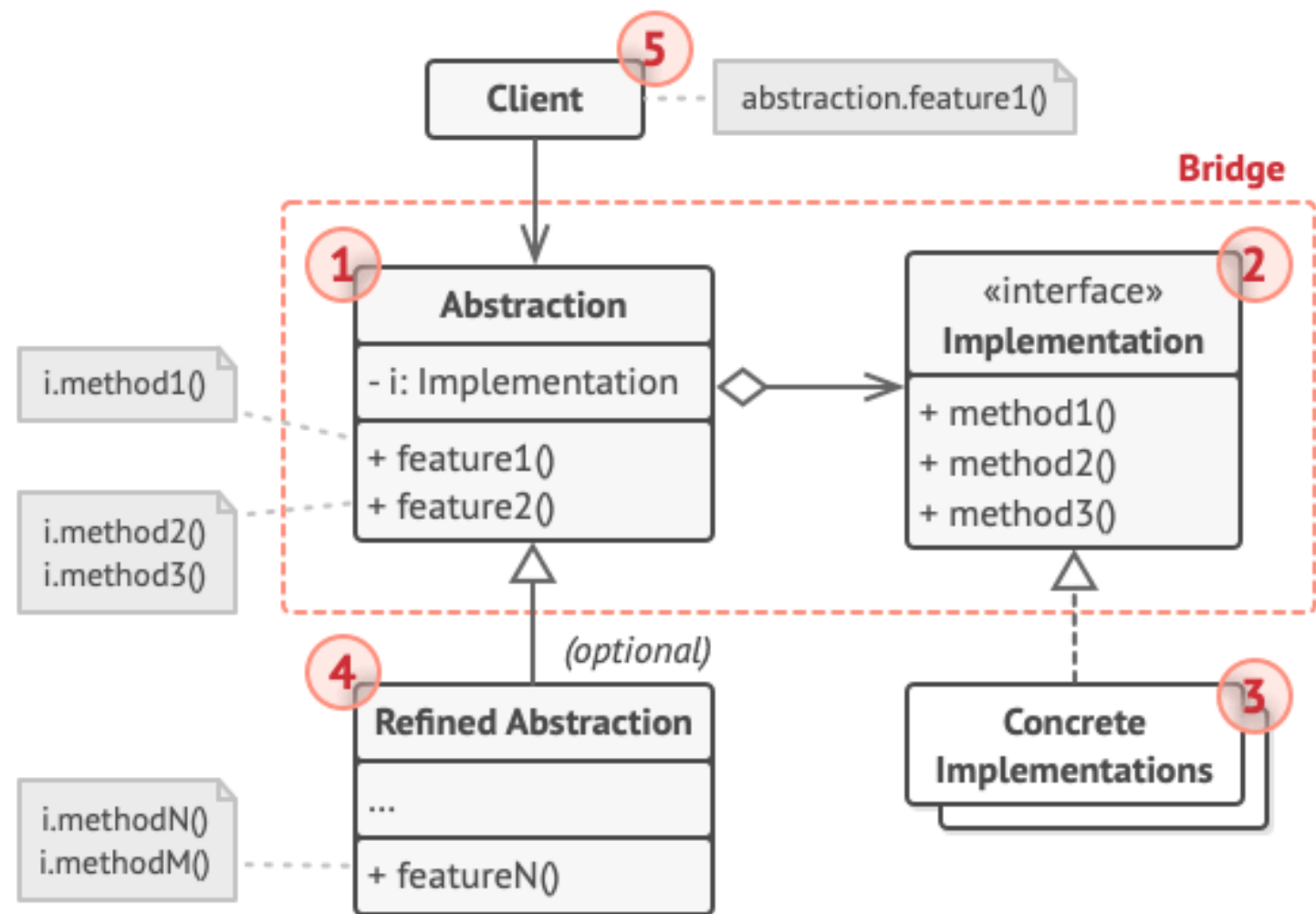
Split a monolithic class into two separate hierarchies—Abstraction and Implementation—allowing them to evolve independently.



THE SCHEMATIC: BRIDGE








```
// The "abstraction" defines the interface for the "control"  
// part of the two class hierarchies. It maintains a reference  
// to an object of the "implementation" hierarchy and delegates  
// all of the real work to this object.
```

```
class RemoteControl is  
    protected field device: Device  
    constructor RemoteControl(device: Device) is  
        this.device = device  
    method togglePower() is  
        if (device.isEnabled()) then  
            device.disable()  
        else  
            device.enable()  
    method volumeDown() is  
        device.setVolume(device.getVolume() - 10)  
    method volumeUp() is  
        device.setVolume(device.getVolume() + 10)  
    method channelDown() is  
        device.setChannel(device.getChannel() - 1)  
    method channelUp() is  
        device.setChannel(device.getChannel() + 1)
```

```
// You can extend classes from the abstraction hierarchy  
// independently from device classes.
```

```
class AdvancedRemoteControl extends RemoteControl is  
    method mute() is  
        device.setVolume(0)
```

```
// The "implementation" interface declares methods common to all  
// concrete implementation classes. It doesn't have to match the  
// abstraction's interface. In fact, the two interfaces can be  
// entirely different. Typically the implementation interface  
// provides only primitive operations, while the abstraction  
// defines higher-level operations based on those primitives.
```

```
interface Device is  
    method isEnabled()  
    method enable()  
    method disable()  
    method getVolume()  
    method setVolume(percent)  
    method getChannel()  
    method setChannel(channel)
```

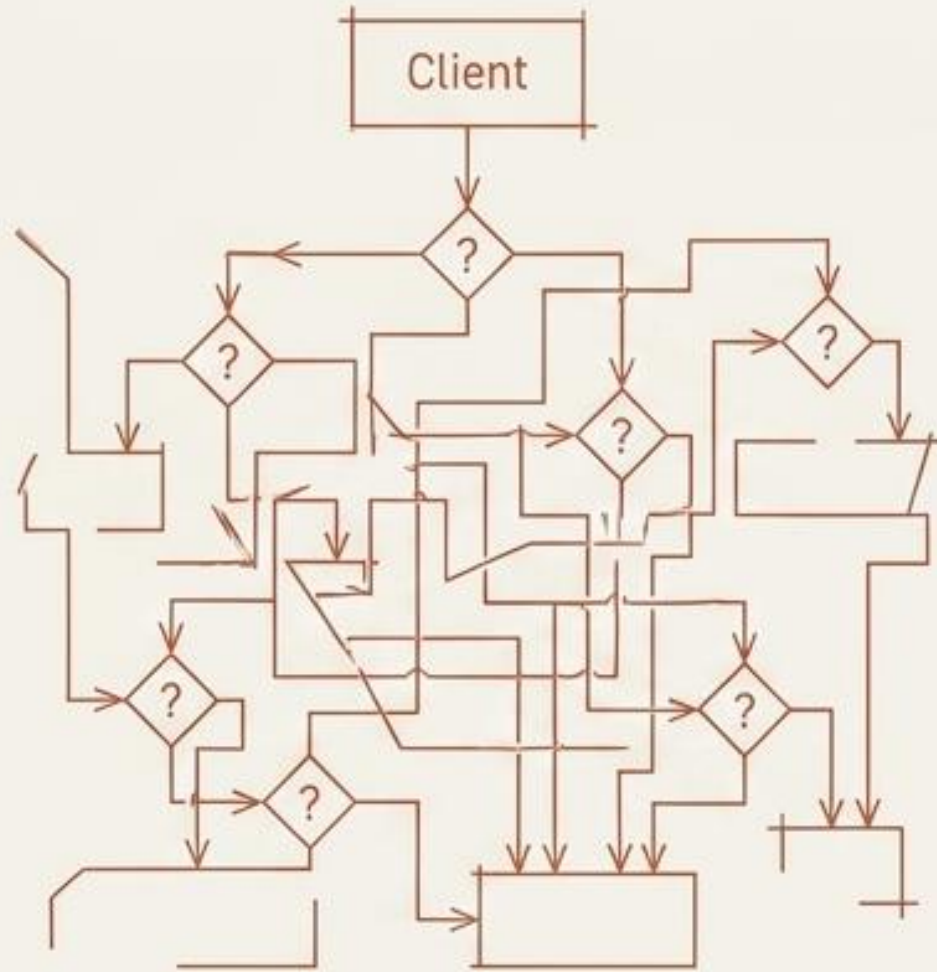
```
// All devices follow the same interface.  
class Tv implements Device is  
    // ...
```

```
class Radio implements Device is  
    // ...
```

```
// Somewhere in client code.  
tv = new Tv()  
remote = new RemoteControl(tv)  
remote.togglePower()  
  
radio = new Radio()  
remote = new AdvancedRemoteControl(radio)
```

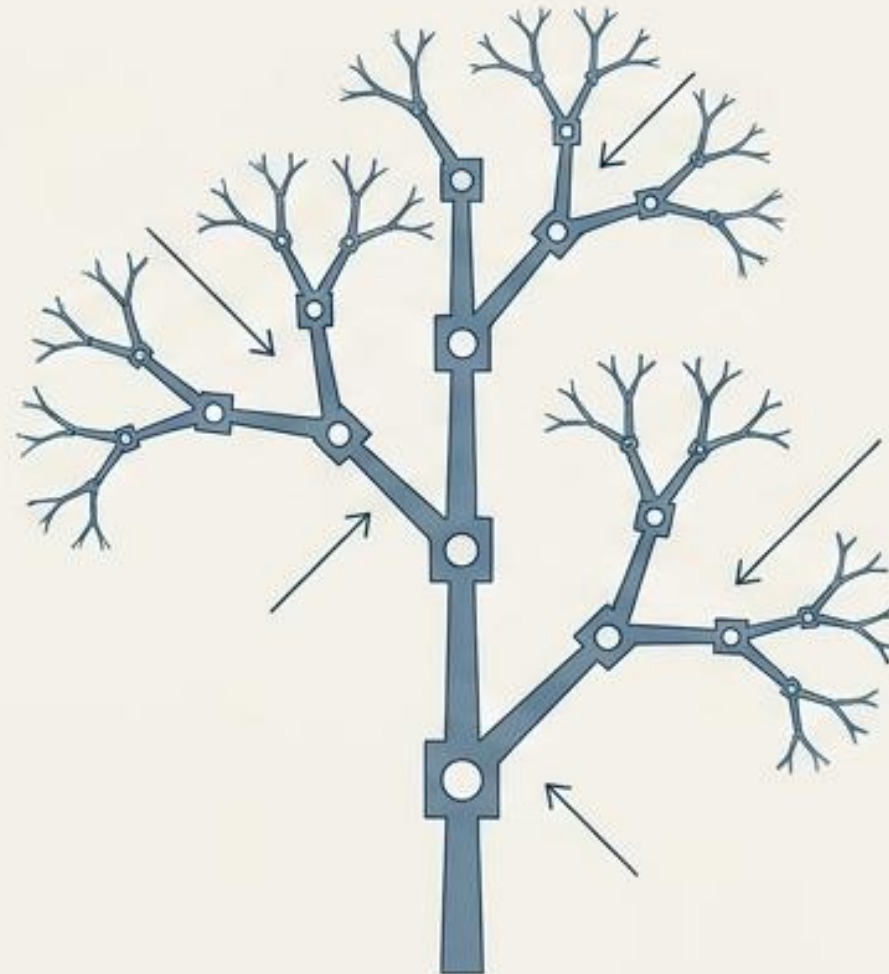

THE FRICTION

Clients must write complex conditional checks to differentiate between single items and groups of items.

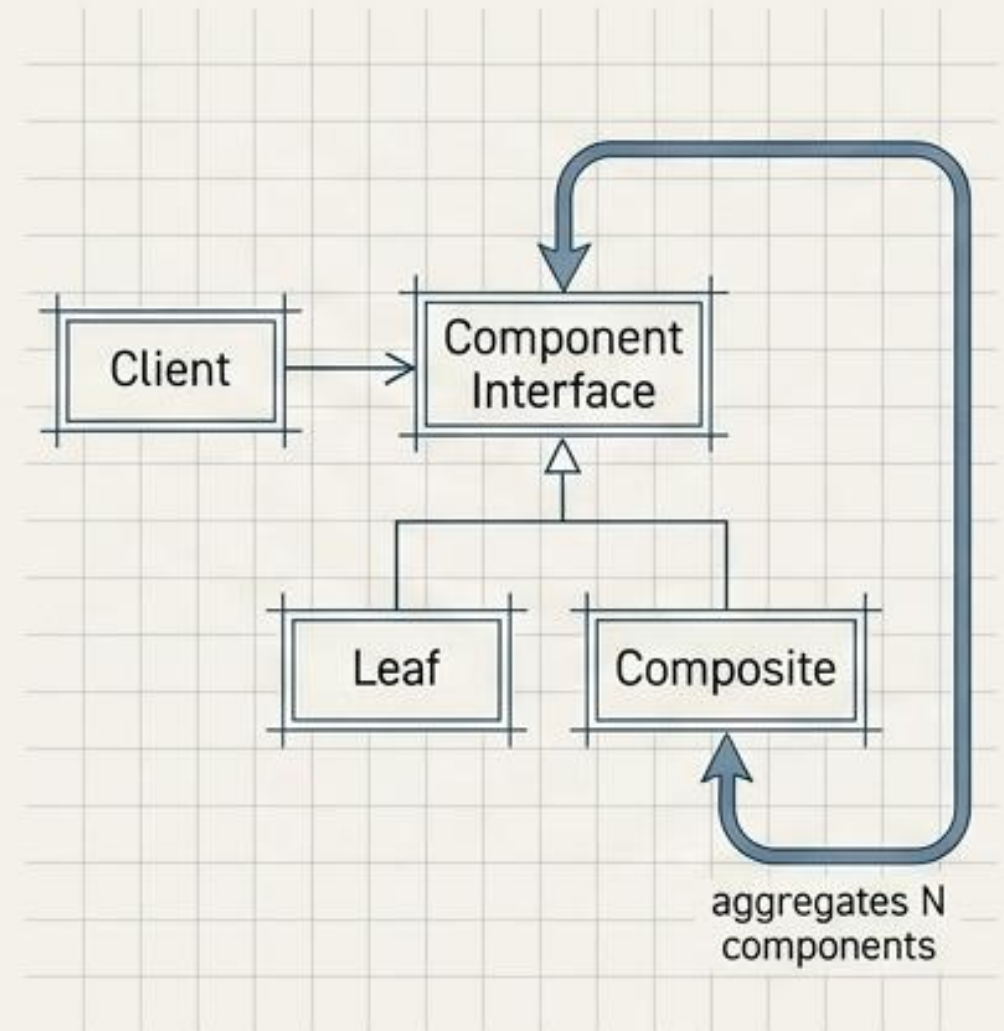


THE MECHANISM

Compose objects into tree structures. Clients treat individual objects (leaves) and compositions (branches) uniformly.

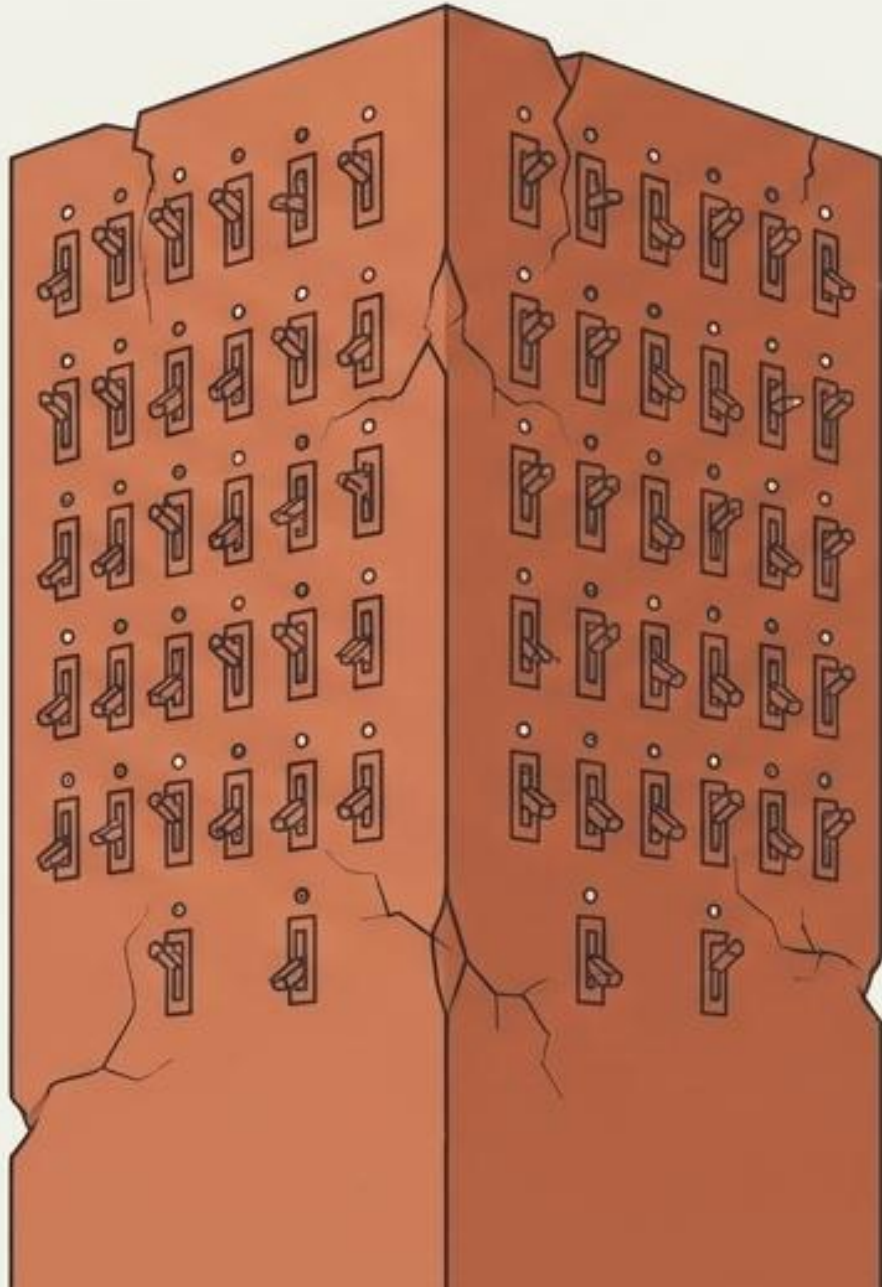


THE SCHEMATIC: COMPOSITE



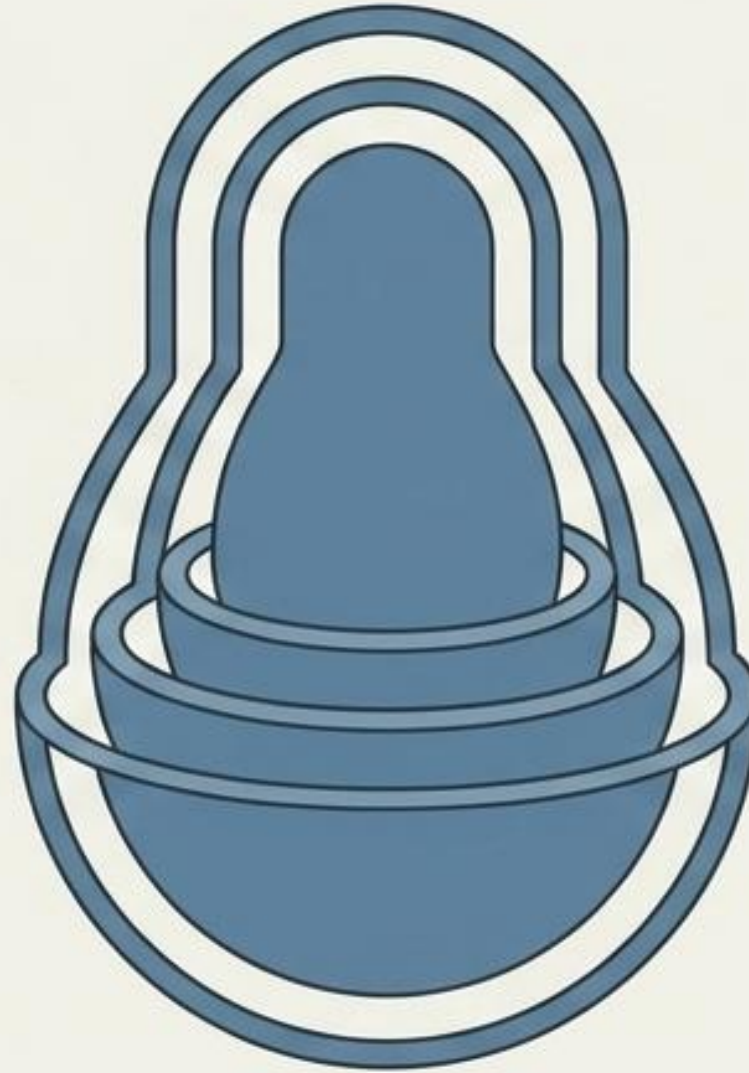
THE FRICTION

Bloated monolithic classes attempting to hardcode every possible combination of optional behaviors.

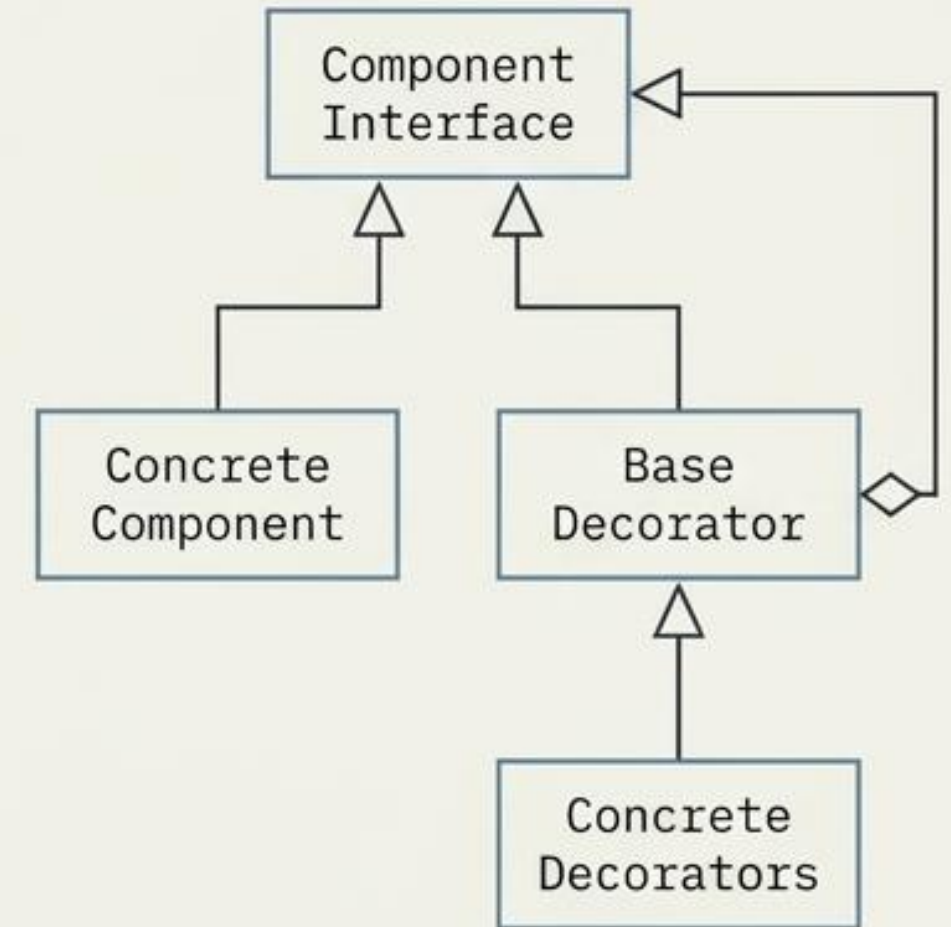


THE MECHANISM

Attach new behaviors dynamically by placing objects inside special wrapper objects. Stack wrappers infinitely.

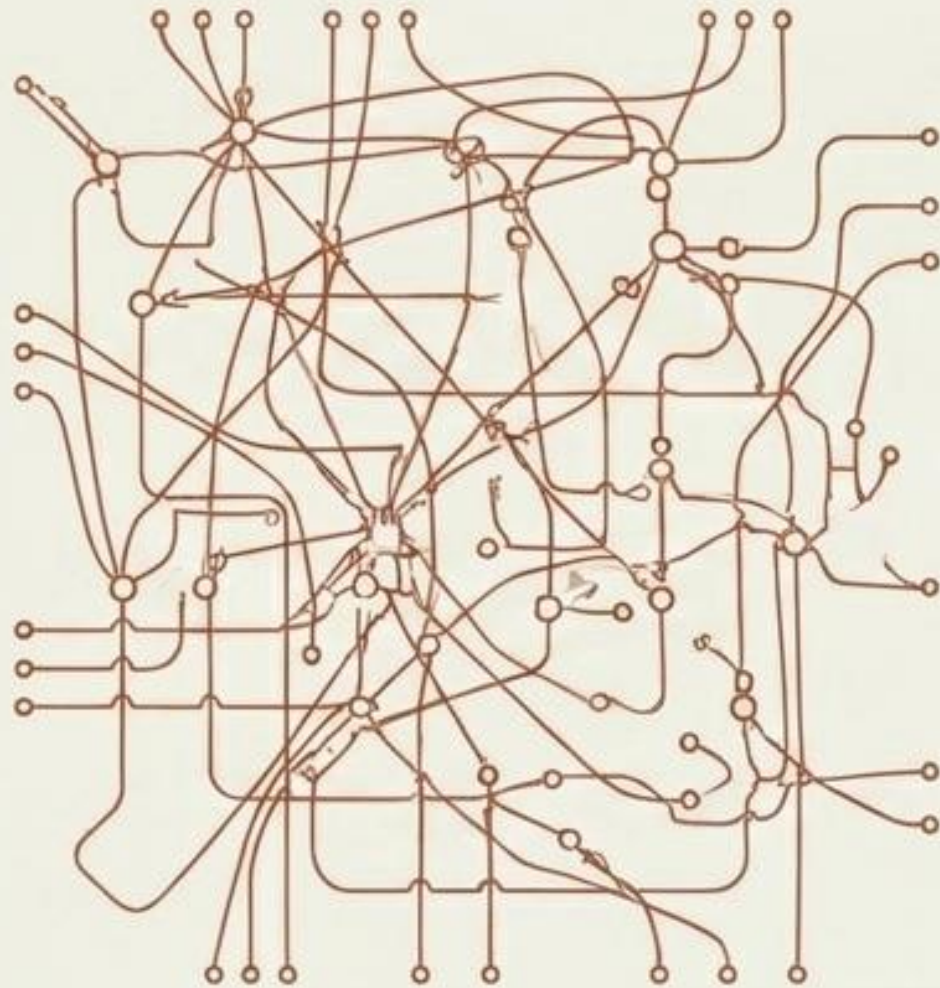


THE SCHEMATIC: DECORATOR



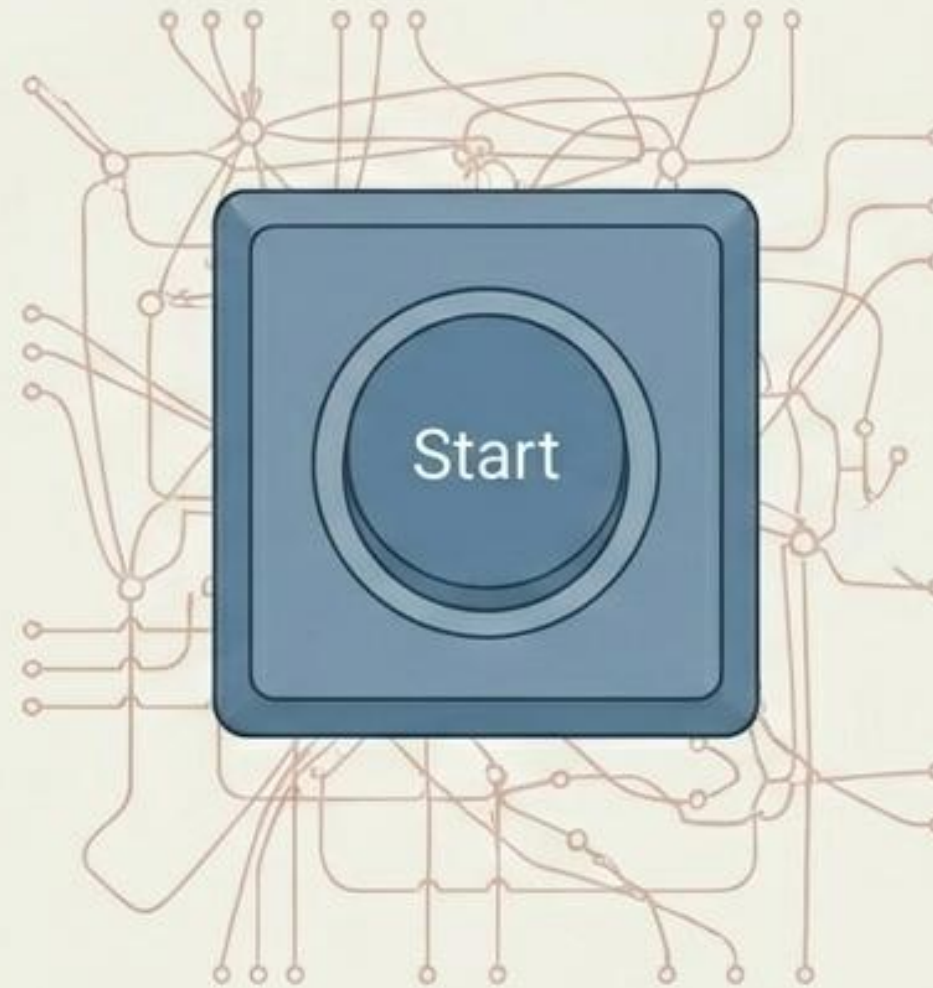
THE FRICTION

Clients are forced to manage the intricate, fragile dependencies of a highly complex subsystem.

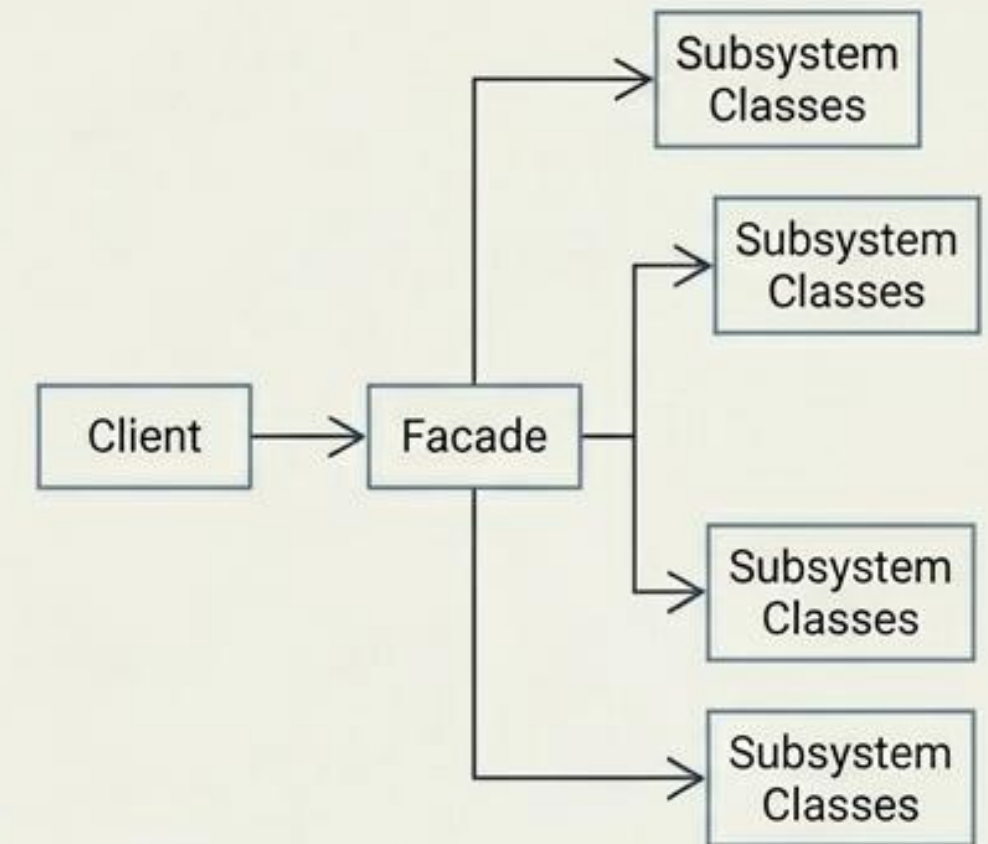


THE MECHANISM

Provide a simplified, higher-level interface to shield the client from the underlying architectural mess.



THE SCHEMATIC: FACADE



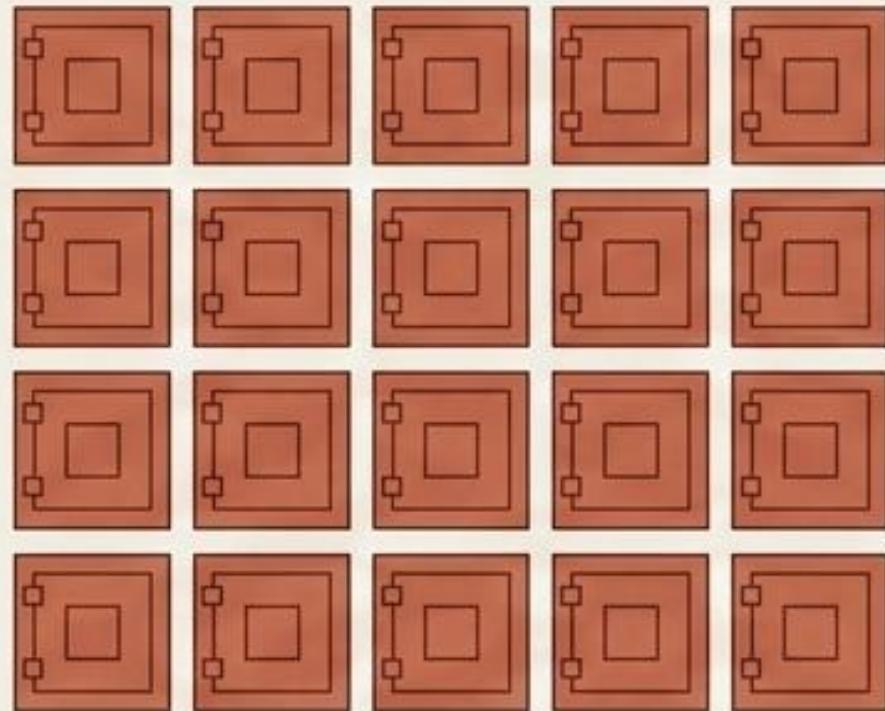
THE FRICTION

System memory exhausts rapidly when millions of identical objects are instantiated with duplicate state data.

SYSTEM RAM CAPACITY

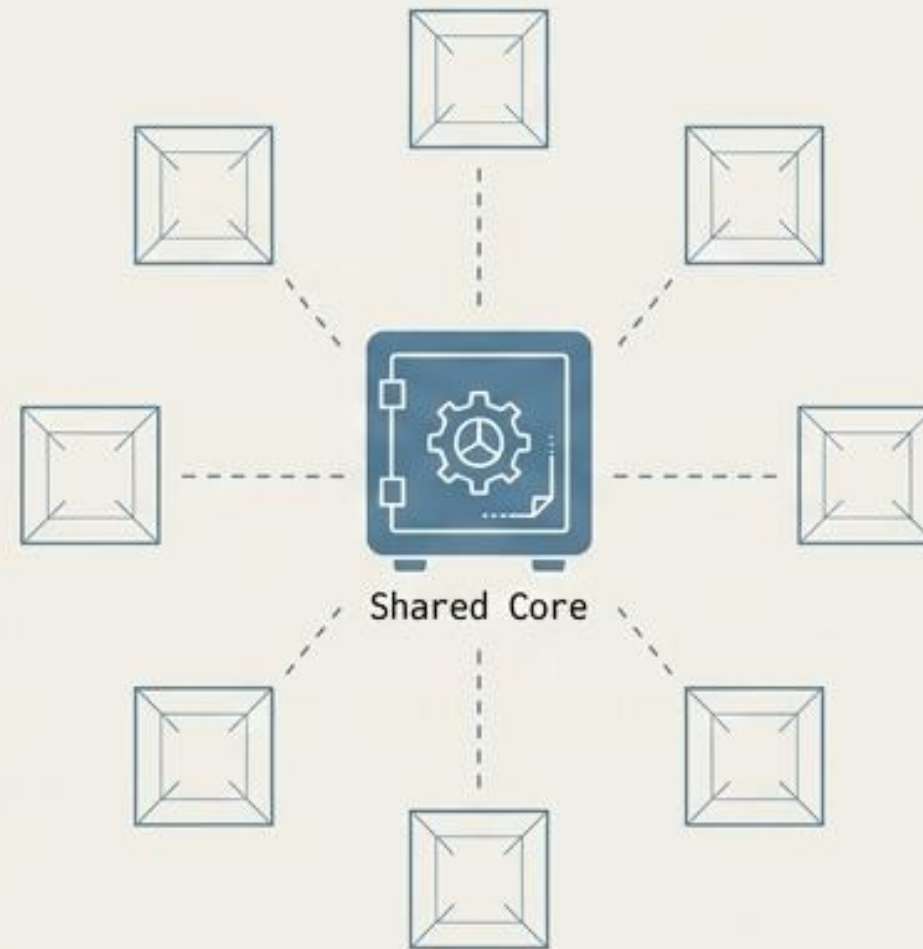


SYSTEM RAM CRITICAL

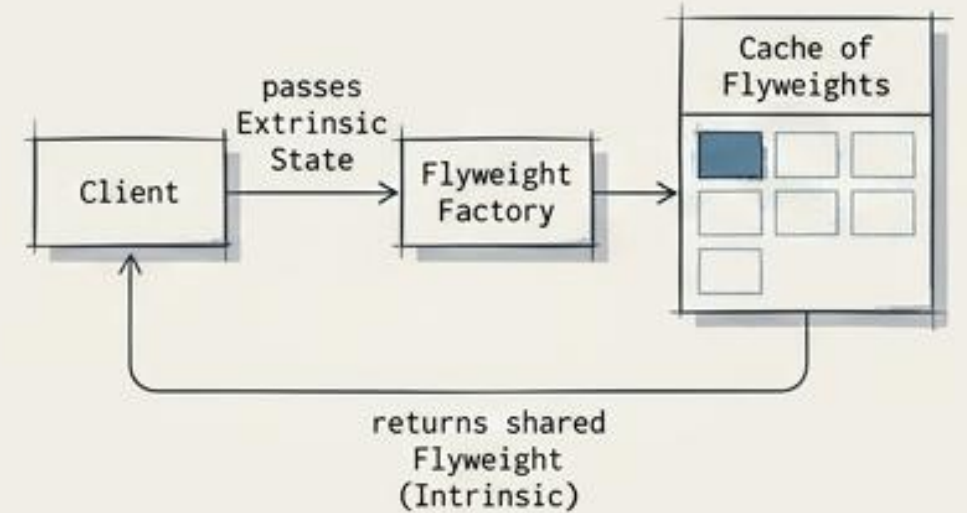


THE MECHANISM

Separate intrinsic (shared) state from extrinsic (contextual) state. Share the core blueprint to reduce memory footprint.



THE SCHEMATIC: FLYWEIGHT



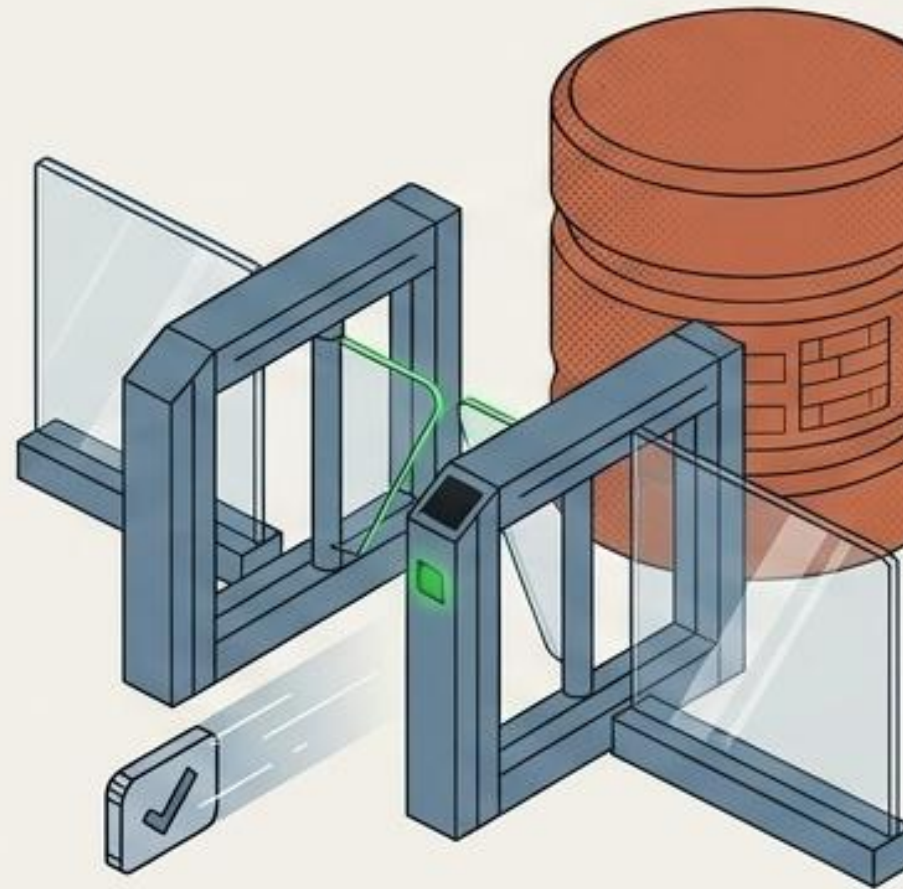
THE FRICTION

Massive, resource-heavy objects initialize immediately upon system boot, bottlenecking performance before they are even needed.

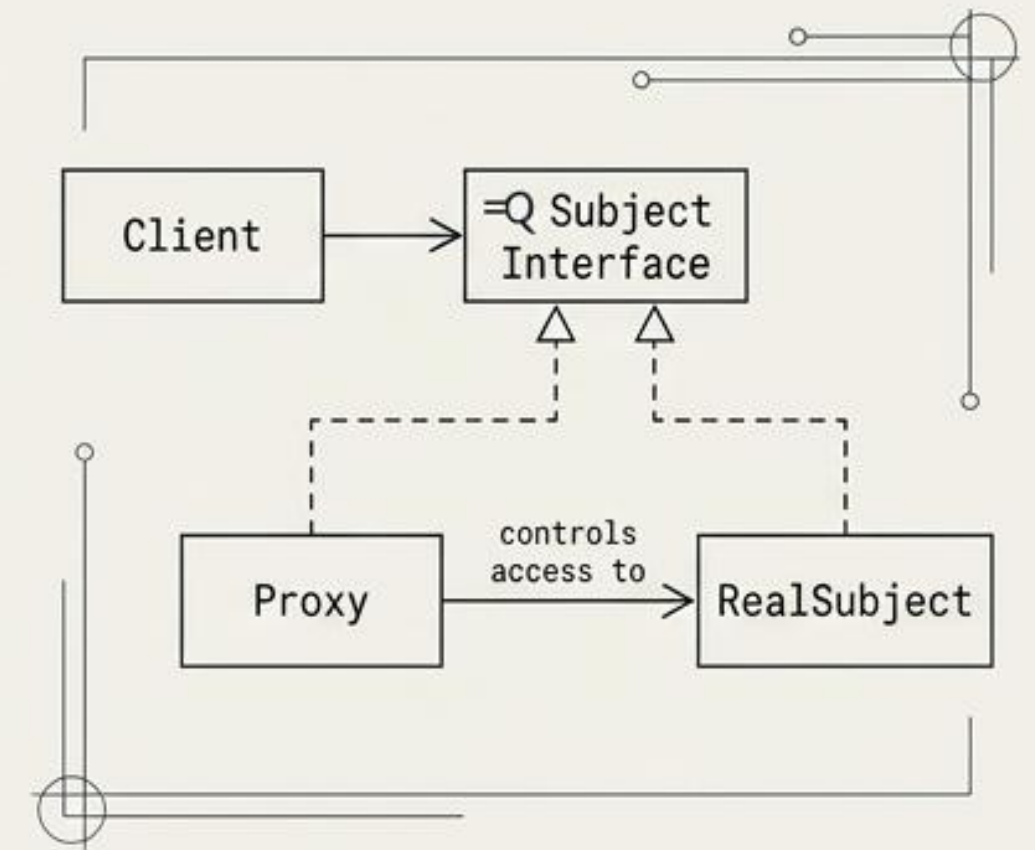


THE MECHANISM

Provide a placeholder or surrogate object to control access, delay initialization, or handle security checks.



THE SCHEMATIC: PROXY

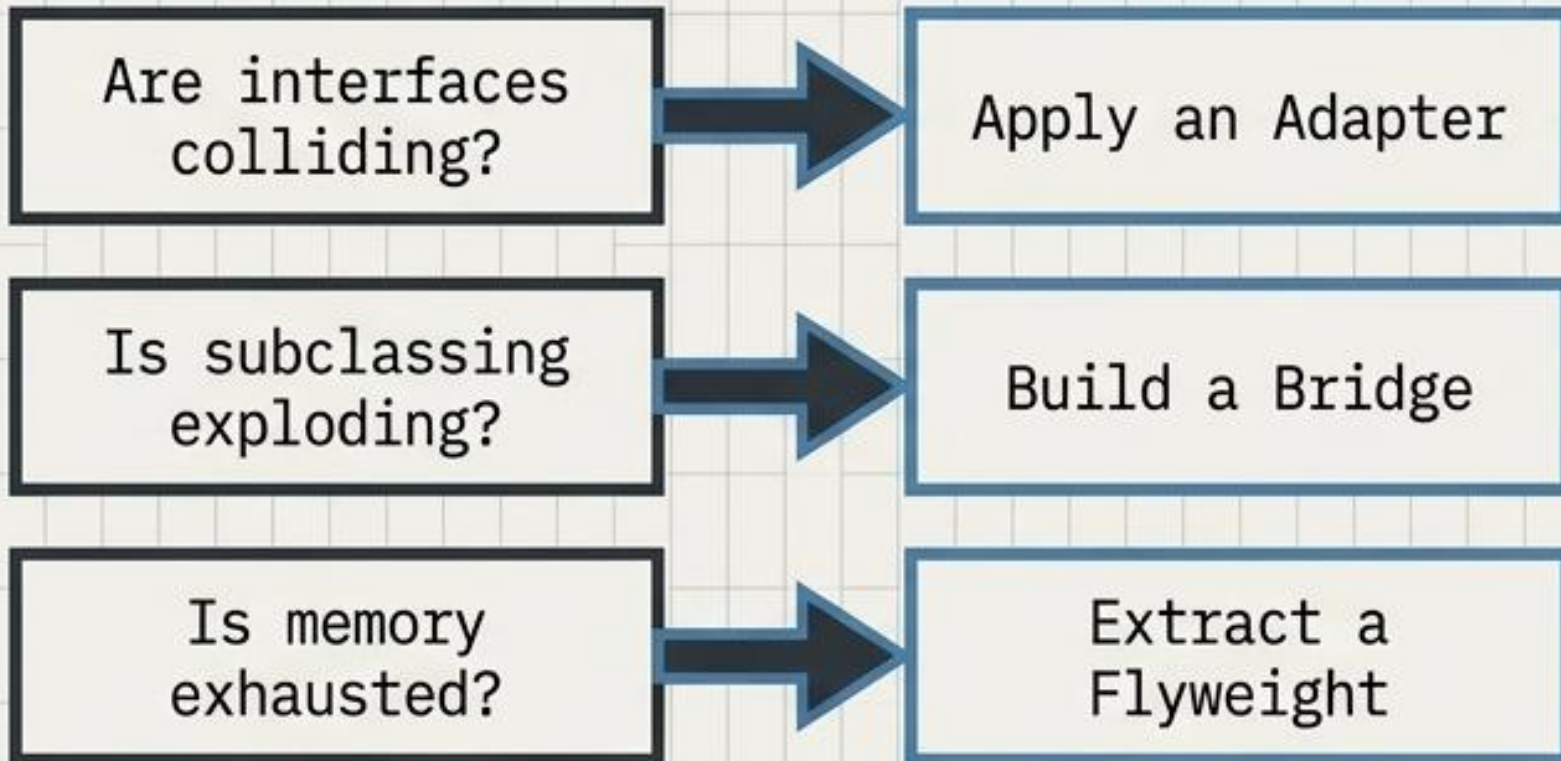


THE ARCHITECT'S DIAGNOSTIC MATRIX

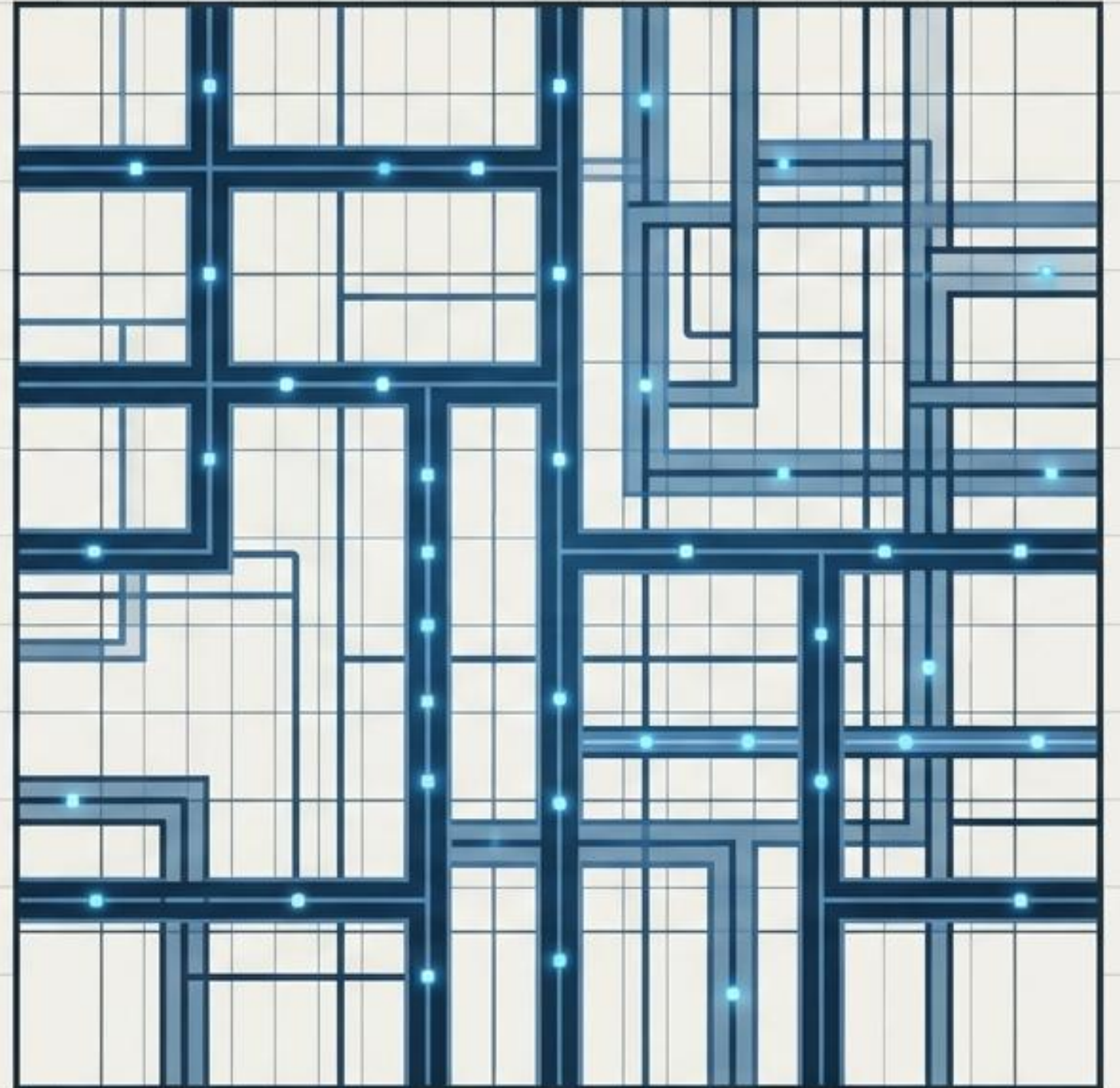
PATTERN	THE CORE INTENT	PRIMARY ANALOGY	STRUCTURAL SIGNATURE
Adapter	Translates incompatible interfaces	Power Plug	Wrapper translating A into B.
Bridge	Decouples orthogonal dimensions	Remote & TV	Two parallel, connected hierarchies.
Composite	Unifies tree structures	Fractal Branches	Recursive interface aggregation.
Decorator	Adds dynamic behavior	Nesting Dolls	Intercepting, stackable wrapper.
Facade	Simplifies subsystem complexity	Control Panel	Single entry point to many.
Flyweight	Conserves memory footprint	Shared Blueprint	Intrinsic vs. Extrinsic state cache.
Proxy	Controls object access	The Gatekeeper	Surrogate intercepting specific requests.

Master the Blueprint, Build the Future

Design patterns are not rigid laws; they are diagnostic tools. Real architectural mastery lies in accurately identifying the friction within your system and applying the precise structural pattern required to resolve it-without introducing unnecessary complexity.

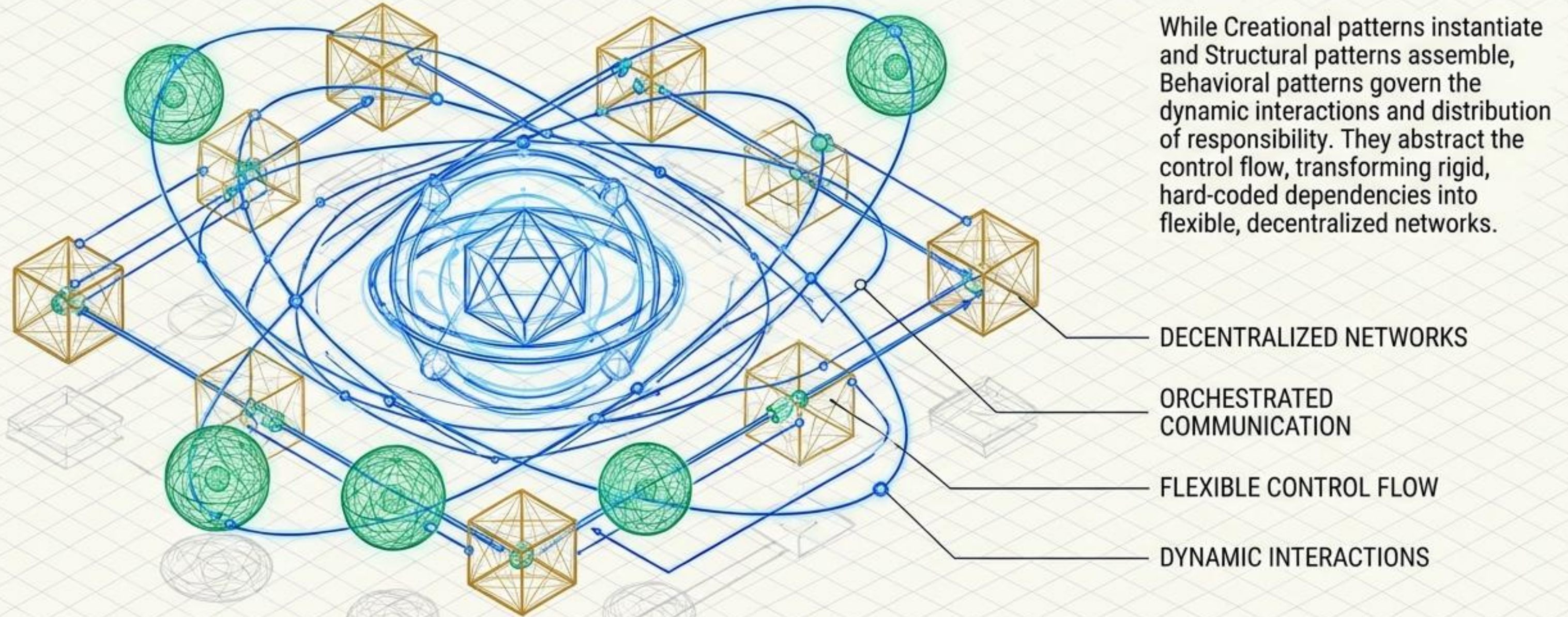


> Apply structure to simplify reality.



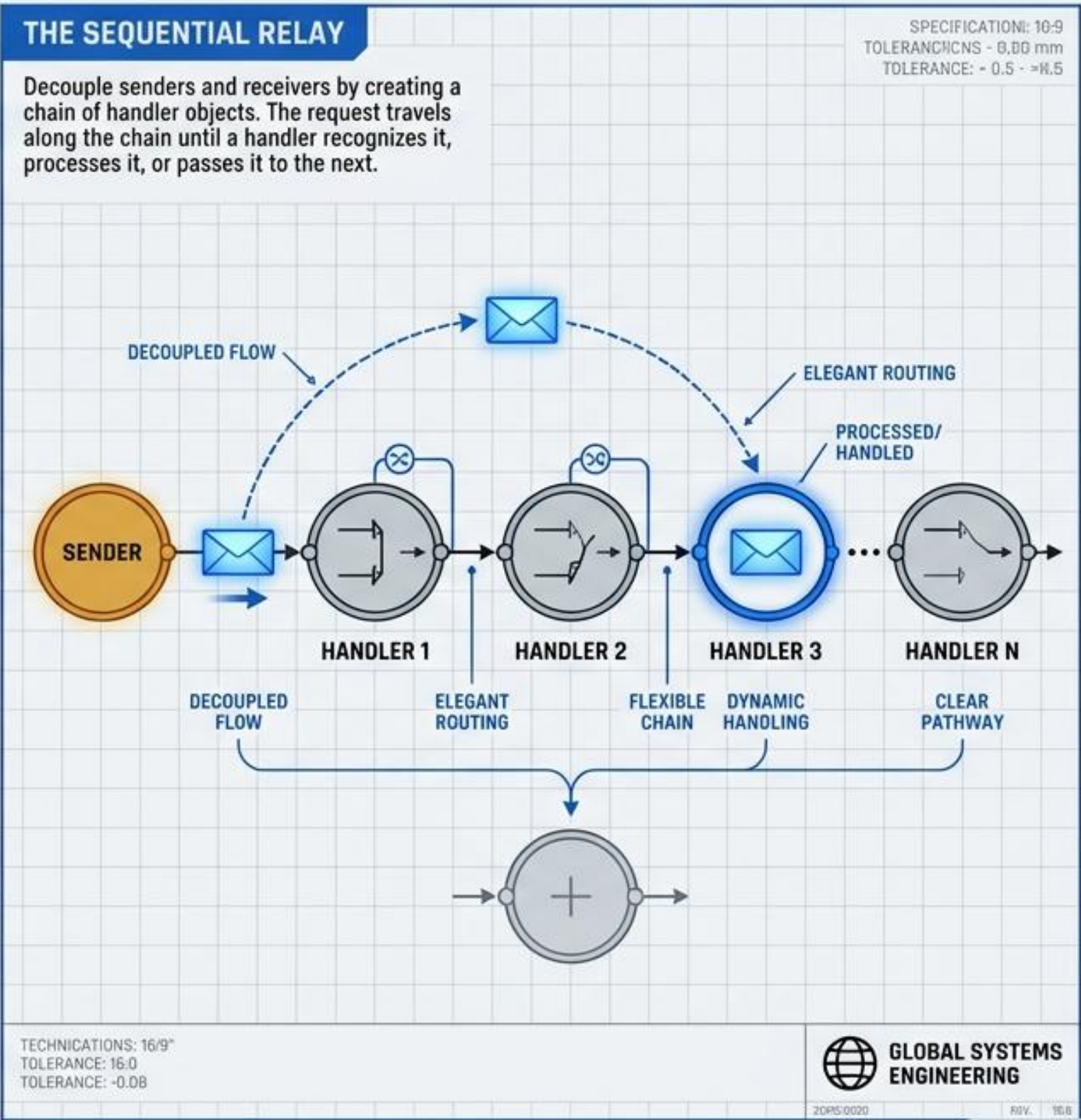
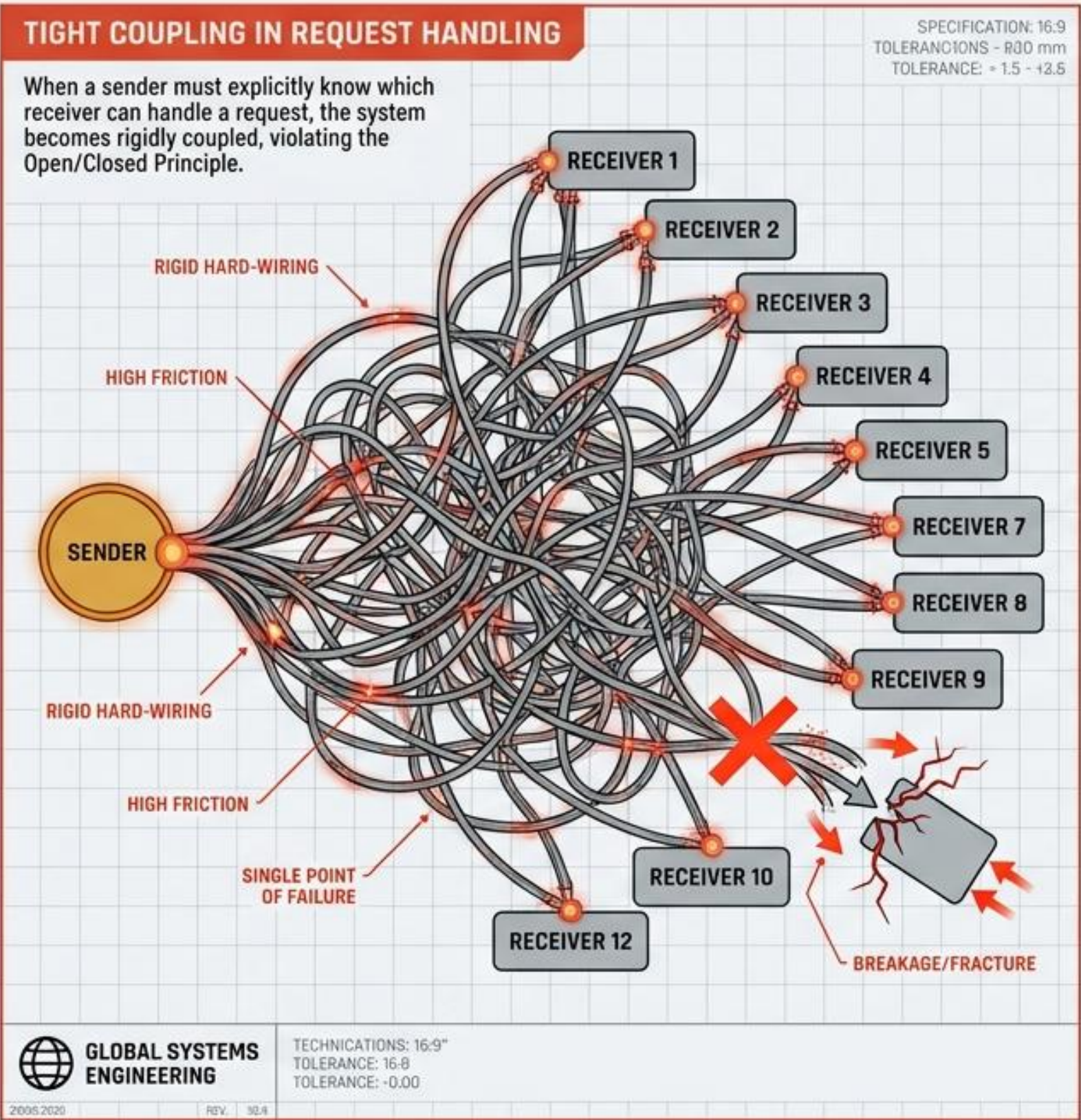
BEHAVIORAL DESIGN PATTERNS

The Blueprint for Object Communication, Choreography, and Control

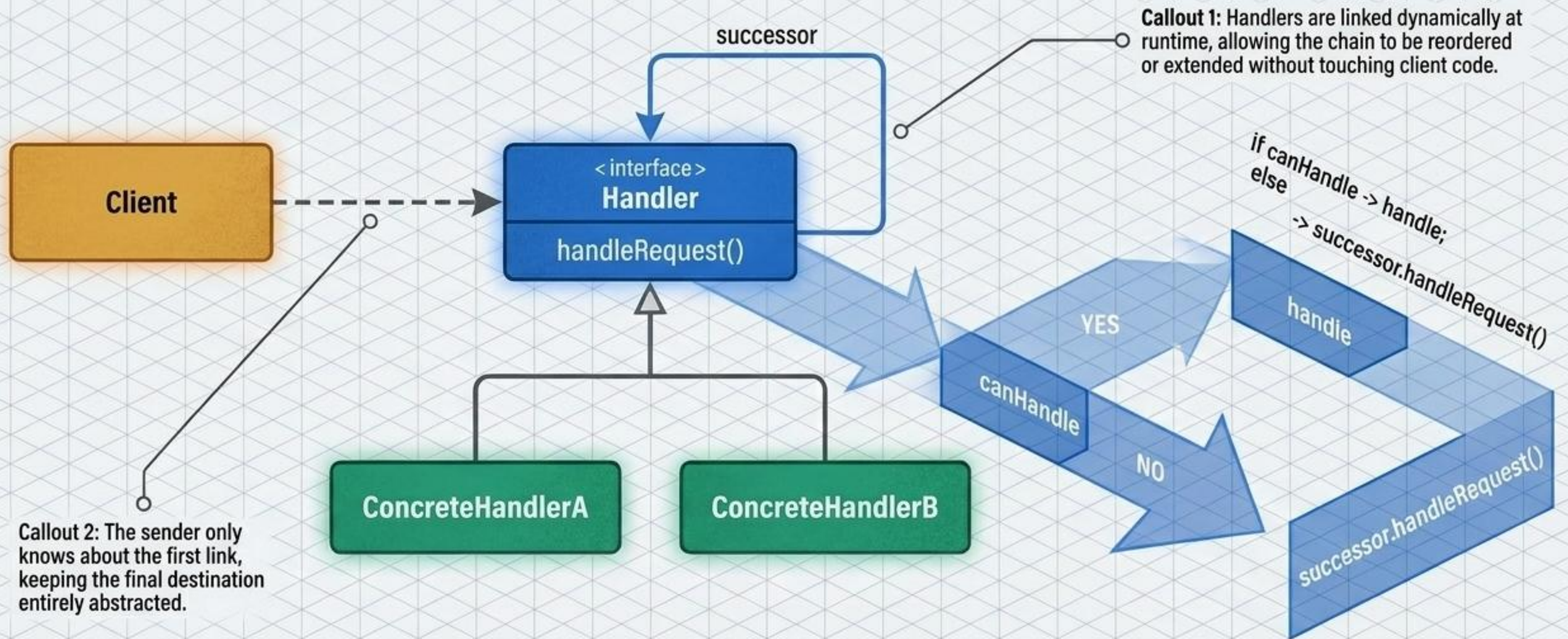


ADVANCED SOFTWARE ARCHITECTURE DEEP-DIVE

CHAIN OF RESPONSIBILITY // TIGHT COUPLING VS. SEQUENTIAL RELAY



CHAIN OF RESPONSIBILITY // INTERNAL STRUCTURE

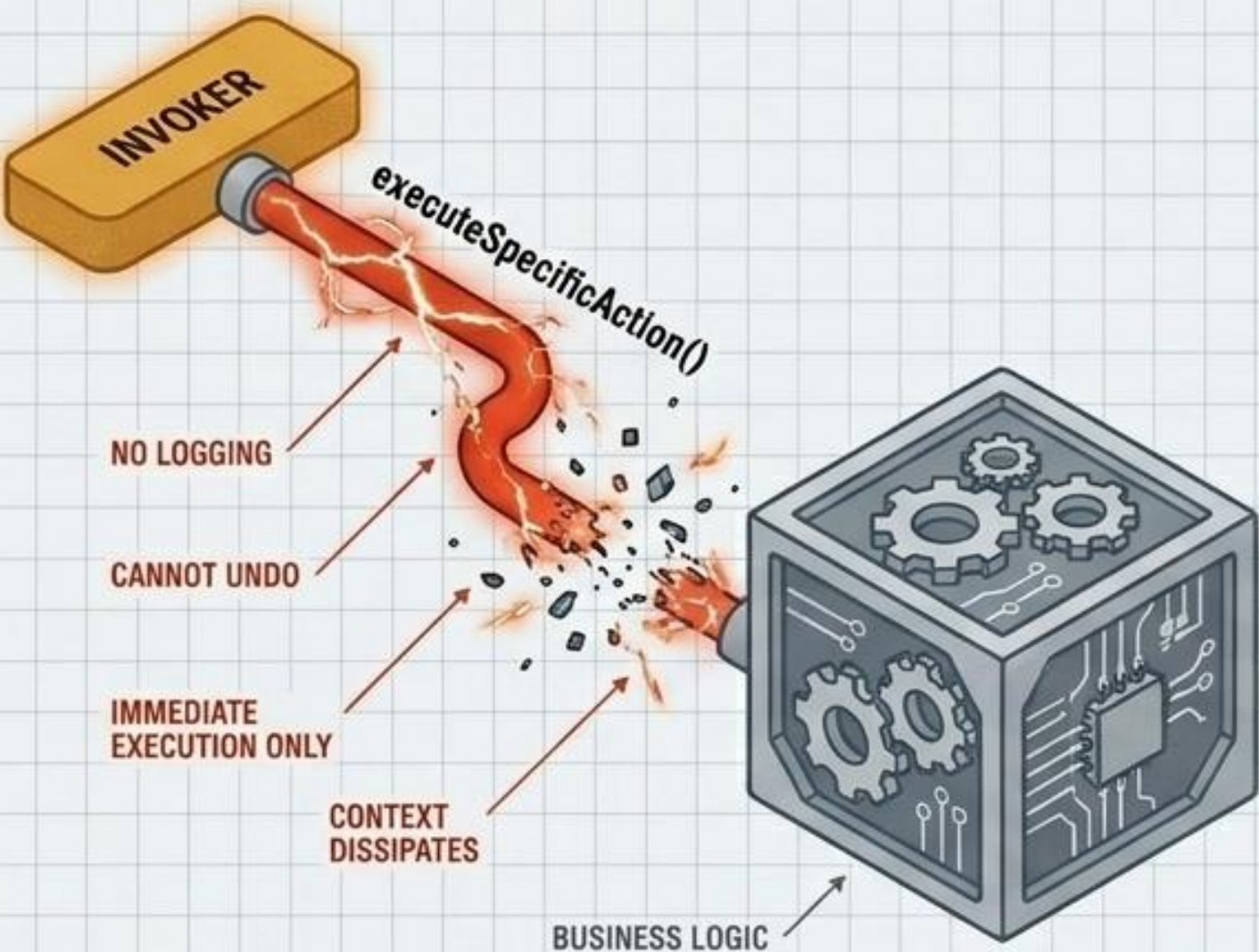


COMMAND // HARDWIRED REQUESTS VS. OBJECTIFIED INTENT

LOST INTENT

Directly coupling an invoker to a receiver makes it impossible to queue, log, undo, or parameterize requests. Once fired, the context vanishes.

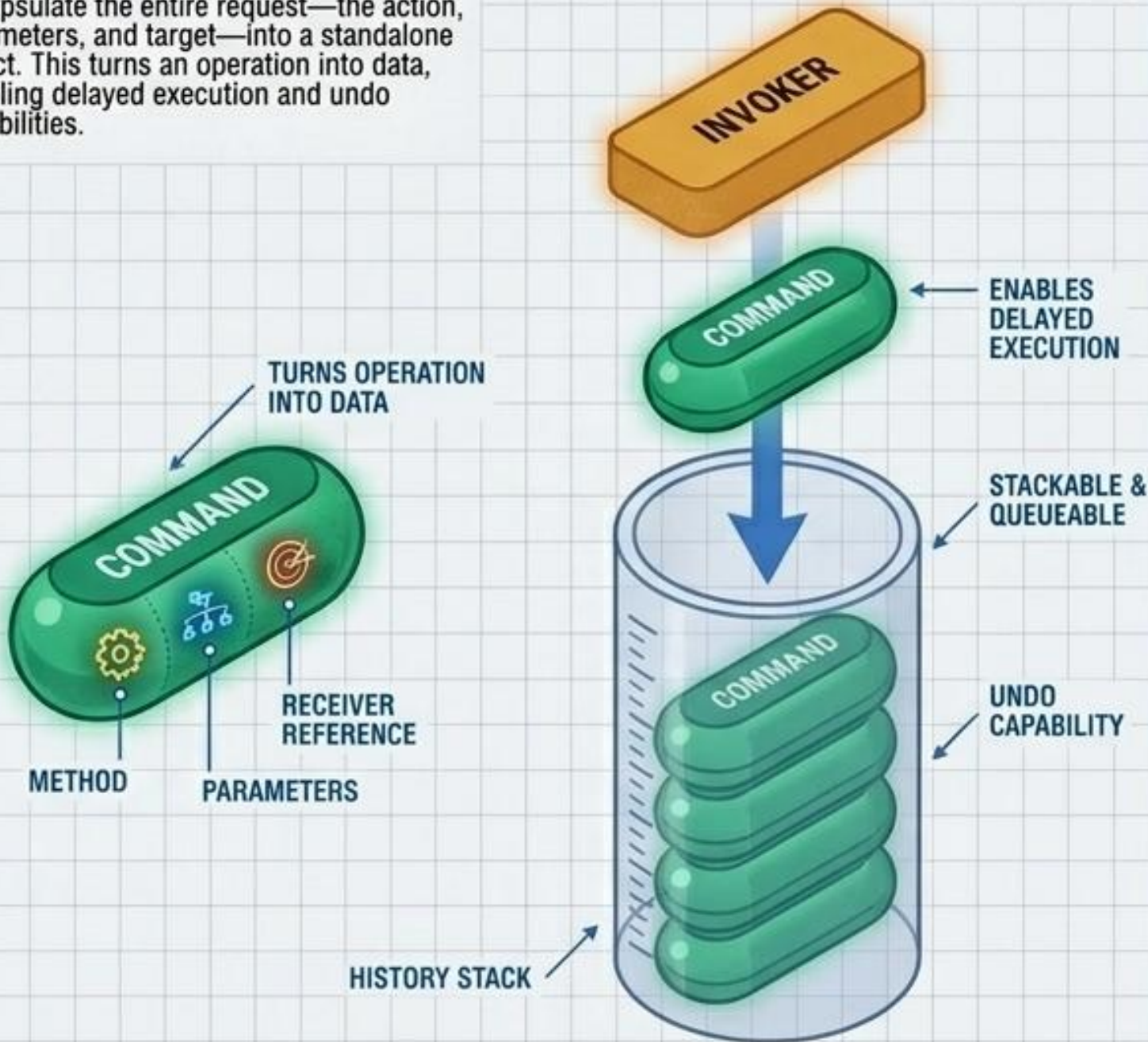
SPECIFICATION: 16:9
TOLERANCES - 0.30 mm
TOLERANCE: - 1.5 - 13.8



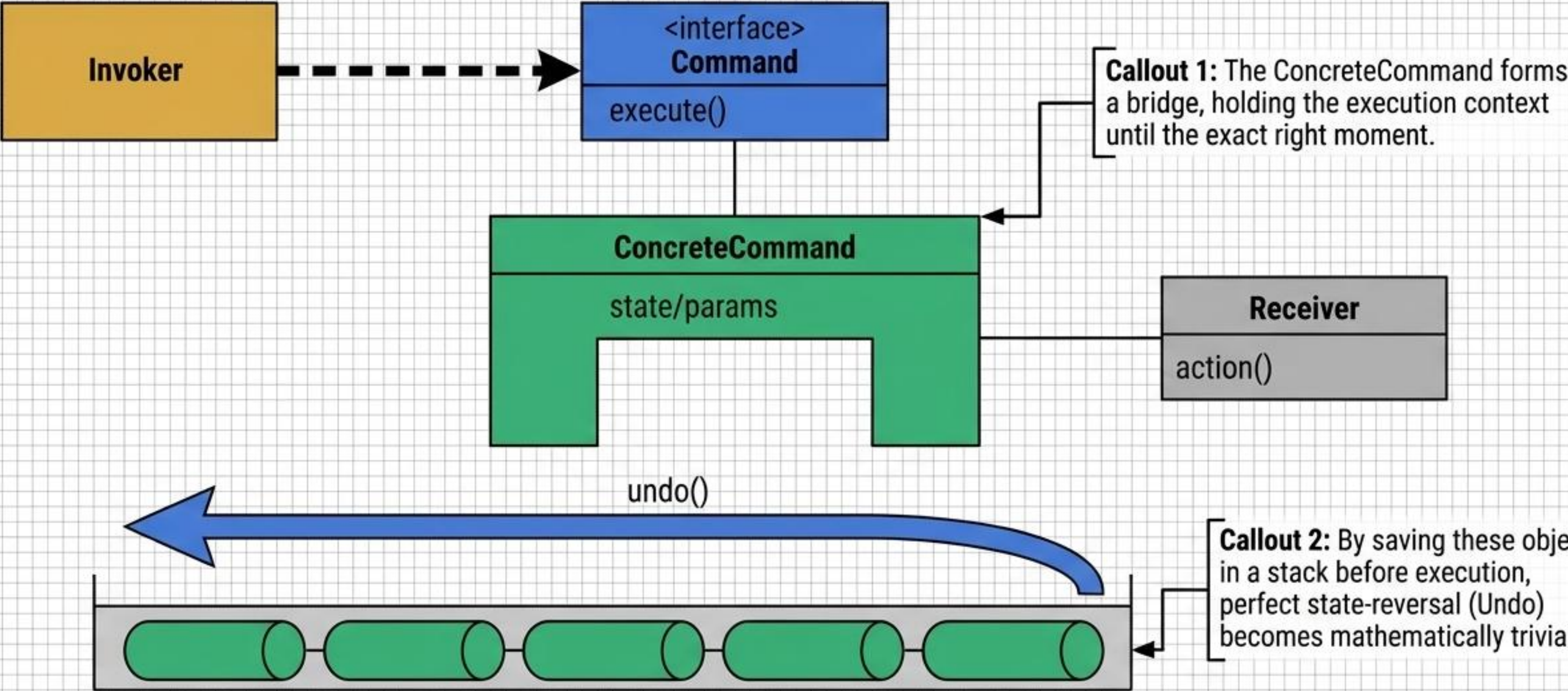
OBJECTIFYING THE REQUEST

Encapsulate the entire request—the action, parameters, and target—into a standalone object. This turns an operation into data, enabling delayed execution and undo capabilities.

SPECIFICATIONS: 10:9
TOLERANCES - 0.30 mm
TOLERANCE: - 0.5 - 14.5



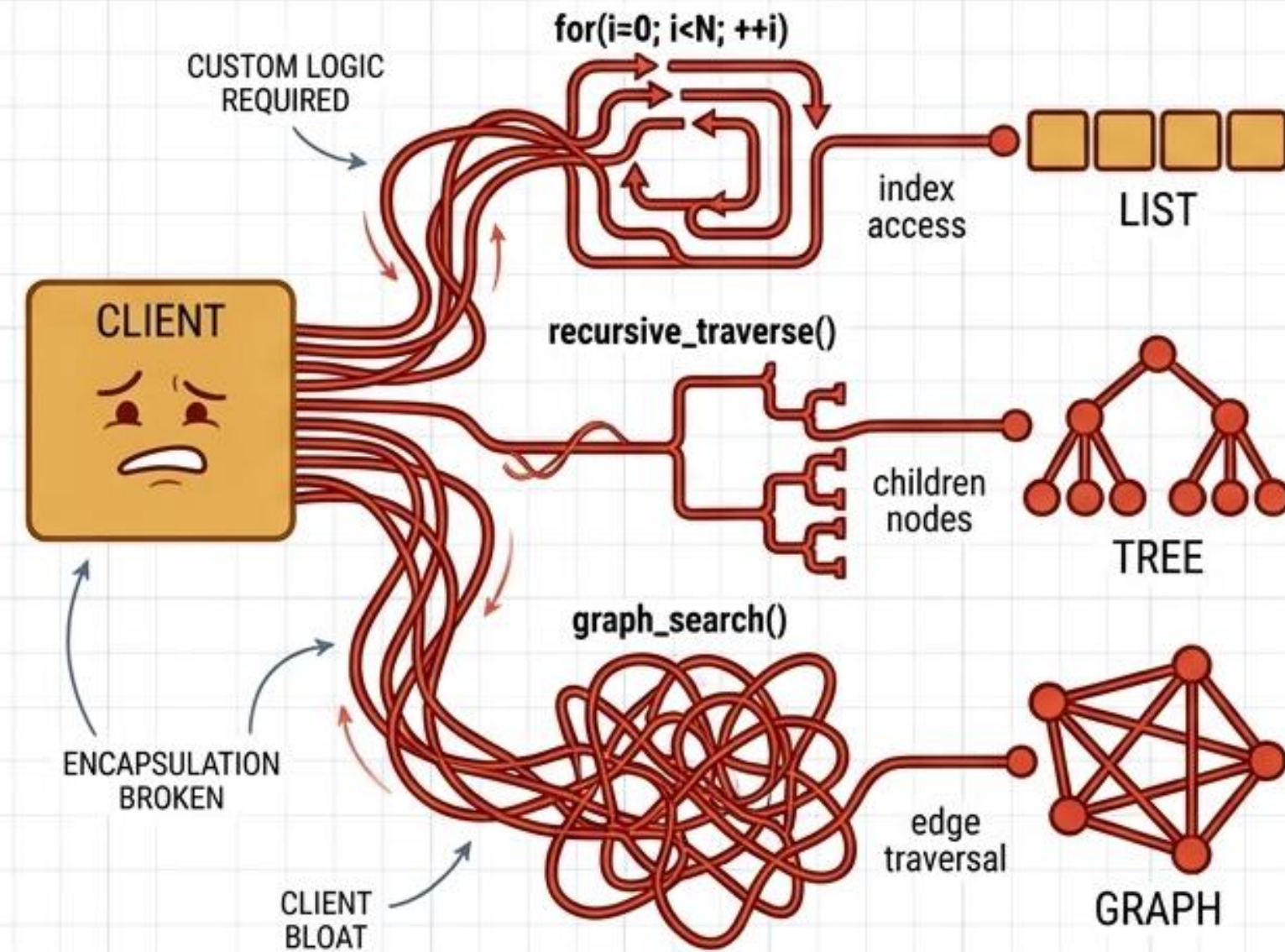
COMMAND // INTERNAL STRUCTURE



ITERATOR // EXPOSED STRUCTURES VS. ABSTRACTED TRAVERSAL

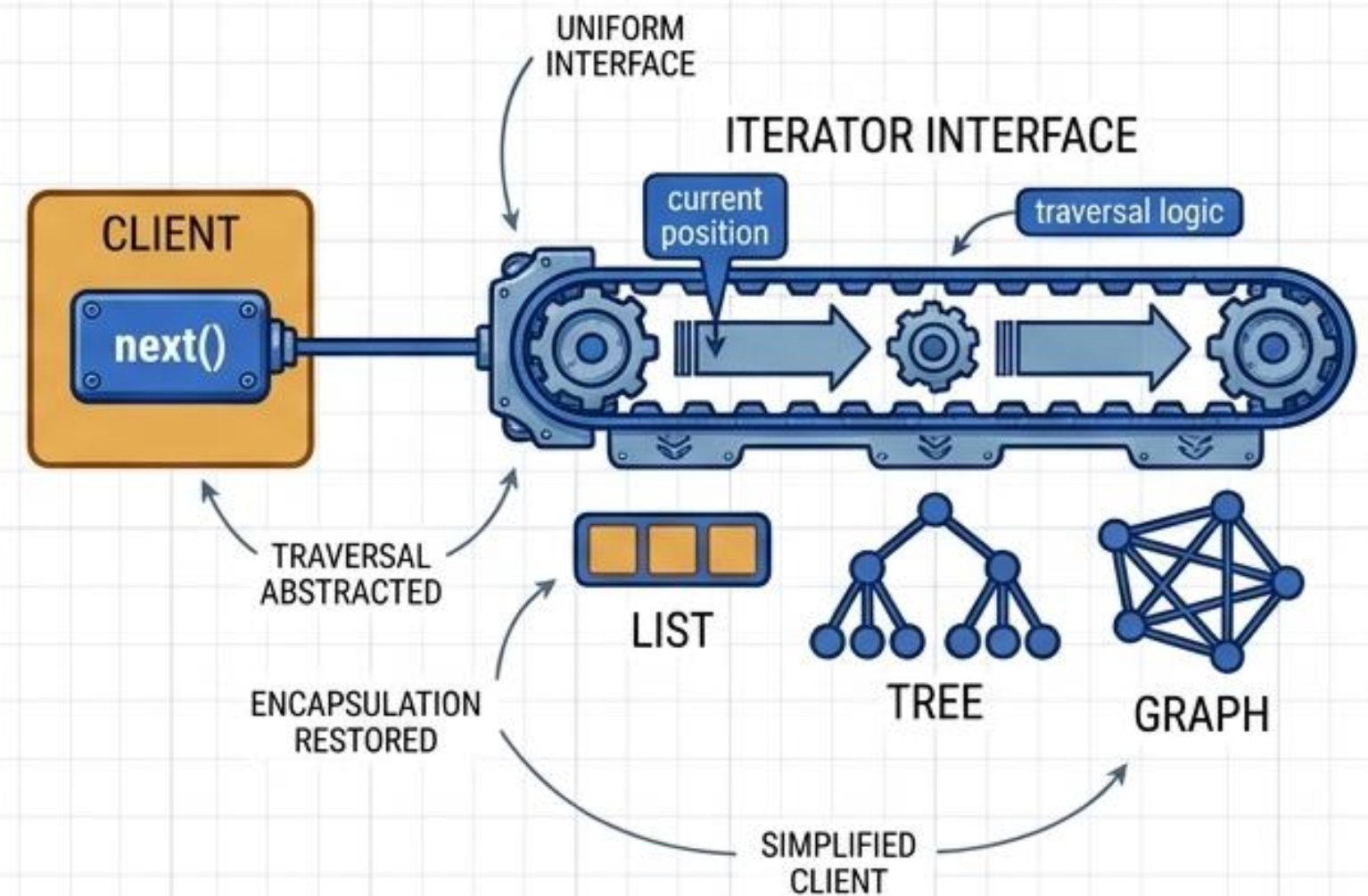
EXPOSED INTERNAL STRUCTURES

Forcing clients to implement distinct traversal logic for lists, trees, and graphs breaks encapsulation and bloats client code.



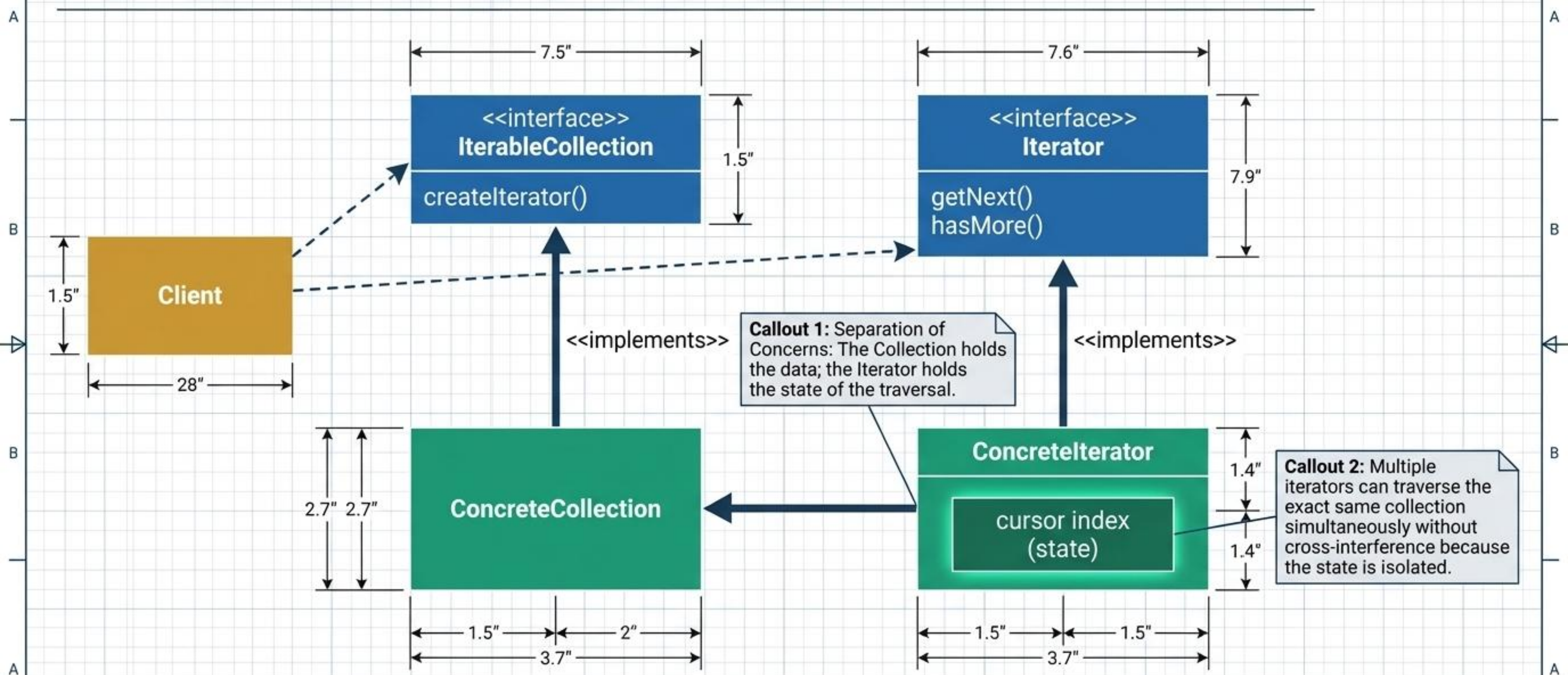
ABSTRACTING THE TRAVERSAL

Extract traversal behavior into a separate iterator object. It tracks position and handles mechanics, providing a uniform interface while hiding the collection's inner workings.



ITERATOR // INTERNAL STRUCTURE

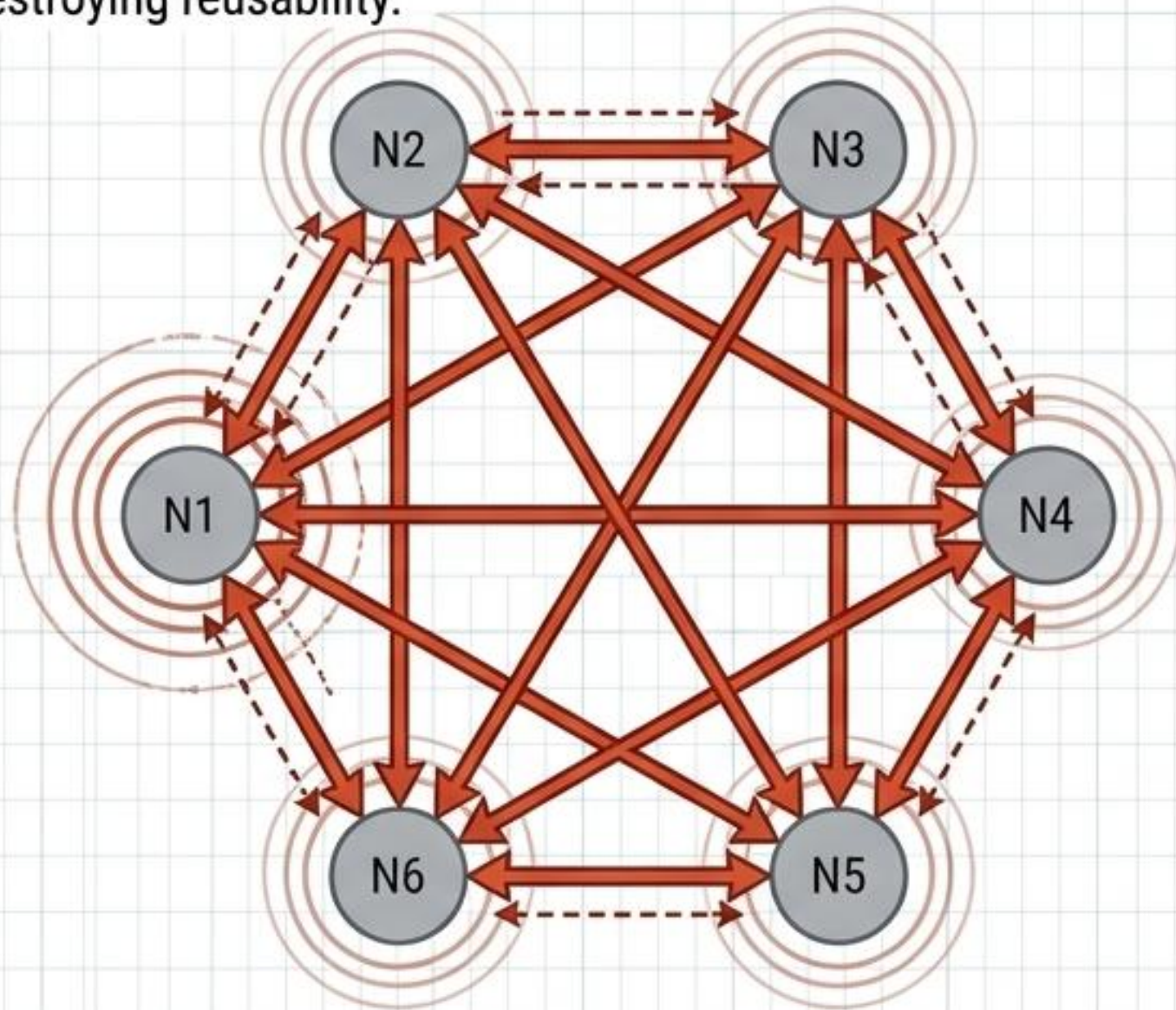
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1	1/13	Stard cont	1



MEDIATOR // THE SPAGHETTI WEB VS. THE CENTRALIZED HUB

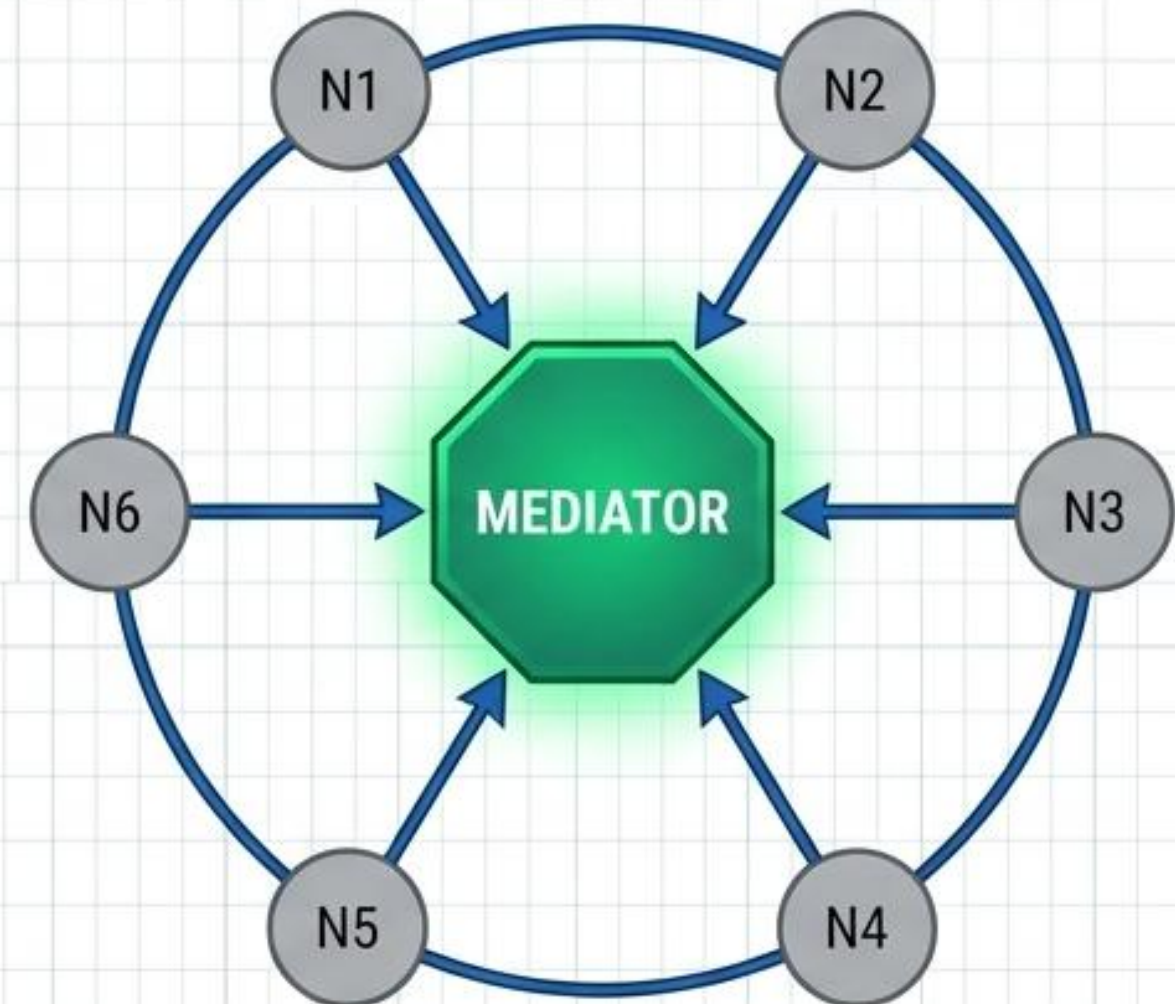
UNMAINTAINABLE INTERDEPENDENCIES

When objects communicate directly, the system devolves into a dense web. Changing one component risks breaking them all, destroying reusability.



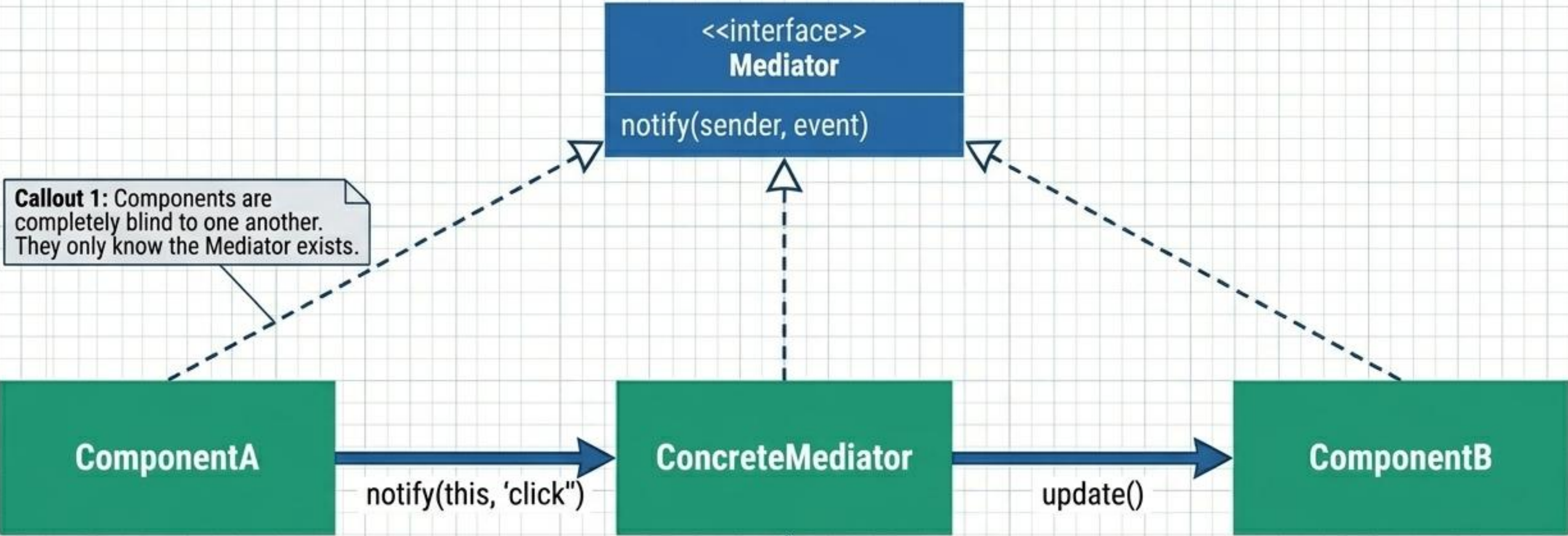
THE CENTRAL HUB

Restrict direct communications. Force components to collaborate only via a central mediator, turning chaotic many-to-many relationships into manageable one-to-many logic.



MEDIATOR // INTERNAL STRUCTURE

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Callout 1: Components are completely blind to one another. They only know the Mediator exists.

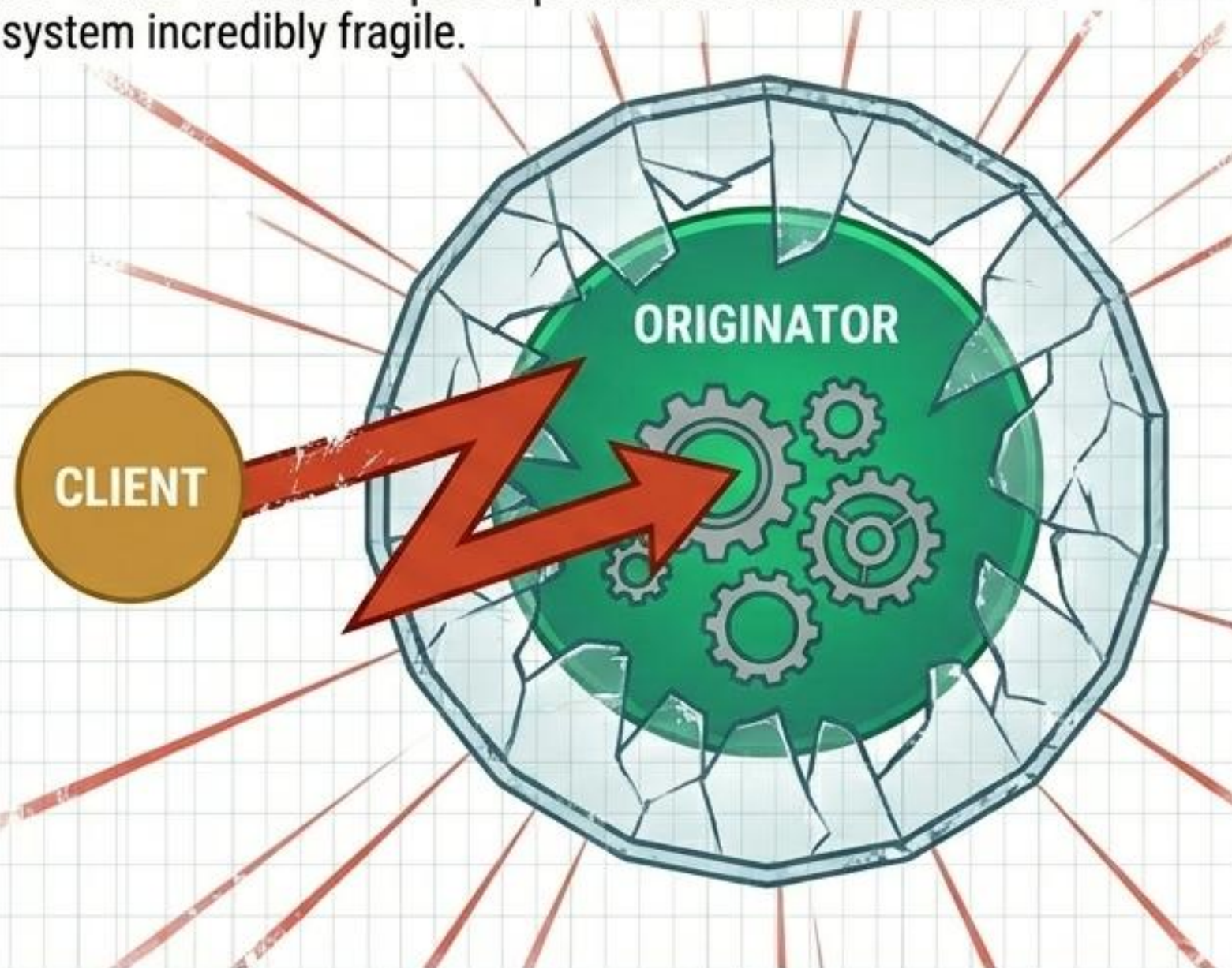
Callout 2: The notify method centralizes complex conditional logic, defining exactly how the system reacts when a specific component changes state.



MEMENTO // THE ENCAPSULATION BREACH VS. THE STATE SNAPSHOT

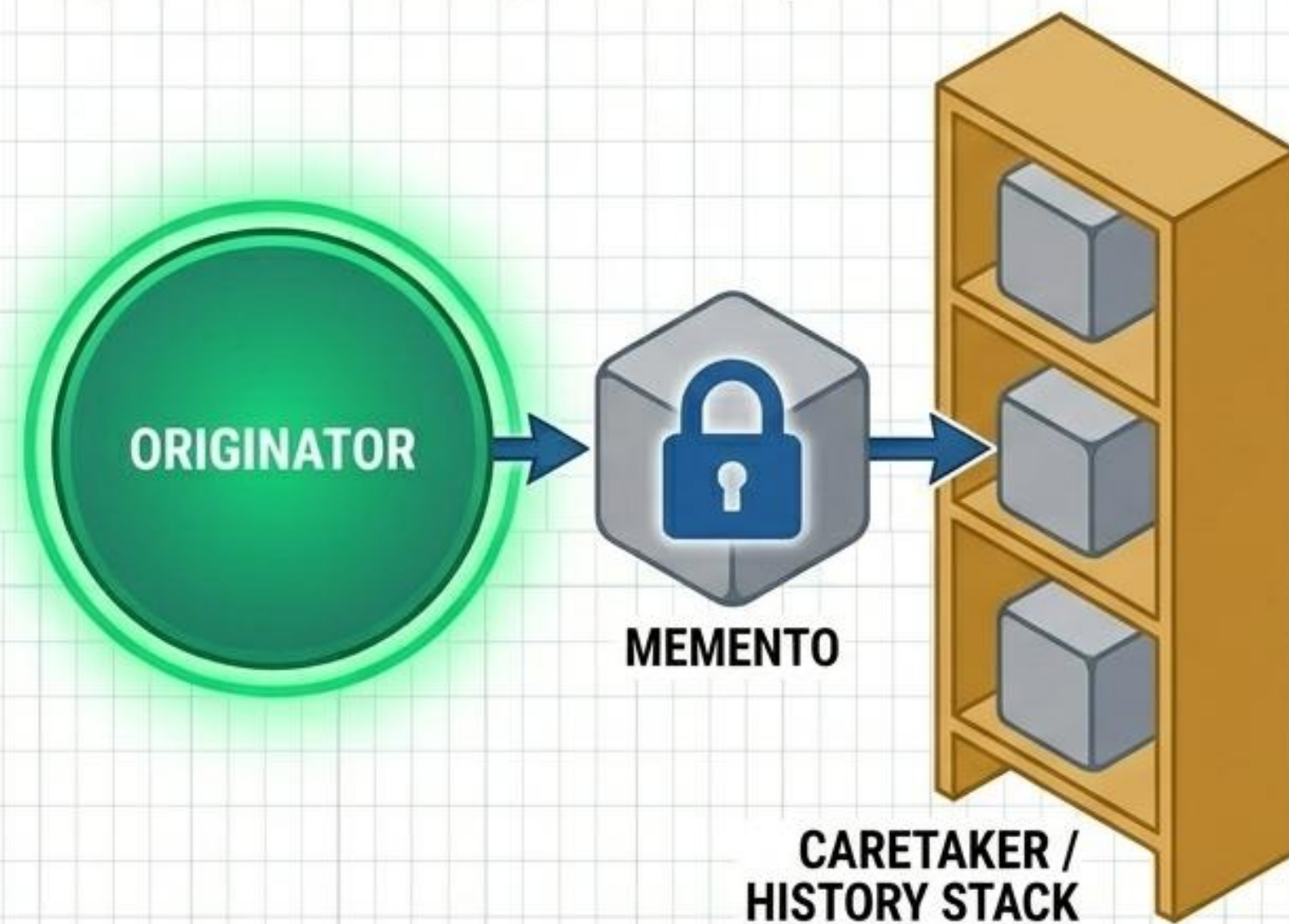
BREAKING THE SHIELD

Allowing external objects to access and record internal state for "Undo" features exposes private fields and makes the system incredibly fragile.



THE OPAQUE SNAPSHOT

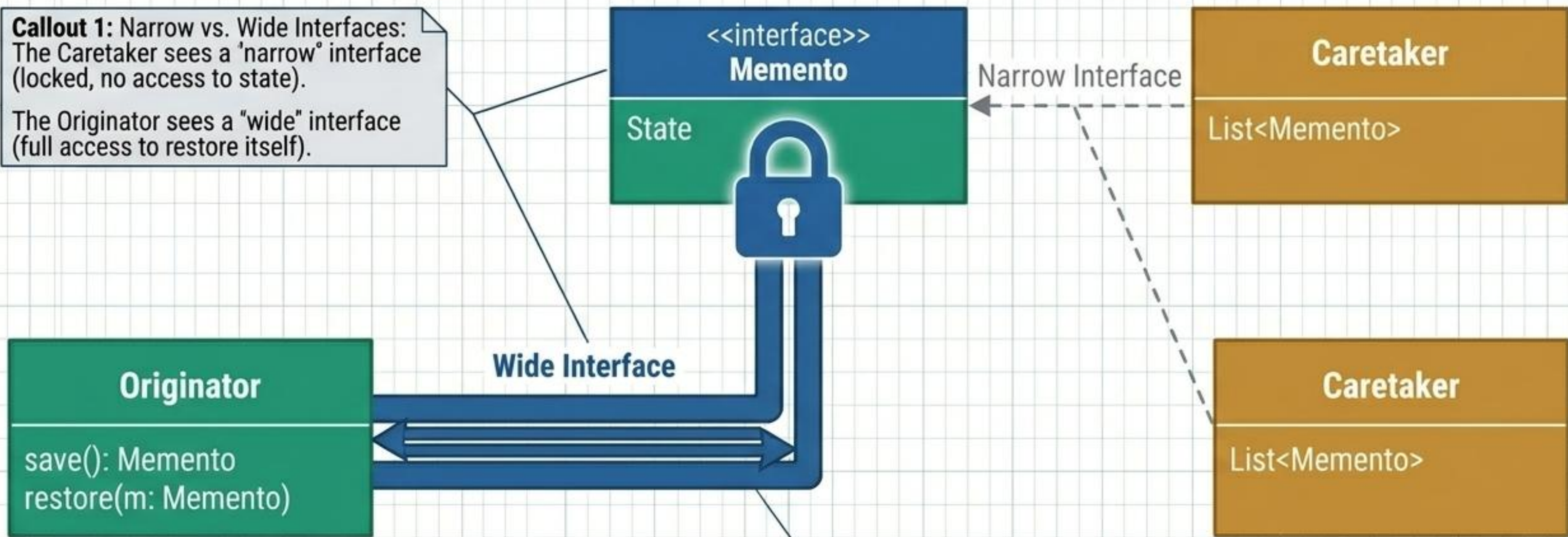
The object creates a snapshot of its state (Memento) and hands it to a caretaker. The caretaker can store and return the snapshot, but cannot inspect or modify its contents.



MEMENTO // INTERNAL STRUCTURE

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Callout 1: Narrow vs. Wide Interfaces:
The Caretaker sees a 'narrow' interface (locked, no access to state).
The Originator sees a "wide" interface (full access to restore itself).



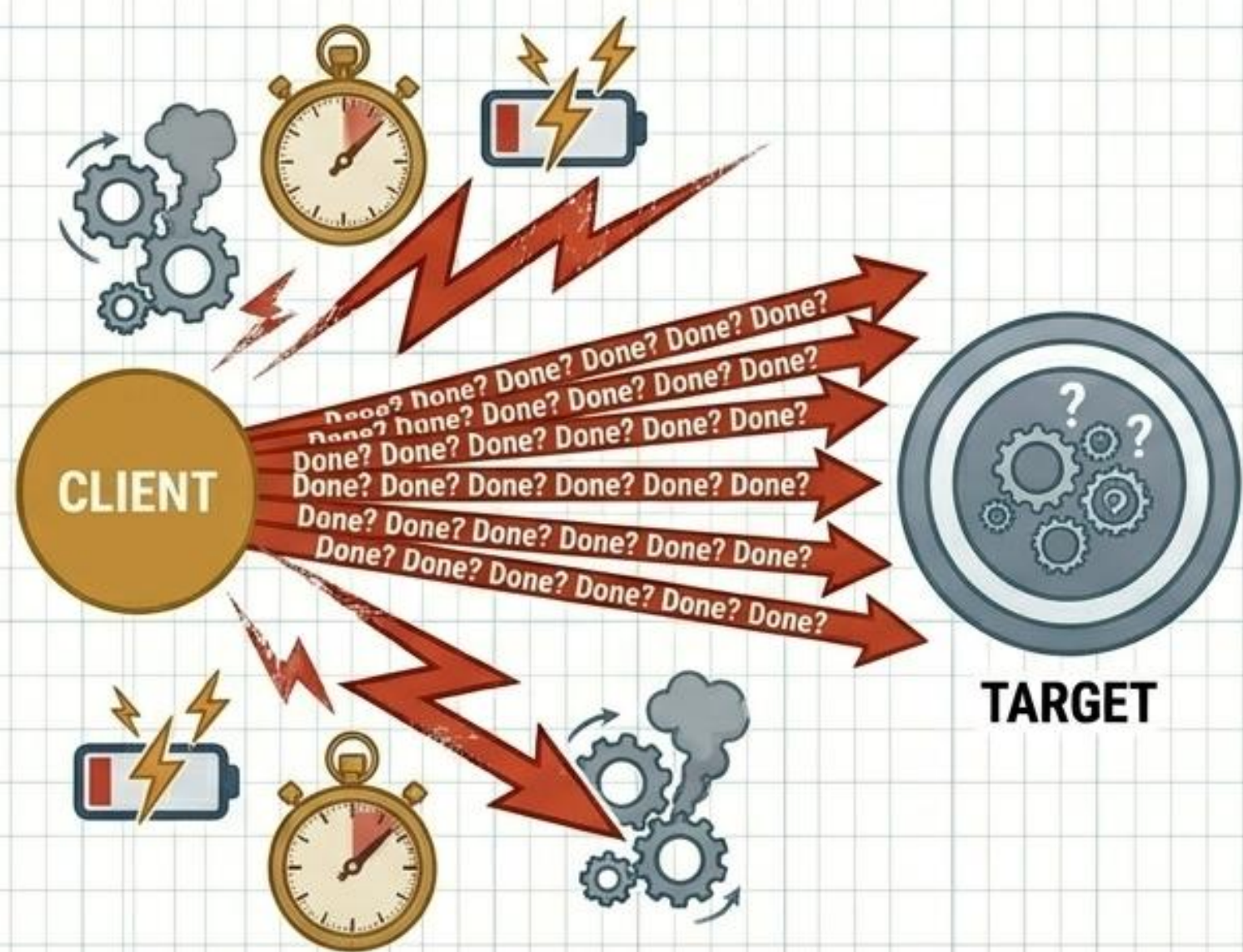
Callout 2: Perfect encapsulation is preserved while still granting the system infinite rollback capabilities.



OBSERVER // INEFFICIENT POLLING VS. PUBLISH-SUBSCRIBE MECHANICS

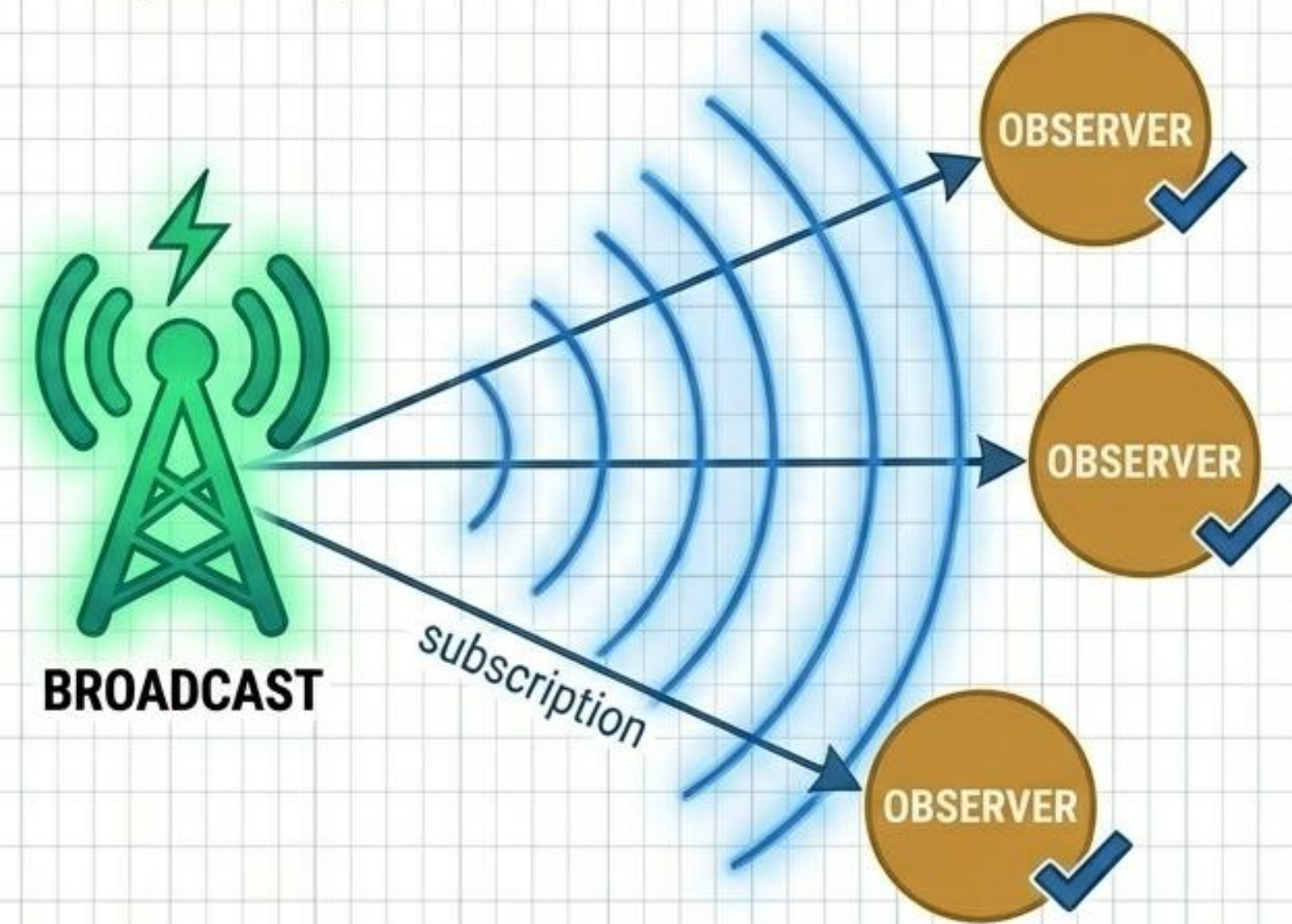
THE POLLING TAX

Constantly checking another object for state changes burns resources, creates tight coupling, and scales terribly.



BROADCAST SUBSCRIPTIONS

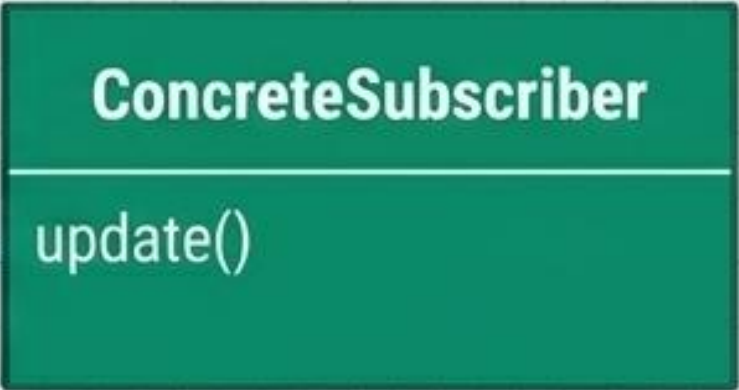
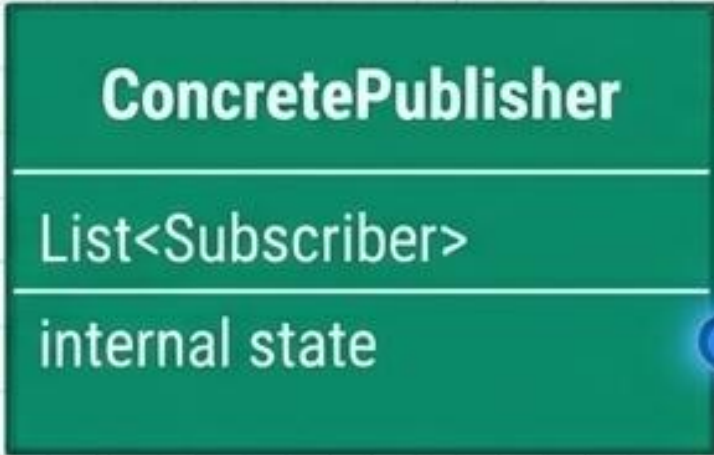
The publisher maintains a list of subscribers. When its state changes, it instantly pushes a broadcast outward. Objects react dynamically with zero wasted resources.



OBSERVER // INTERNAL STRUCTURE

Ut no.	Specifications	Revision	Rev.
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Callout 1: The Publisher does not need to know the concrete classes of its subscribers, only that they implement the generic update() interface.



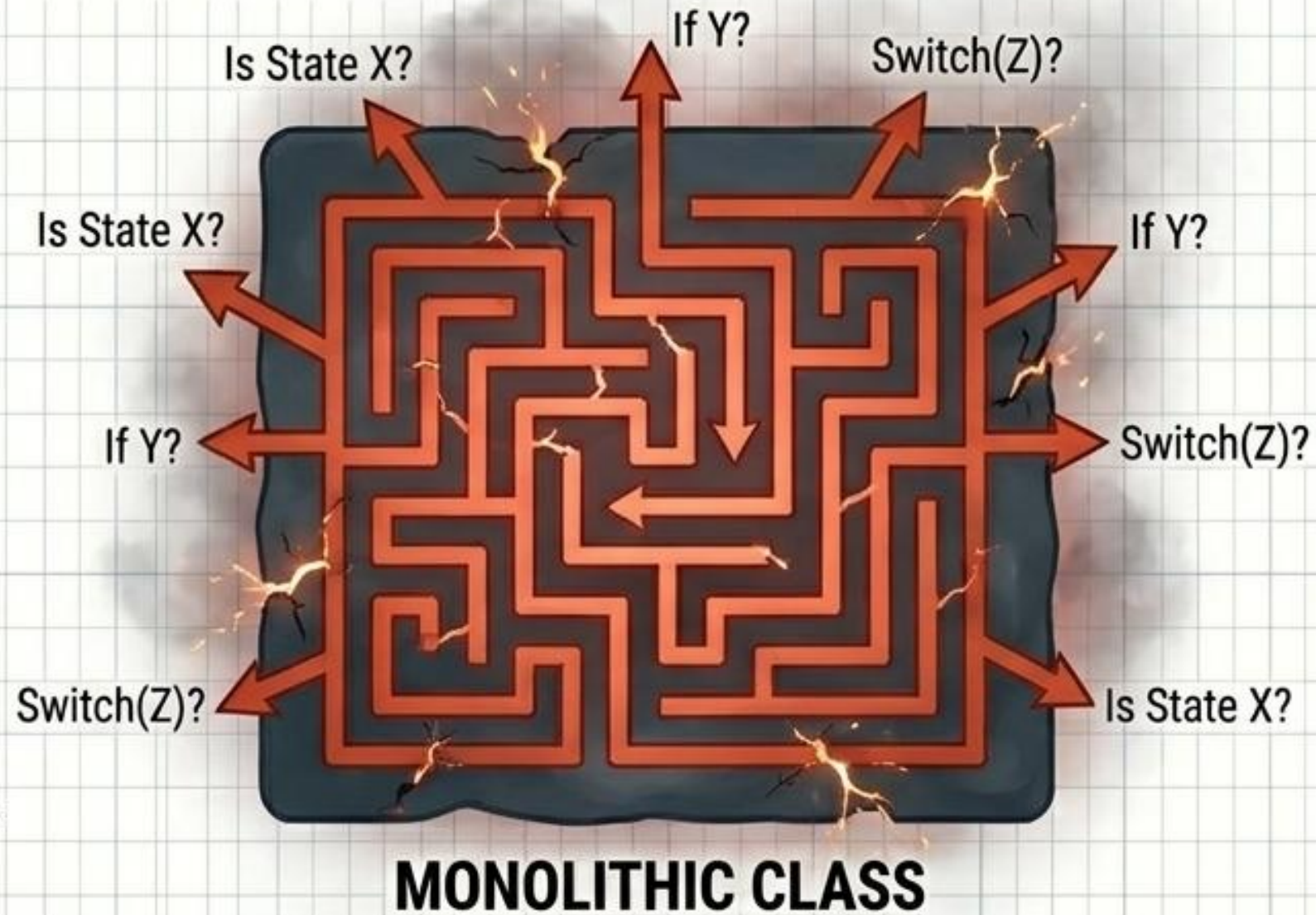
Callout 2: Enables massive, decoupled event-driven architectures where new subscribers can be added at runtime.



STATE // MONSTROUS CONDITIONALS VS. OBJECTIFYING CONTEXT

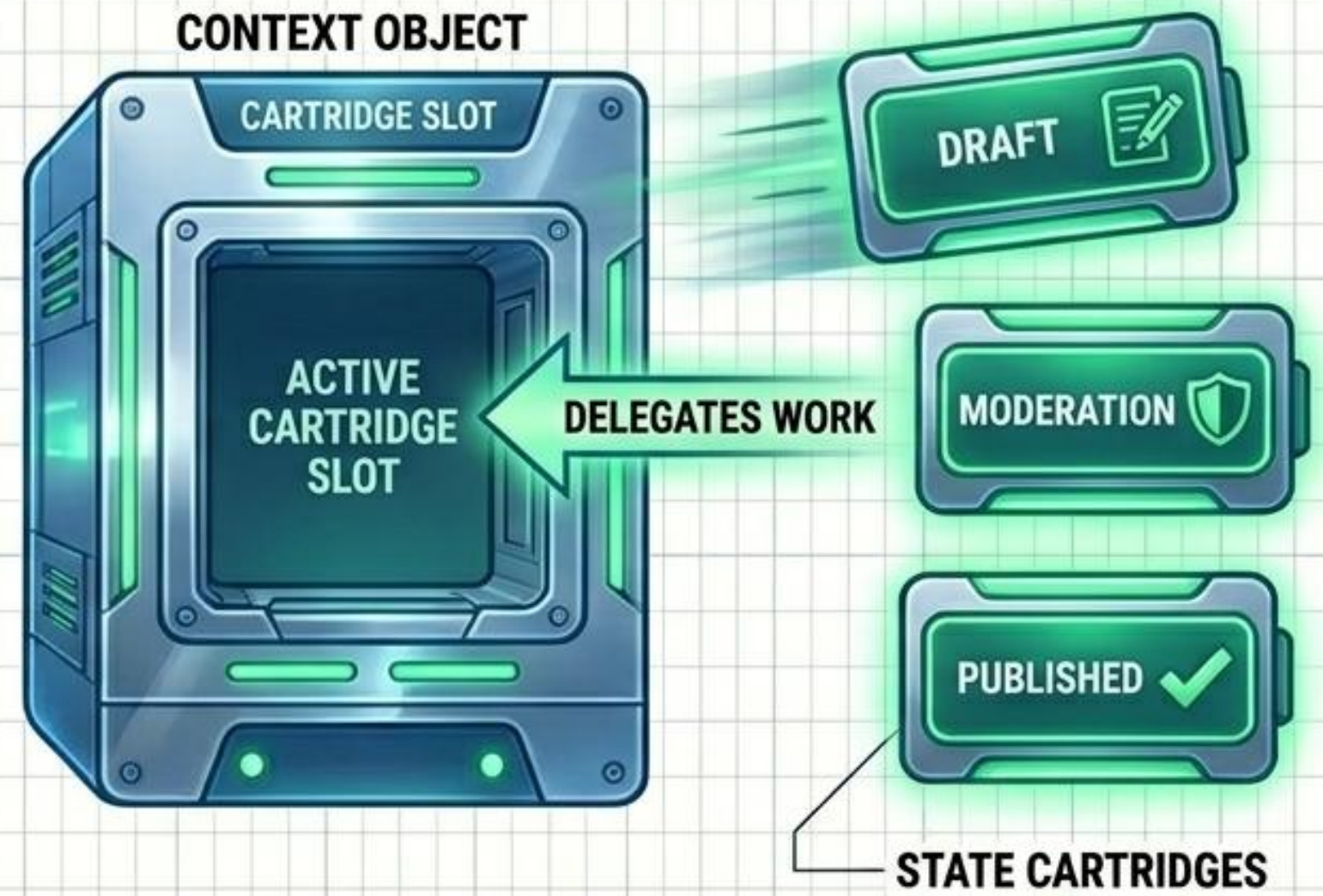
CONDITIONAL LABYRINTHS

When behavior heavily depends on internal state, relying on massive conditional statements creates fragile, unmaintainable monoliths.



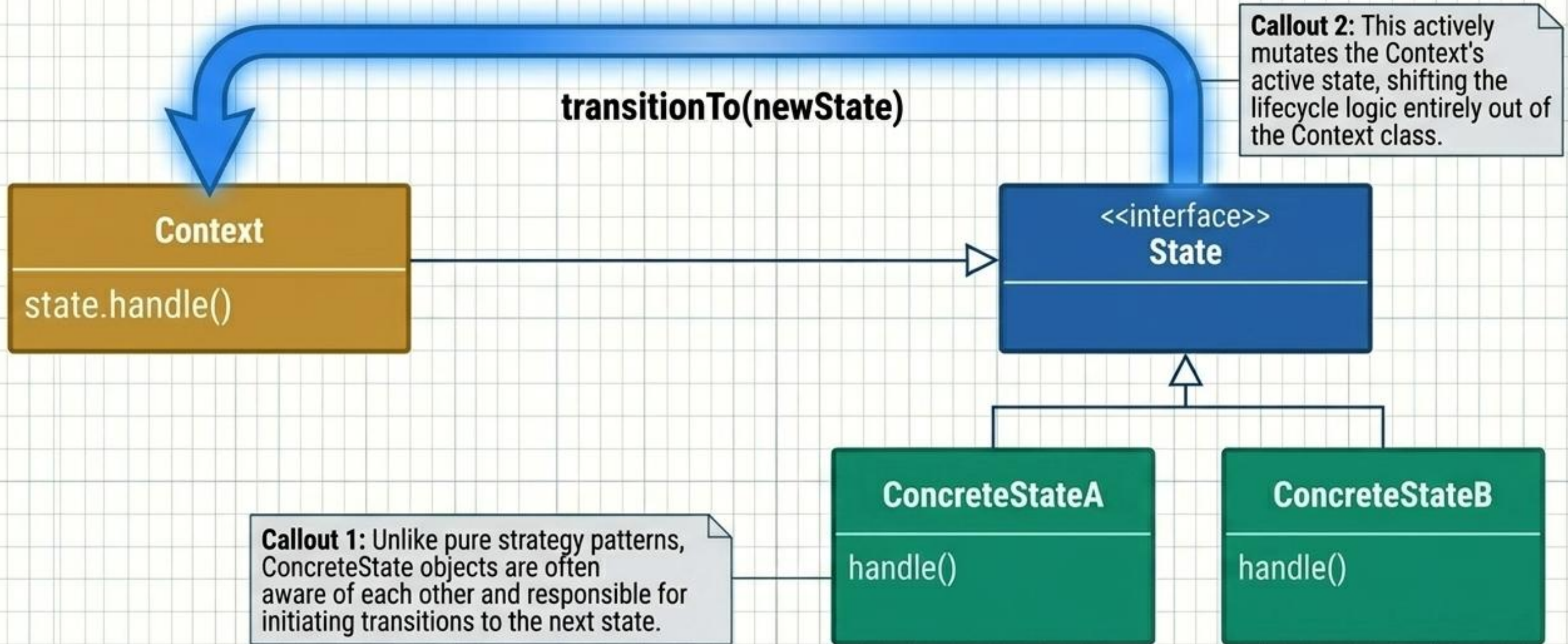
SWAPPABLE STATE CARTRIDGES

Extract state-specific behaviors into distinct classes. The Context object delegates work to the active state cartridge. Changing state simply means swapping the cartridge.



STATE // INTERNAL STRUCTURE

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STRATEGY // ALGORITHMIC CLUTTER VS. SWAPPABLE ALGORITHMS

ALGORITHMIC CLUTTER

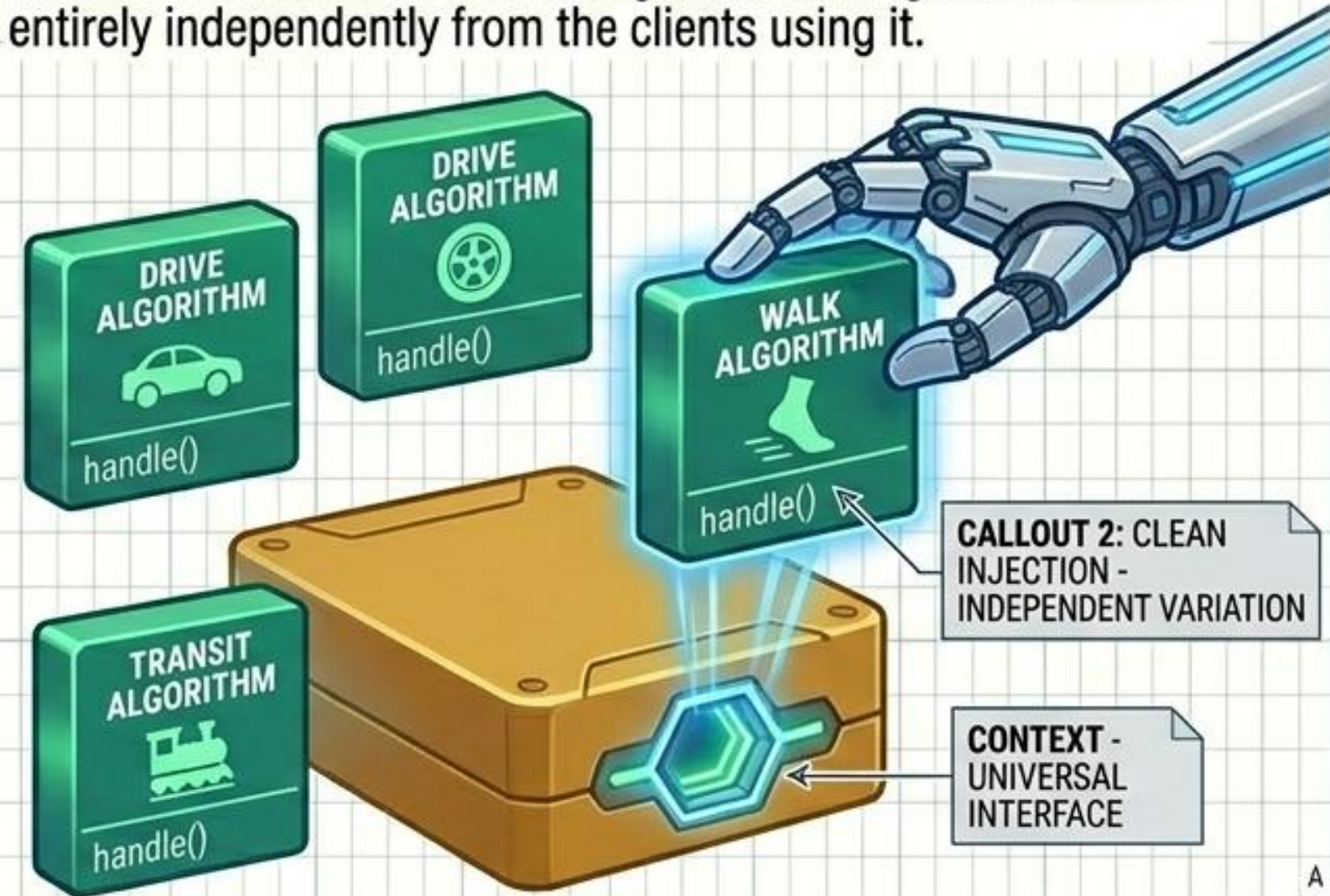
Housing multiple variations of an algorithm inside a single class leads to bloated codebases and severe merge conflicts.



MONOLITHIC CLASS

MODULAR INJECTION

Define a family of algorithms, encapsulate each in a separate class, and make them interchangeable. The algorithm varies entirely independently from the clients using it.



NAVIGATION CONTEXT



GLOBAL SYSTEMS
ENGINEERING

Specification
GEHHA SYSTEMS

Specification

Specification: DCPI/BISTAE, MISN and BRVIMC007..43'2021

Description

Rev.

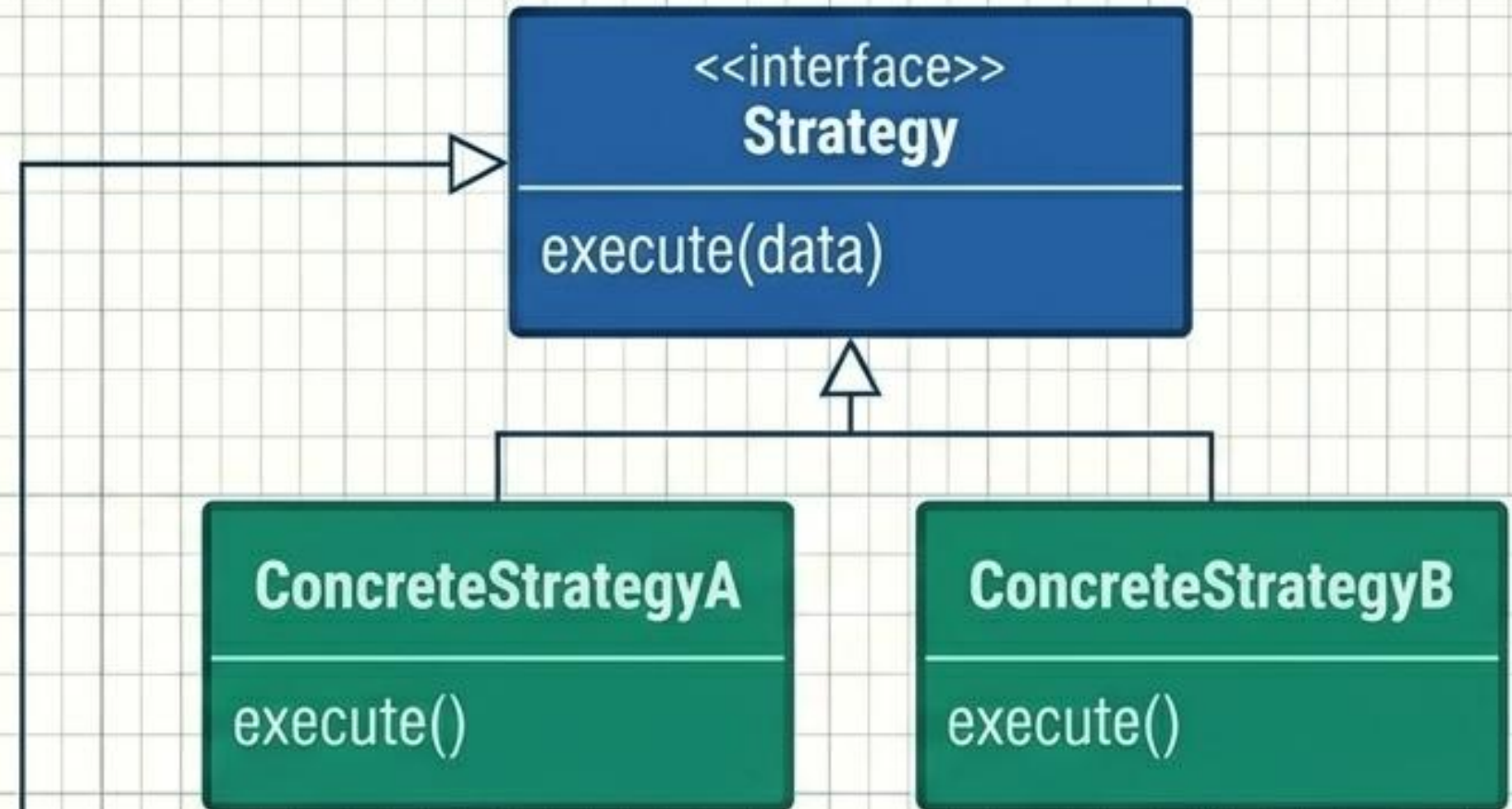
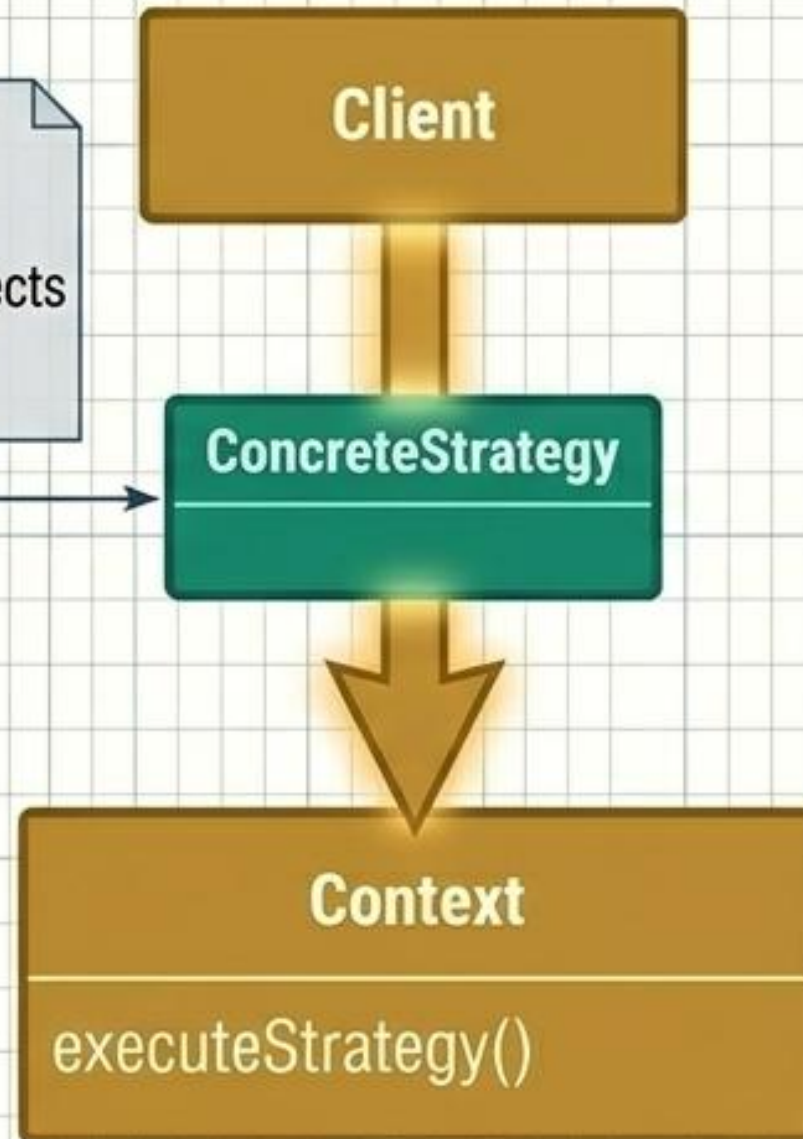
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STRATEGY // INTERNAL STRUCTURE

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Callout 1: The Client is in total control. It assesses the environment and injects the specific strategy the Context needs.



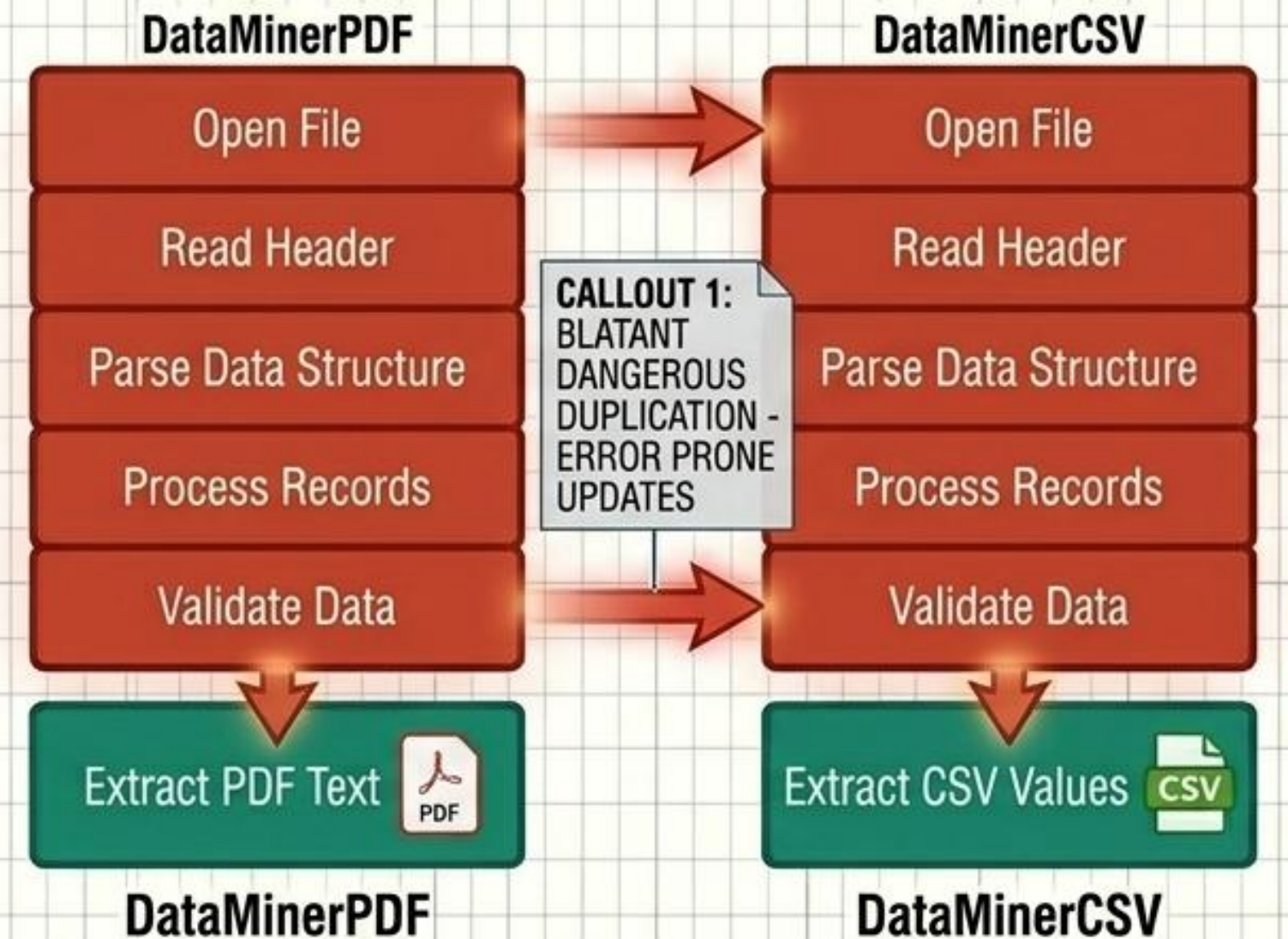
Callout 2: Strategies generally do not know about each other and do not trigger transitions, distinguishing this sharply from the State pattern.



TEMPLATE METHOD // DUPLICATED LOGIC VS. THE ALGORITHMIC SKELETON

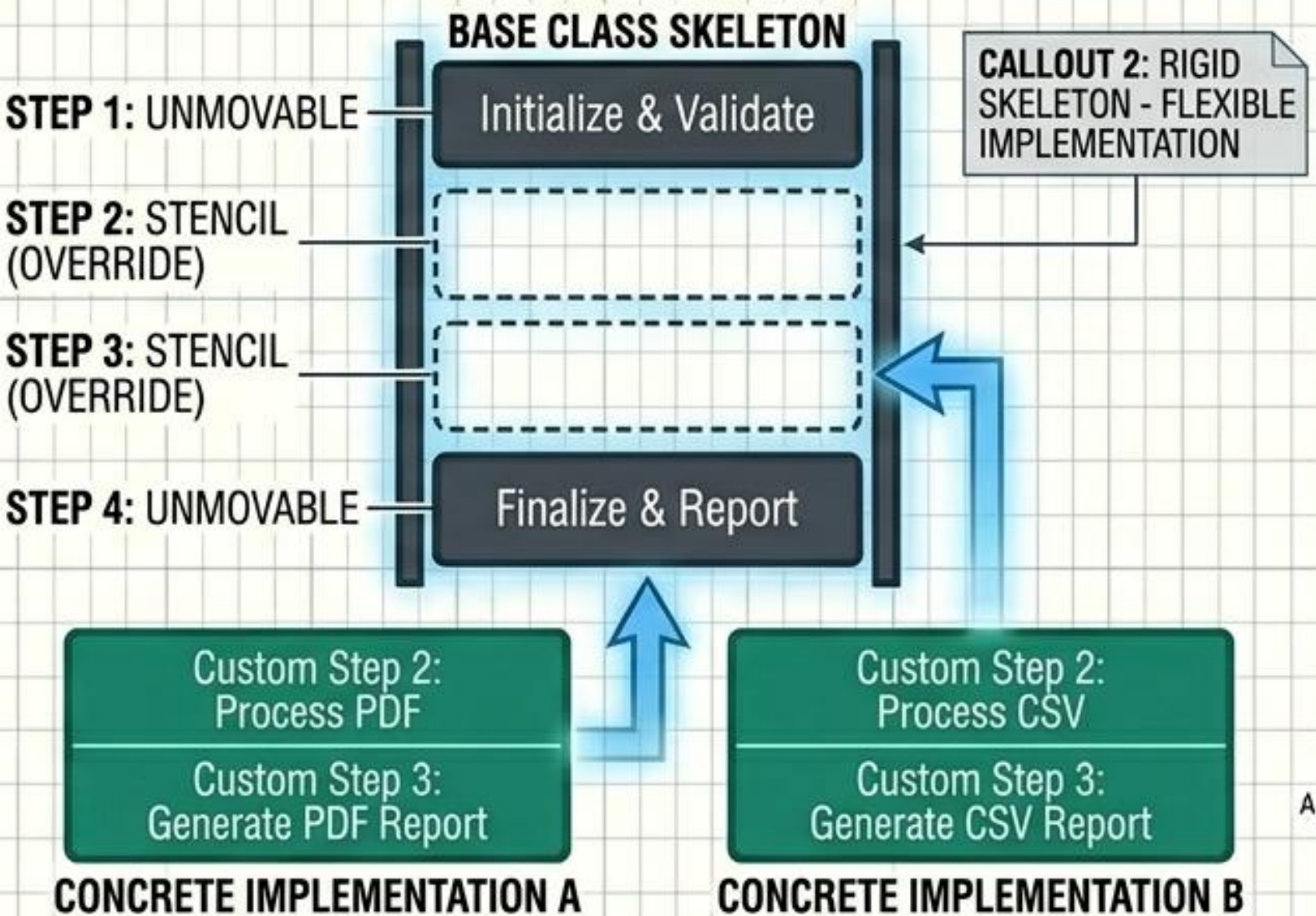
STRUCTURAL DUPLICATION

When several classes share identical overarching steps but differ in details, structural code is duplicated, making systemic updates incredibly error-prone.

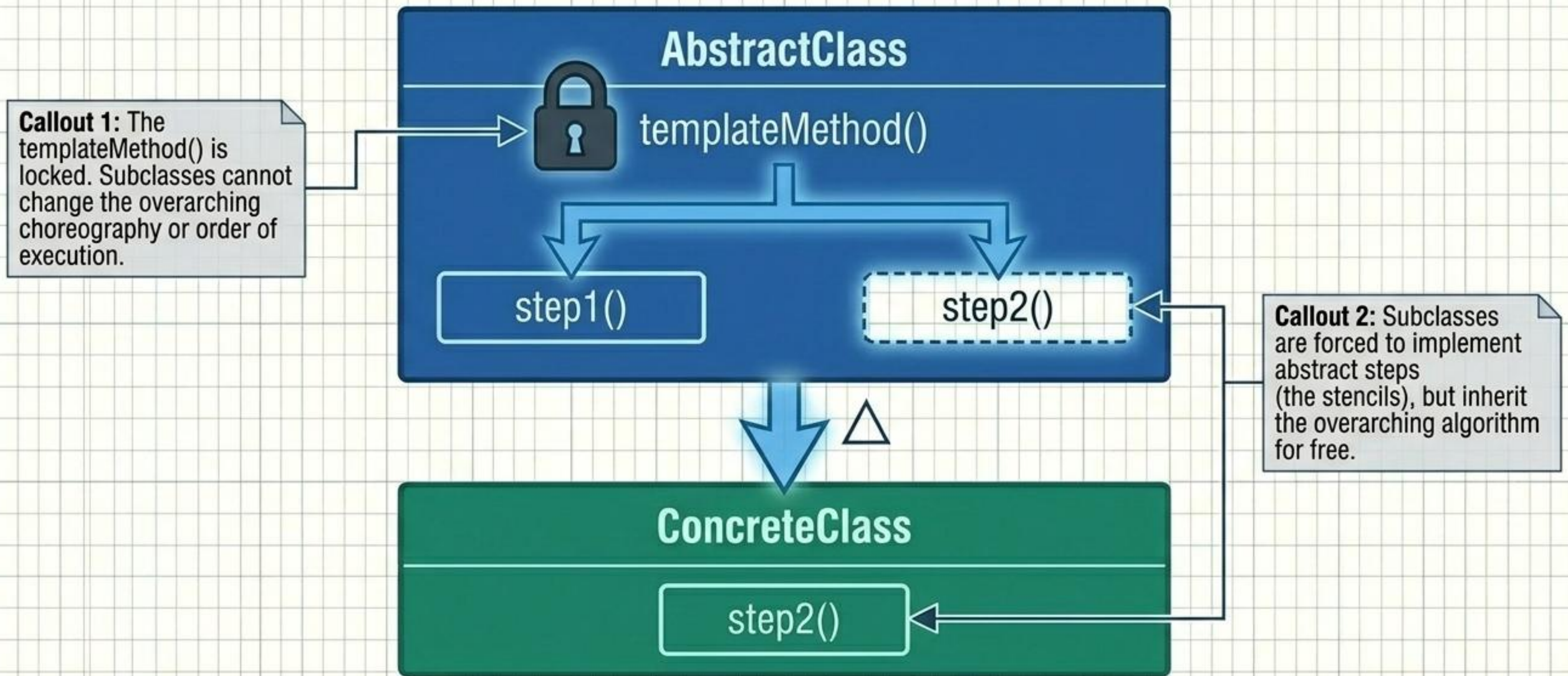


THE SKELETON OUTLINE

Define the skeleton of an algorithm in a base class method. Let subclasses override specific steps to customize behavior while the base class firmly dictates the sequence.

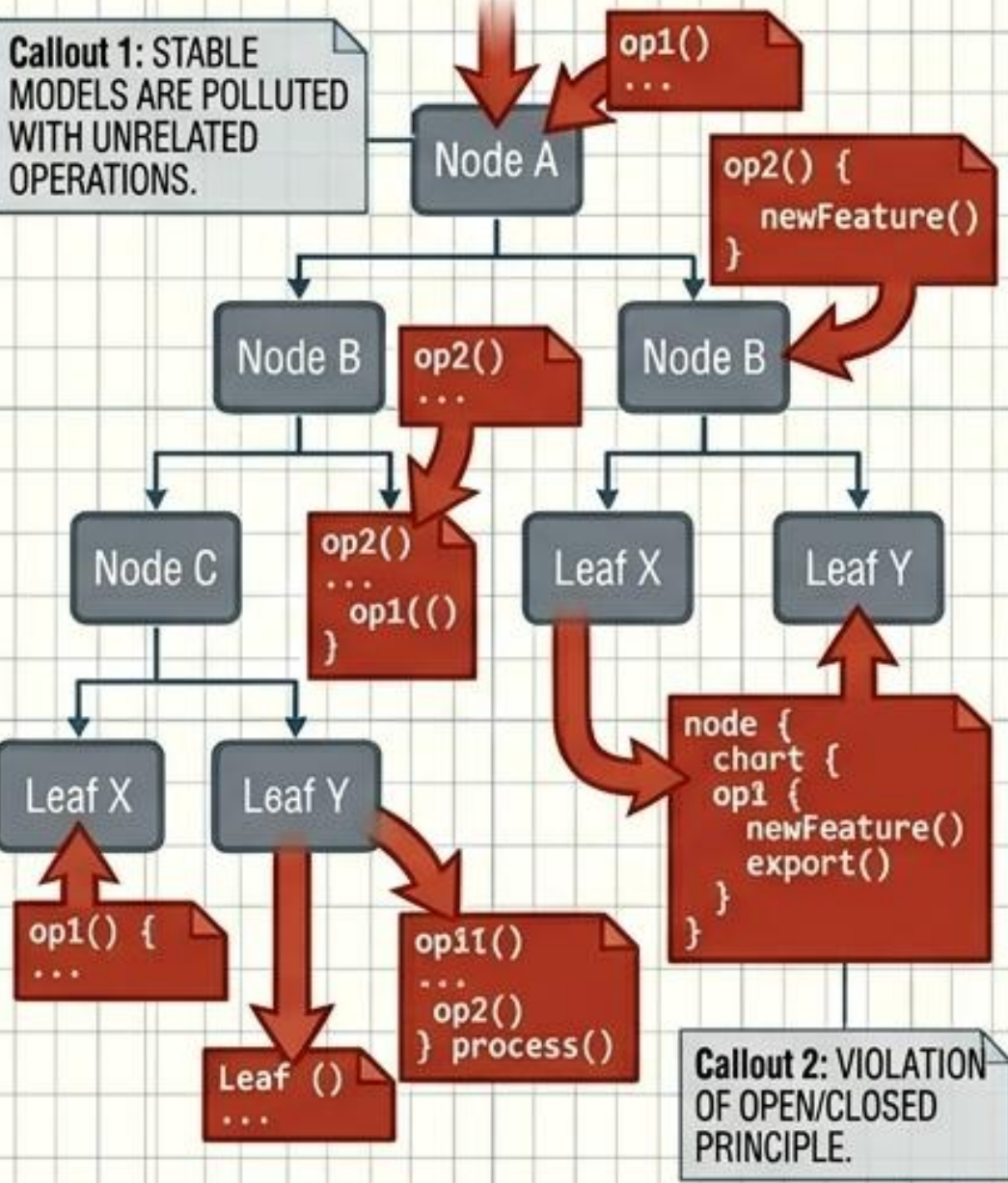


TEMPLATE METHOD // INTERNAL STRUCTURE

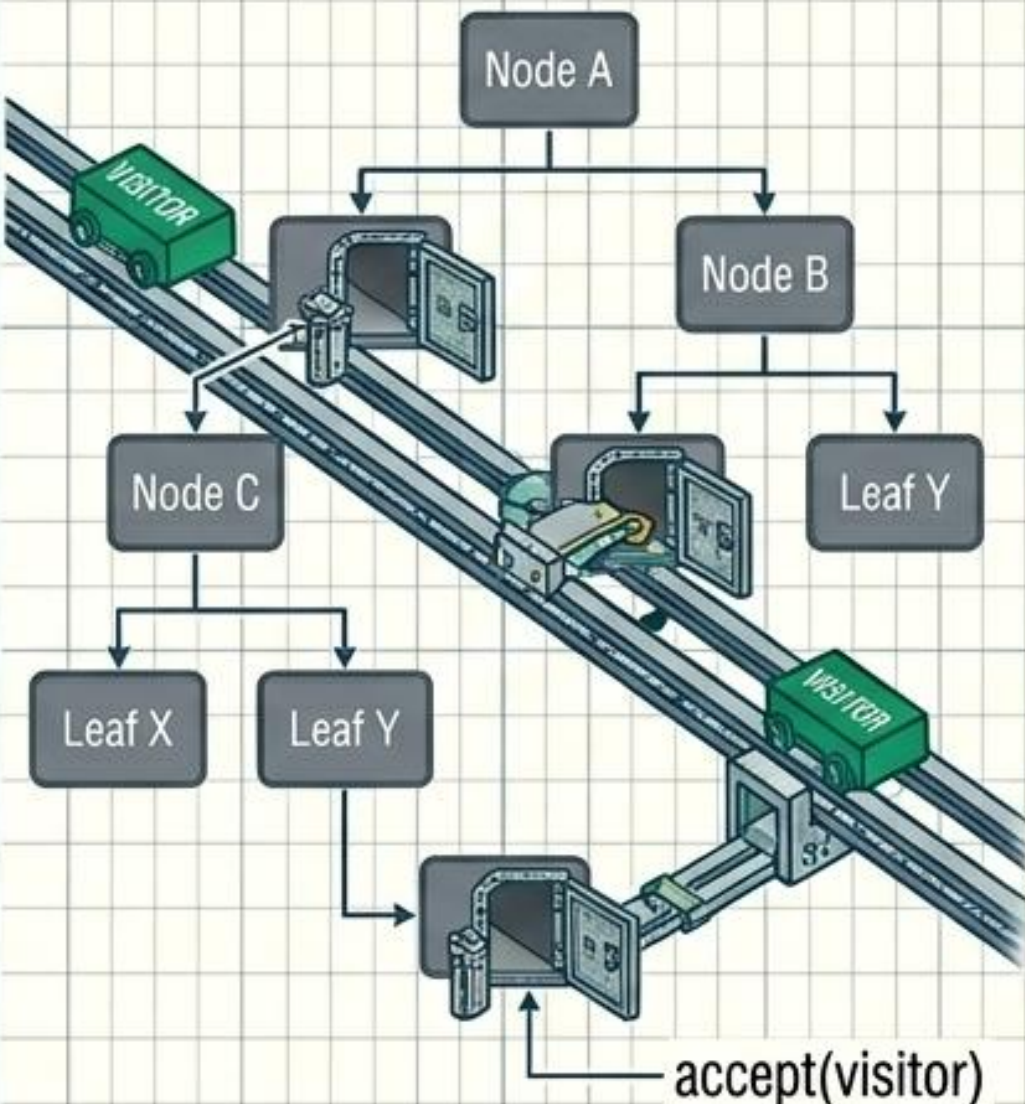


VISITOR // POLLUTING STRUCTURES VS. DOUBLE DISPATCH ROUTING

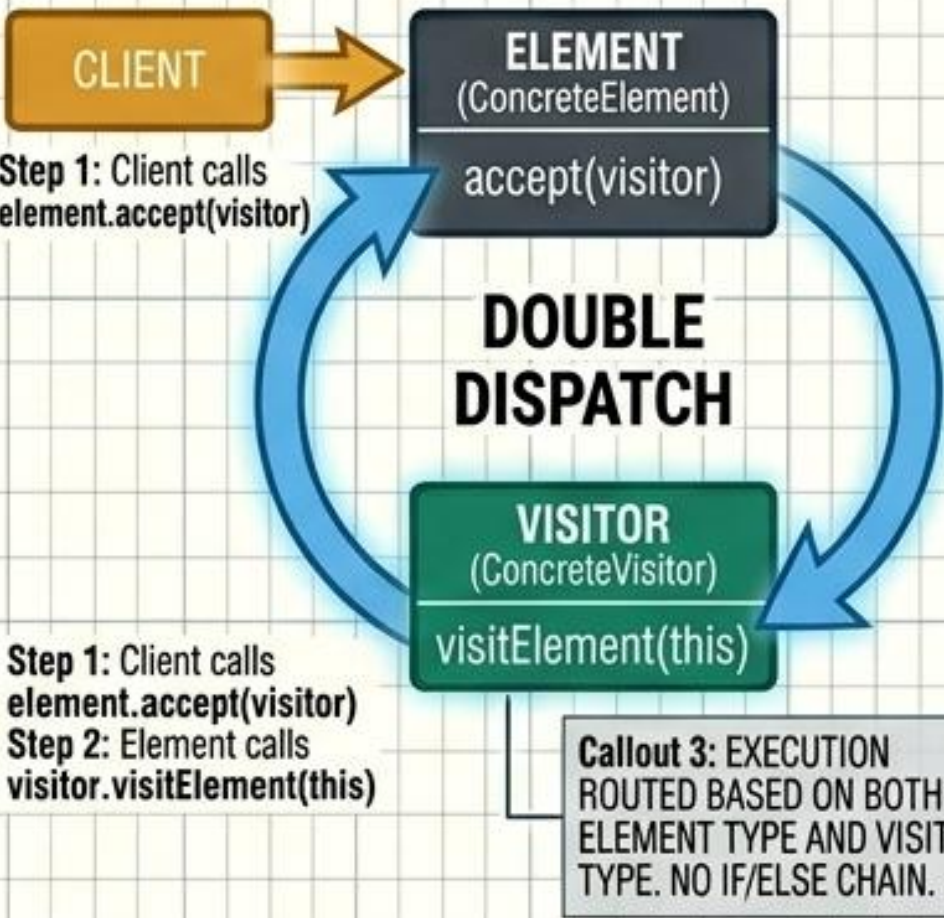
THE PROBLEM: Adding new operations to a complex object structure pollutes highly stable domain models.



THE SOLUTION: Extract the behavior into a Visitor object. Elements implement an accept(visitor) method.



DOUBLE DISPATCH: Step 1: Client routes to the Element. Step 2: Element routes back to the Visitor's specific method, guaranteeing the correct type is processed without massive if/else type-checking.



GLOBAL SYSTEMS ENGINEERING

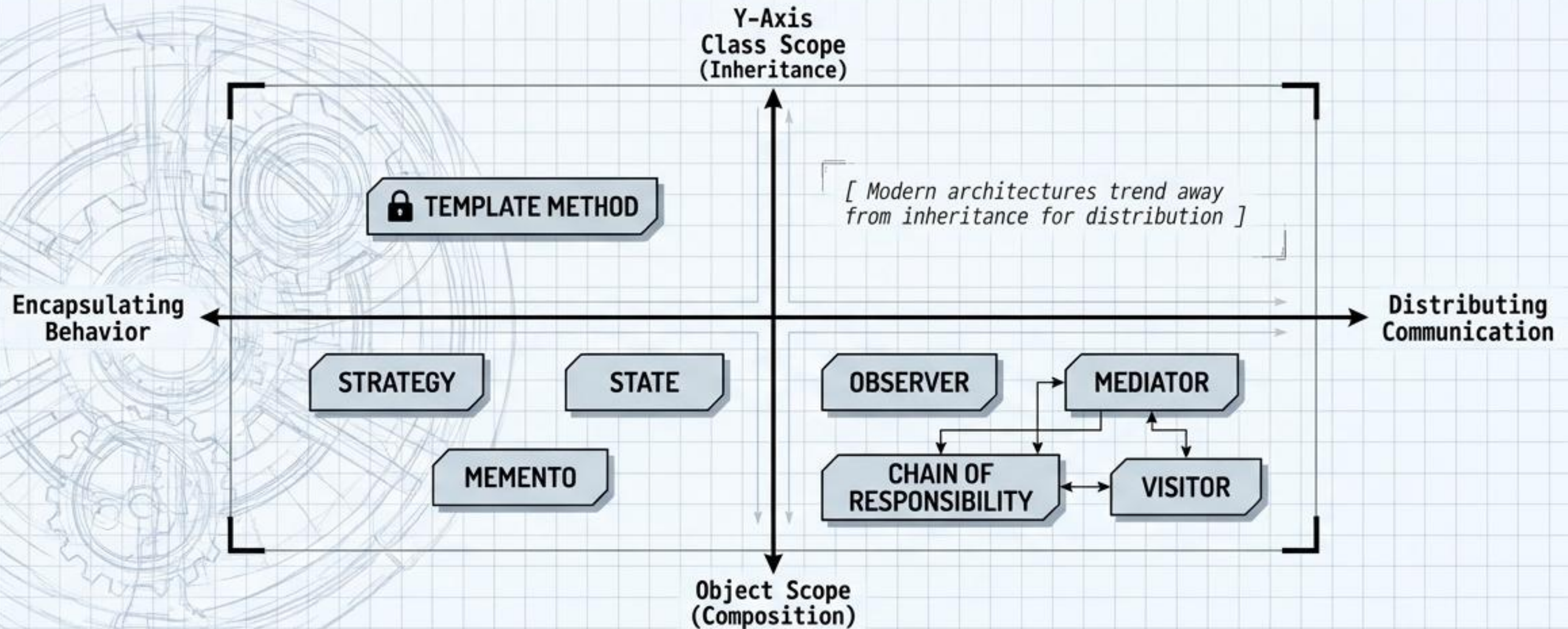
Specification: DESIGN PATTERNS // VISITOR
Specification Number: VSTR/DOUBDISP.REV01.2024

Description:
VISITOR PATTERN AND DOUBLE
DISPATCH MECHANISM

Rev. A Page. 1

SYNTHESIZING CHOREOGRAPHY // THE BEHAVIORAL DIAGNOSTIC MATRIX

Master this matrix to choose the precise architectural choreography your system requires. While Template Method locks behavior at compile-time via inheritance, most modern solutions leverage object composition for dynamic runtime flexibility.



“Refactoring transforms a mess into clean code and simple design.”
– Alexander Shvets, Refactoring.Guru